

Reducing Curtailment on the Irish Energy Grid

Team 3: Isabel Costello & Matthew Blakeney

September 11, 2026

Inertia on the Grid



Figure 1: Irish Energy Grid Power Sources, based on all-island transmission system in WP2033 - the winter-peak 2033 study case from EirGrid and SONI's Ten Year Transmission Forecast Statement 2024.

Effects to Avoid

- Frequency deviation from 50 Hz. (Must be within a few Hz.)
- Large rates of change of frequency. (Must average over every half-second to less than 0.5 Hz/s.)

Moment of inertia in rotational kinetic energy of synchronous generators minimises both!

Curtailment to maintain frequency

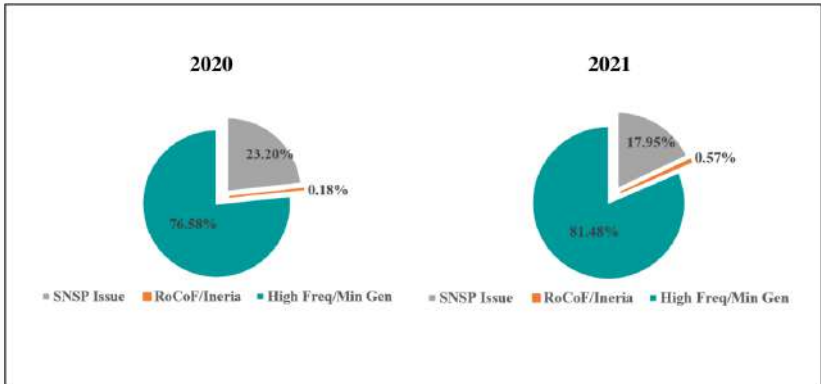


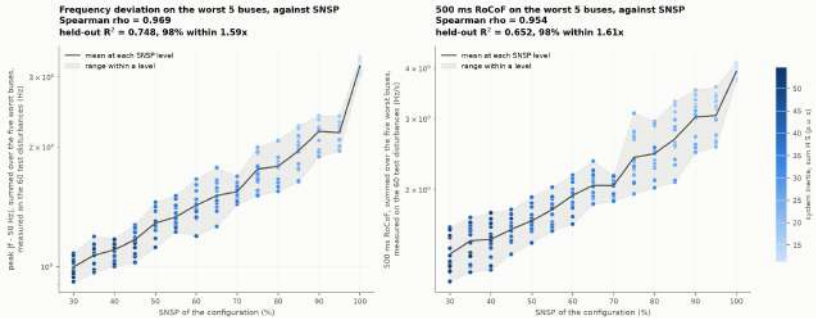
Figure 2: Reason code (in %) when curtailment was active, from *Analysis of Wind Energy Curtailment in the Ireland and Northern Ireland Power Systems* by Hurtado et al. Feb. 2023

Kuramoto model:

$$m_i \ddot{\theta}_i + \gamma_i \dot{\theta}_i = P_i + \sum_j K_{ij} \sin(\theta_j - \theta_i), \quad K_{ij} = a V_i V_j Y_{ij}. \quad (1)$$

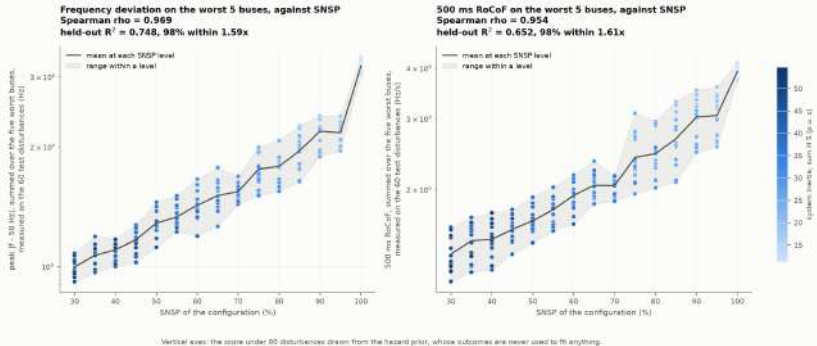
$$SNSP = \frac{\text{Asynchronous Power Generation}}{\text{Total Power Consumption}} \quad (2)$$

$$SNSP = \frac{\text{Asynchronous Power Generation}}{\text{Total Power Consumption}} \quad (2)$$



Vertical axes: the score under 60 disturbances drawn from the hazard prior, whose outcomes are never used to fit anything.

$$SNSP = \frac{\text{Asynchronous Power Generation}}{\text{Total Power Consumption}} \quad (2)$$



Initial intuition: variance around trend due to local swings in frequency.
 Measured: Only $\sim 12\%$ of variance explained by this.

Better Predictor of Risk

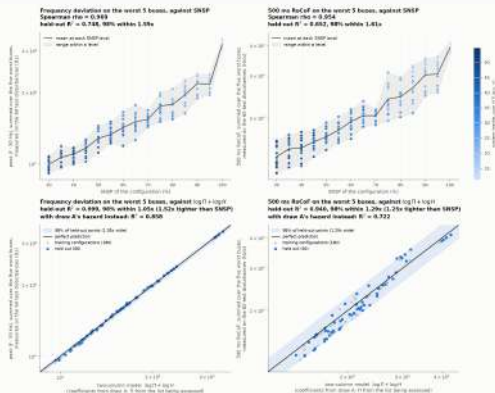
$$\left. \frac{df_{C.O.I.}}{dt} \right|_{t=0^+} = \frac{\sum_{\text{sites}} \Delta P_{(\text{site } i)}}{H_{C.O.I.}} \quad (3)$$

Use this as risk predictor:

Better Predictor of Risk

$$\left. \frac{df_{C.O.I.}}{dt} \right|_{t=0^+} = \frac{\sum_{\text{sites}} \Delta P_{(\text{site } i)}}{H_{C.O.I.}} \quad (3)$$

Use this as risk predictor:



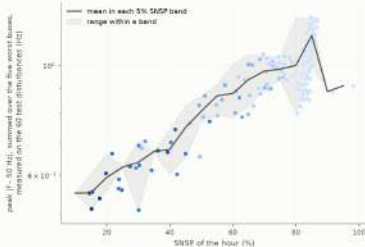
Vertical axis throughout: the y-axis is the log of the number of sites in the test-out set, against the log of the number of sites in the test-out set. The x-axis is the log of the number of sites in the test-out set.

Better Predictor of Risk - Irish Grid

Frequency deviation on the worst 5 buses, against SNSP

Spearman rho = 0.819

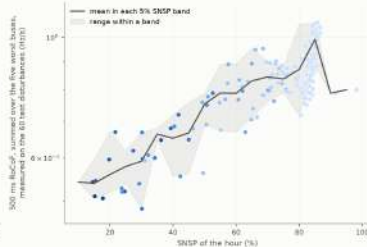
held-out $R^2 = 0.755$, 98% within 1.37x



500 ms RoCoF on the worst 5 buses, against SNSP

Spearman rho = 0.808

held-out $R^2 = 0.630$, 98% within 1.37x

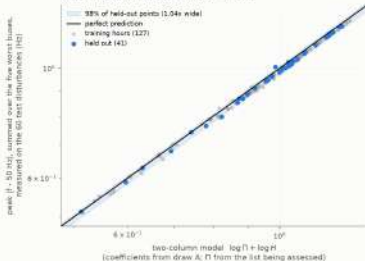


Frequency deviation on the worst 5 buses, against $\log \Pi + \log H$

held-out $R^2 = 0.995$, 98% within 1.04x (1.32x tighter than SNSP)

contiguous-block split: 0.993, against SNSP's 0.577

with draw A's hazard instead: $R^2 = 0.733$



500 ms RoCoF on the worst 5 buses, against $\log \Pi + \log H$

held-out $R^2 = 0.948$, 98% within 1.14x (1.21x tighter than SNSP)

contiguous-block split: 0.959, against SNSP's 0.545

with draw A's hazard instead: $R^2 = 0.591$

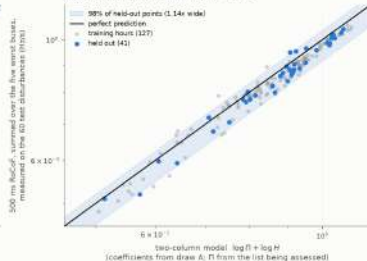




Figure 3: Gas turbine



Figure 4: Synchronous Condenser (SC)

S.C.s: Replacing Conventional Inertia



Figure 3: Gas turbine



Figure 4: Synchronous Condenser (SC)

Investment cost of a synchronous condenser ranges from €40m to €80m for a 2000MVAs energy storage. (AFRY financial report 2024)

Optimal location of reinforced inertia to stabilize power grids

Sangjoon Park,¹ Cook Hyun Kim,¹ and B. Kahng¹¹CCSS, KI for Grid Modernization, Korea Institute of Energy Technology, Naju, Jeonnam 58320, Korea
(Date: July 8, 2025)

The increasing adoption of renewable energy sources has significantly reduced the inertia in the modernized power grid, making the system more vulnerable. One way to stabilize the grid is to add extra inertia from unused turbines, called the fast frequency response (FFR), to the existing grid. However, reinforcing inertia can cause unintended consequences, such as more significant avalanche failures. This phenomenon is known as the Braess paradox. Here, we propose a method to find the optimal position of FFR. This method is applied to the second-order Kuramoto model to find an effective position to mitigate cascading failures. To address this, we propose a method to evaluate a ratio between the positive effects of mitigation and the negative consequences. Through this analysis, we find that the peripheral area of the network is a seemingly effective location for inertia reinforcement across various reinforcement scales. This strategy provides essential insights for enhancing the stability of power grids in a time of widespread renewable energy usage.

I. INTRODUCTION

The power grid is essential for sustaining modern lives, including transportation, healthcare instruments, and food supply. Increasing environmental concerns have recently accelerated the adoption of renewable energy-based power generation systems [1–6]. While this transition reduces dependence on fossil fuels, it has significantly decreased the inertia of power grids, leading to a desperate need for reinforcement strategies [7, 8]. Inertia plays a crucial role in reducing frequency fluctuations before control systems can respond, helping to prevent system instability. Without sufficient inertia, the grid becomes more vulnerable to frequency variations, resulting in large-scale blackouts [9–11].

Various methods have been proposed to enhance the stability of power grids, such as increasing transmission line capacity or connecting new lines. However, these studies have shown that reinforcement may sometimes cause overloads, ultimately reducing grid stability rather than improving it [12–16]. Similarly, while expected to enhance stability, inertia reinforcement may paradoxically exacerbate cascading failures, leading to unintended consequences.

This study investigates how the location and strength of inertia reinforcement affect cascading failure sizes in the swing equation, a sort of the second-order Kuramoto model [17]. Our findings reveal that reinforcement can mitigate cascading failures but may also increase their failure size under certain situations. This underscores the importance of identifying optimal reinforcement locations. To evaluate the effects of reinforcement, we measure the ratio of mitigation to adverse outcomes. This ratio decreases as the severity of the Braess paradox increases; however, it increases when mitigation effects are substantial.

By analyzing this ratio across various locations and reinforcement scales, we demonstrate that a topological metric, defined as the combination of mean shortest path length and degree, is a key indicator for identifying opti-

mal reinforcement locations. The results indicate that this metric exhibits a high Pearson correlation coefficient [18] with the ratio, further supporting its relevance in reinforcement placement. To validate the effectiveness of this topological metric, we classify power plants into three groups based on their metric values: central group, peripheral group, and other group. We then analyze the distribution of the most effective reinforcement locations among these groups to determine which category contains the most optimal reinforcement sites.

The remainder of this paper is organized as follows: Section II introduces the second-order Kuramoto model and the avalanche process. The distribution of cascading failure sizes is obtained for various frequency thresholds. Section III defines inertia reinforcement strategies and examines their mitigation and adverse effects by measuring cascading failures. Section IV proposes a metric for evaluating reinforcement strategies and presents an optimized approach. Finally, Section V provides conclusions and discusses the implications of this study.

II. SECOND-ORDER KURAMOTO MODEL WITH AVALANCHE PROCESS

The generalized second-order Kuramoto model [17] for phase and frequency synchronizations is expressed as follows:

$$m_i \ddot{\theta}_i + \gamma_i \dot{\theta}_i = P_i + \sum_{j \in \mathcal{N}(i)} K_{ij} \sin(\theta_j - \theta_i), \quad (1)$$

where θ_i and $\dot{\theta}_i$ represent the phase and angular velocity of bus i , respectively. The index i denotes the bus index and runs $i = 1, \dots, N$. The parameters m_i and P_i denote the inertia of bus i , respectively. P_i can be positive for generators and negative for consumers. γ_i is the dissipation coefficient, and K_{ij} is the coupling strength between neighboring buses. As time passes, the system reaches a steady state where the power at each bus balances with the interaction term.

S.C. Placement - Ireland

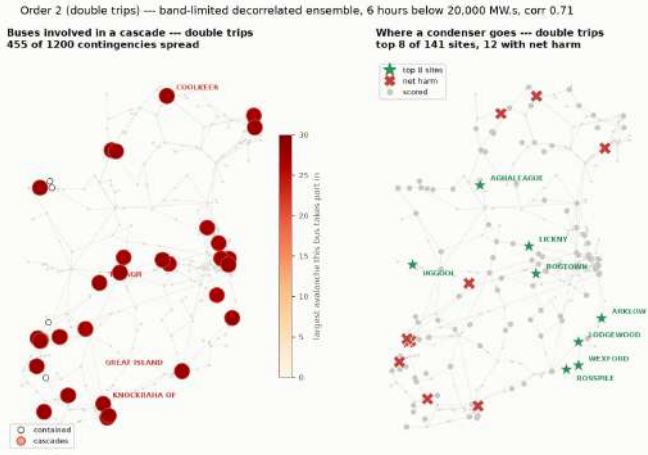
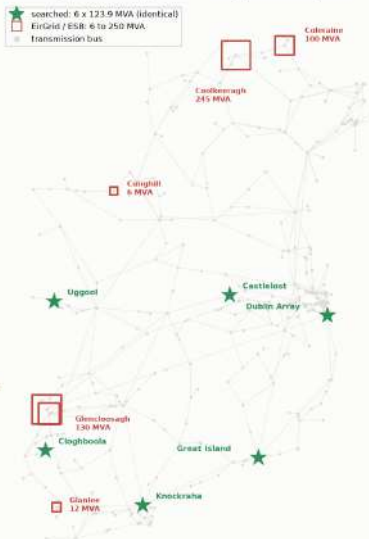


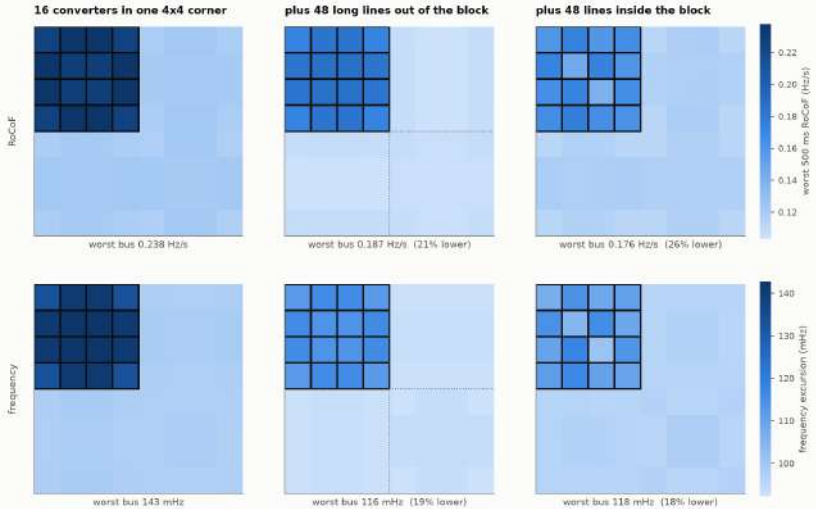
Figure 6: Cascades (left) and appropriate SC sites (right) for double trips.

Optimal Placement of 6 SC's on the Irish Grid

Where six synchronous condensers should go
searched placement against the real procurement, same 12,138 MW.s of inertia

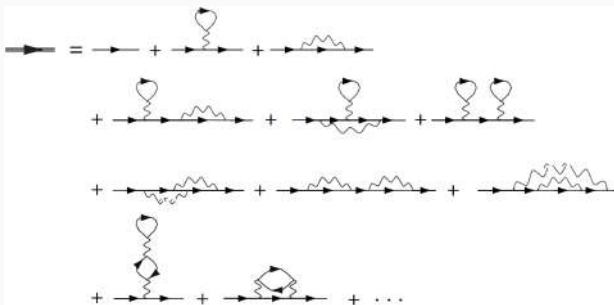


Additional lines reduce risk: Intuition

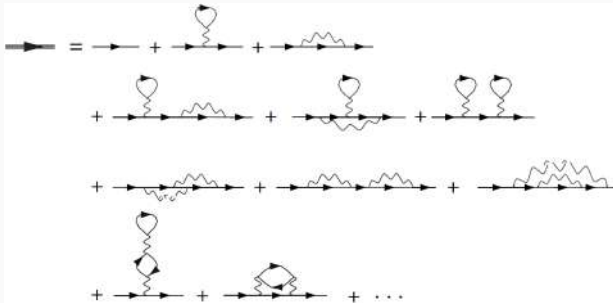


$$\Delta\theta_i = \sum_j G_{i,j} \Delta P_j \quad (4)$$

'Renormalised Mass' & Form Factors



'Renormalised Mass' & Form Factors



In Fermi liquid theory, this produces a 'renormalised mass' - different to the mass originally written into the theory, and dependent on the energy at which it is measured.

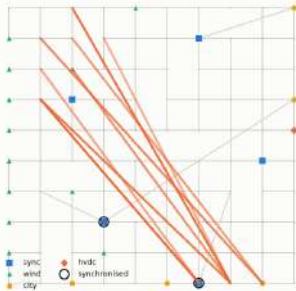
$$C_{\text{freq}}(a, b) = \max_{\omega \in [\omega_1, \omega_2]} \sigma_{\max}(\omega \Pi G_{\text{new}}(i\omega) B)$$

$$C_{\text{acc}}(a, b) = \max_{\omega} \sigma_{\max}(\omega^2 \Pi G_{\text{new}}(i\omega) B).$$

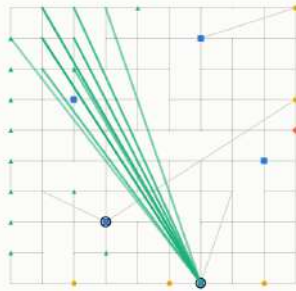
(Maximum singular value as norm.)

Additional Lines to Reduce Risk

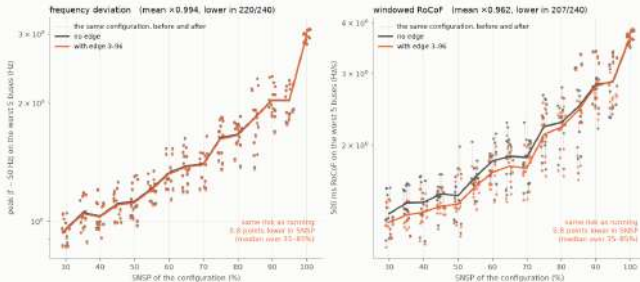
best B by frequency deviation



best B by windowed RoCoF



Additional Lines to Reduce Risk



Further Applications of Greens' Function Approach?

- Greens' function-based risk predictor?
- Greens' function-based cascade predictor?

(Linearised) Kuramoto model in Laplace space:

$$M s^2 \Delta\theta + \Gamma s \Delta\theta + L \Delta\theta = \Delta P \quad (5)$$

(Linearised) Kuramoto model in Laplace space:

$$M s^2 \Delta\theta + \Gamma s \Delta\theta + L \Delta\theta = \Delta P \quad (5)$$

$$A(s) = Ms^2 + \Gamma s + L, \text{ and } G = A^{-1}. \quad (6)$$

Extra: 'Renormalised Mass'

(Linearised) Kuramoto model in Laplace space:

$$M s^2 \Delta\theta + \Gamma s \Delta\theta + L \Delta\theta = \Delta P \quad (5)$$

$$A(s) = Ms^2 + \Gamma s + L, \text{ and } G = A^{-1}. \quad (6)$$

$$\hat{A}_{ii} = A_{ii} + \sum_{j,k} A_{i,j} A_{j,k}^{-1} A_{k,i} \quad (7)$$

Extra: 'Renormalised Mass'

(Linearised) Kuramoto model in Laplace space:

$$M s^2 \Delta\theta + \Gamma s \Delta\theta + L \Delta\theta = \Delta P \quad (5)$$

$$A(s) = Ms^2 + \Gamma s + L, \text{ and } G = A^{-1}. \quad (6)$$

$$\hat{A}_{ii} = A_{ii} + \sum_{j,k} A_{i,j} A_{j,k}^{-1} A_{k,i} \quad (7)$$

The term in $\hat{A}_{ii}$ that comes with s^2 is the renormalised mass. This is the full mass that opposes fluctuations at i as a result of power injection at i .

Stochastic Constraint Grouping through Simulated Annealing



Trinity
College
Dublin

The University of Dublin



Seán Dignam, Cian Gregg, Saoirse Serna
Team 11

$$P_{\text{acc}} = \begin{cases} 1, & \Delta E \leq 0, \\ \exp\left(-\frac{\Delta E}{T}\right), & \Delta E > 0, \end{cases}$$

$$T \rightarrow \alpha T$$

Dispatch downs

Dispatch Down

1. Curtailment
2. Constraint
3. Surplus



How its done.

Other Countries

1. Nodal Pricing/Locational
Marginal Pricing

2. Zonal pricing



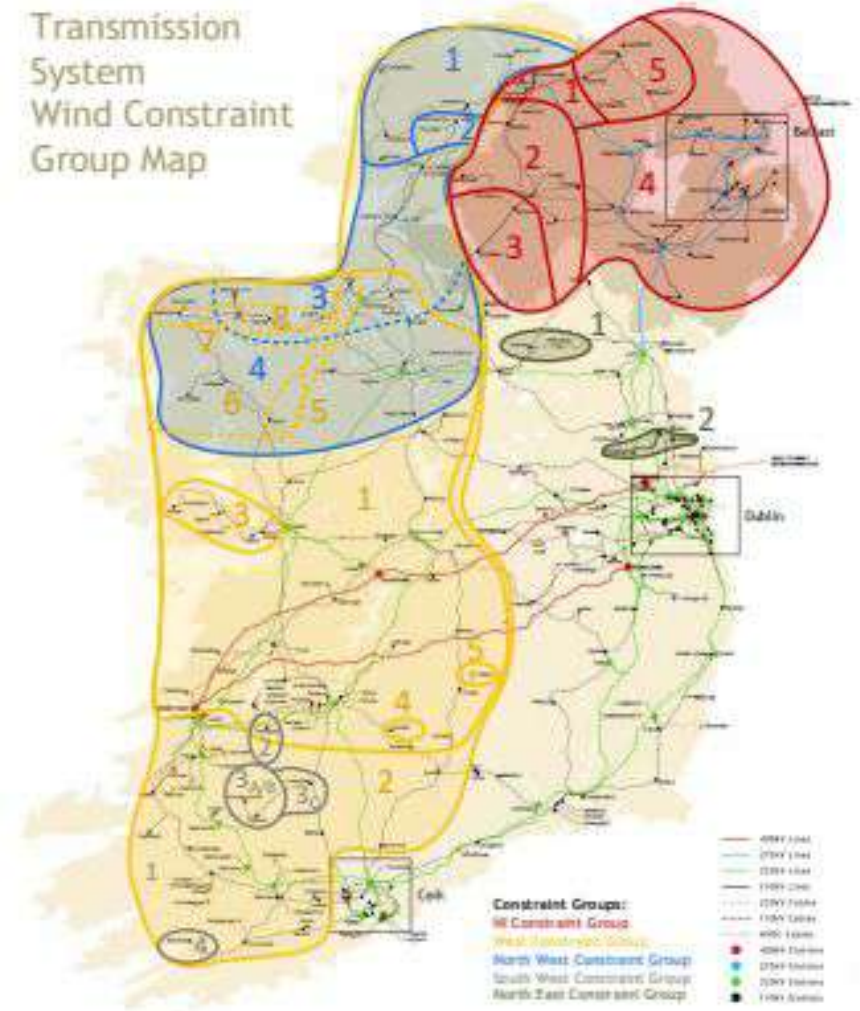
Why not for ireland

- Ireland has a single electricity market where every generator is paid the same wholesale price, meaning the market cannot be used to constrain generation.
- Instead the TSO is in charge of dispatching down generators to satisfy the network's constraints.



Constraint Groups

- Constraint Groups are constructed out of multiple generators which feed into the same transition lines.
- For market fairness all generators in a group are dispatched down equally.



Presentation of the Problem

Static Constraint groups overestimate the amount of the grid which needs to be dispatched down.

Our hypothesis was, that dynamically changing and stochastically spreading the groups will decrease the amount of necessary Dispatch Down.

To do this two things are needed: a pseudo-model of the grid, and an optimisation of the groups.

Modelling Ireland's Grid

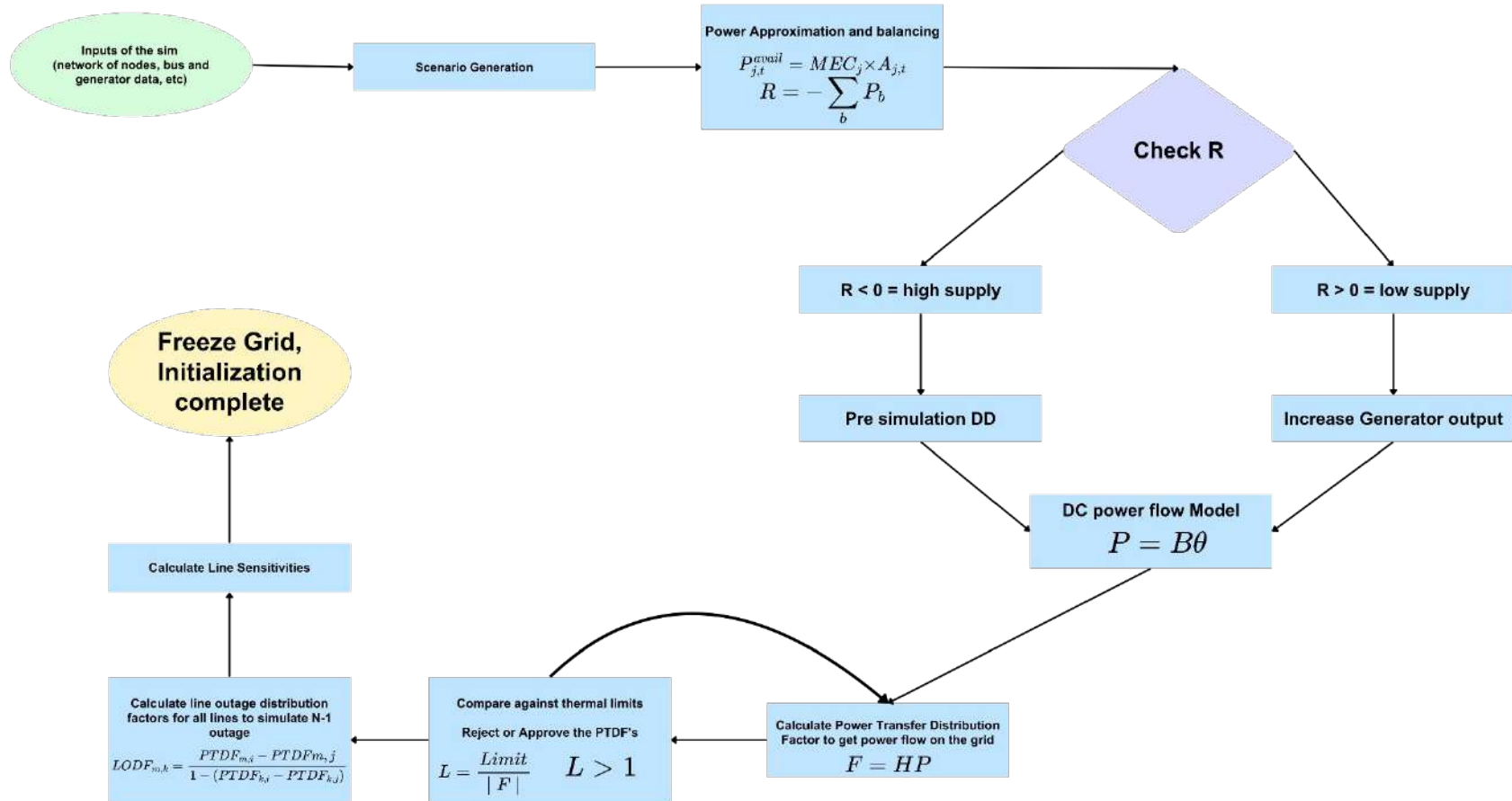
The Grid

The parts of the grid which are important for optimising constraint groups are the renewable energy sector of the grid, and the edges that connect them all together

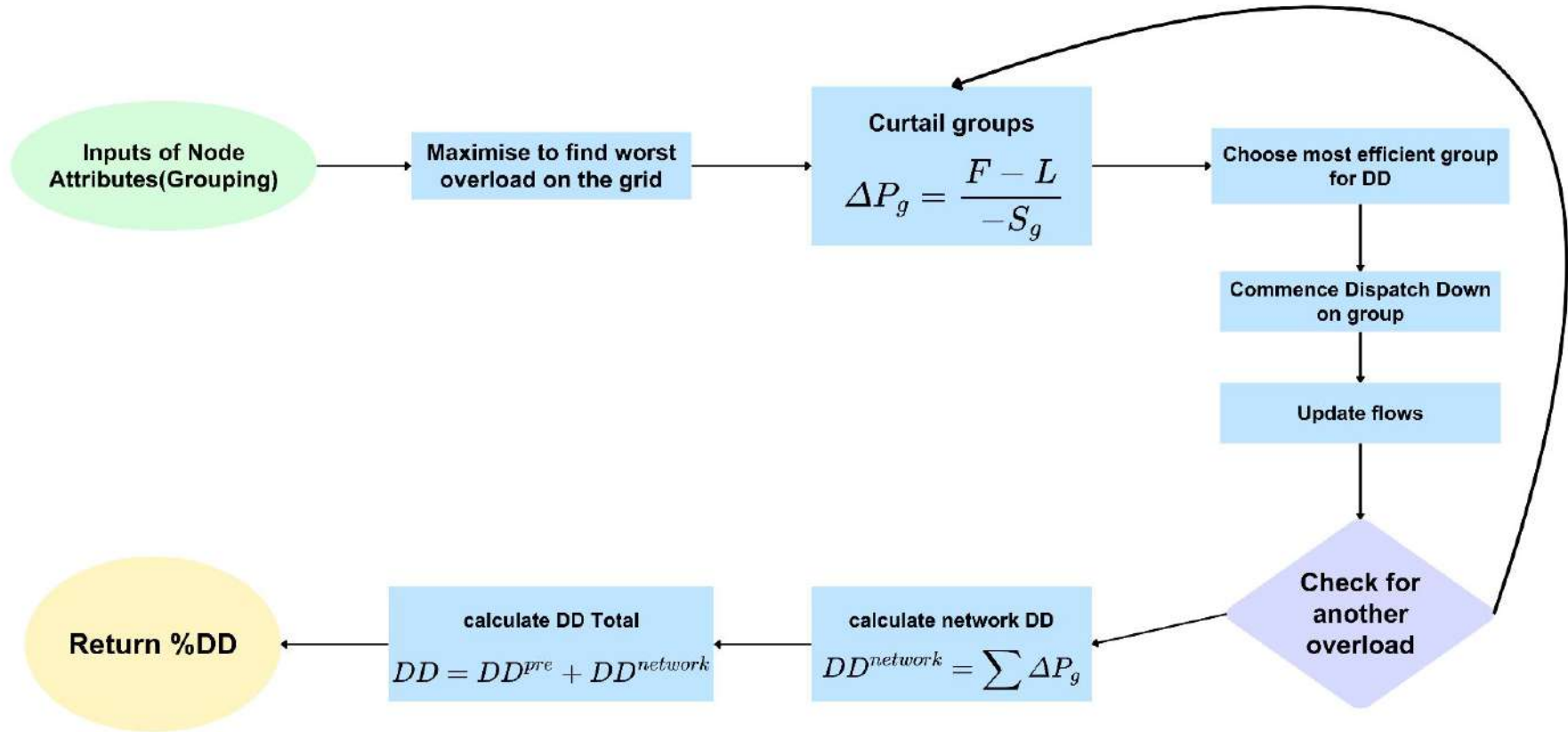
Demand is needed to simulate a surplus in energy as well as a high amount of generation.

We also need some details about the transmission, such as phase angle, voltage, and thermal rating

The initialization of the Simulation, A simplification



The running of the Simulation, A simplification



Assumptions

**Dispatch-down reduction is treated as
improved renewable utilisation, not
electrical-efficiency improvement**

No inertia constraints

N-1 only

DC power flow

Interconnectors Ignored*

Shared/common wind profile assumption

Simplified solar daylight profile

**Each renewable asset belongs to exactly one
constraint group**

*the connections between northern ireland and the republic are included

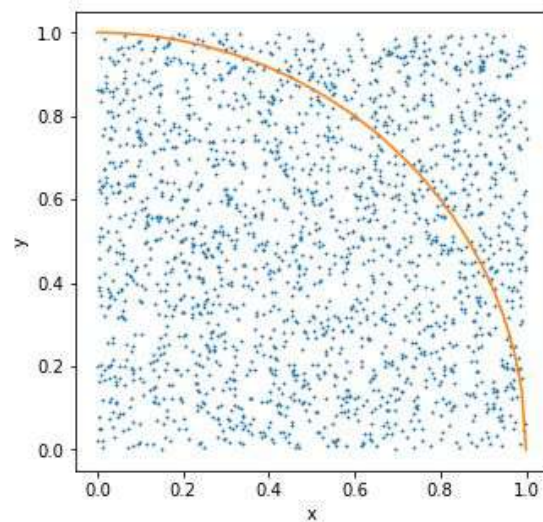
Screened N-1 contingency analysis	DC power flow	10,000 Monte Carlo runs reweight those 168 snapshots
Security screen frozen from the baseline case	Fixed voltage magnitudes	Simplified conventional merit-order dispatch
Security guard is baseline-relative	Reactive power ignored	Fixed conventional minimum output
N-1 only	Small-angle / linear power-flow approximation	No start-up constraints
No dynamic stability modelling	Resistance ignored in the actual power-flow calculation	No minimum up/down-time constraints
No inertia constraints	Thermal MVA ratings approximated as MW limits	Largest conventional generator bus used as the balancing bus
No reserve constraints	Transformer tap-ratio effects omitted	Proportional curtailment within each selected group
No system-services constraints	Transformer phase-shift effects omitted	Each renewable asset belongs to exactly one constraint group
No voltage-security modelling	Only the largest connected AC component is retained	Constraint-group sizes are fixed during annealing
No reactive-power security modelling	HVDC flows are fixed	Total number of constraint groups is fixed
Dispatch-down percentage is the optimisation objective	Synthetic renewable weather profiles	Greedy worst-constraint relief
No ramp-rate constraints	Shared/common wind profile assumption	Maximum number of group-curtailment actions per snapshot
Dispatch-down reduction is treated as improved renewable utilisation, not electrical-efficiency improvement	Simplified solar daylight profile	168 underlying demand snapshots

Simulated Annealing.

Monte Carlo Methods

$$\sigma_n(f(x)) = \frac{1}{n} \sum_i^n f(x^{(i)})$$

$$x^{(i)} \in p(x)$$



Markov Chains

$$P(X_{n+1} = x \mid X_1 = x_1, X_2 = x_2, \dots, X_n = x_n) = P(X_{n+1} = x \mid X_n = x_n)$$

“The next state only depend on the previous state, not the history on how we got there”

$$\pi(\sigma)P(\sigma'|\sigma) = \pi(\sigma')P(\sigma|\sigma')$$

Splitting the transition probability $P(\sigma|\sigma')$ into a proposal distribution $g(\sigma'|\sigma)$, the probability of *proposing* σ' given σ , and an acceptance rate $A(\sigma'|\sigma)$ the probability of *accepting* σ' , gives us

$$\pi(\sigma)g(\sigma'|\sigma)A(\sigma'|\sigma) = \pi(\sigma')g(\sigma|\sigma')A(\sigma|\sigma')$$

Detailed balance holds if we define,

$$A(\sigma'|\sigma) = \min \left(1, \frac{\pi(\sigma')g(\sigma|\sigma')}{\pi(\sigma)g(\sigma'|\sigma)} \right)$$

Now inserting the Boltzmann distribution for π

$$\begin{aligned} A(\sigma'|\sigma) &= \min \left(1, e^{-\beta(E(\sigma')-E(\sigma))} \frac{g(\sigma|\sigma')}{g(\sigma'|\sigma)} \right) \\ &= \min \left(1, e^{-\beta\Delta E} \underbrace{\frac{g(\sigma|\sigma')}{g(\sigma'|\sigma)}}_{=1} \right) \end{aligned}$$

$$A(\sigma'|\sigma) = \min \left(1, e^{-\beta\Delta E} \right)$$

1. Choose an initial grouping

$$G^{(0)} = \{g_1^{(0)}, g_2^{(0)}, \dots, g_n^{(0)}\}, \quad g_i^{(0)} \in \{1, \dots, K\}.$$

2. Propose G' by swapping the groups of two nodes,

$$g_i = A, \quad g_j = B \quad \longrightarrow \quad g'_i = B, \quad g'_j = A.$$

3. Define the energy as the simulated dispatch-down,

$$E(G) = D(G).$$

4. Accept G' according to

$$P_{\text{acc}} = \begin{cases} 1, & \Delta E \leq 0, \\ \exp\left(-\frac{\Delta E}{T}\right), & \Delta E > 0, \end{cases} \quad \Delta E = E(G') - E(G).$$

5. Update

$$G^{(m+1)} = \begin{cases} G', & \text{accepted,} \\ G^{(m)}, & \text{rejected.} \end{cases}$$

6. Repeat while cooling,

$$T_{r+1} = \alpha T_r.$$

7. Record the lowest energy found,

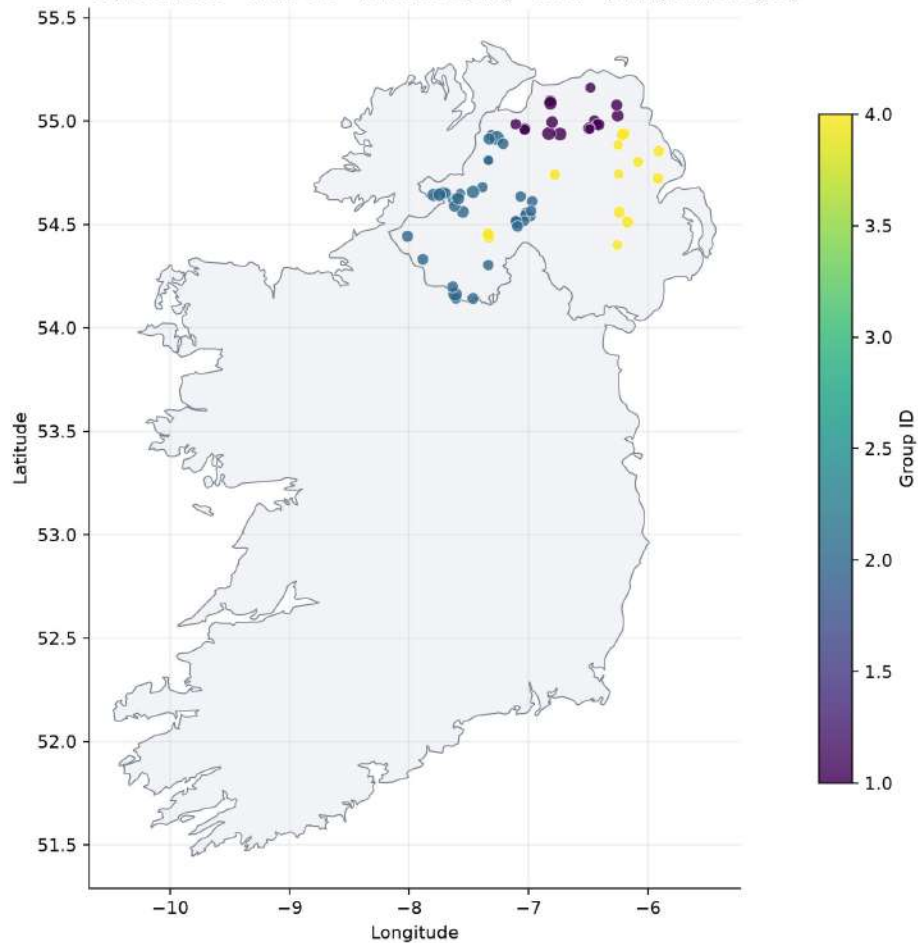
$$E_{\text{best}} = \min_m E(G^{(m)}),$$

and compare it with the baseline grouping.

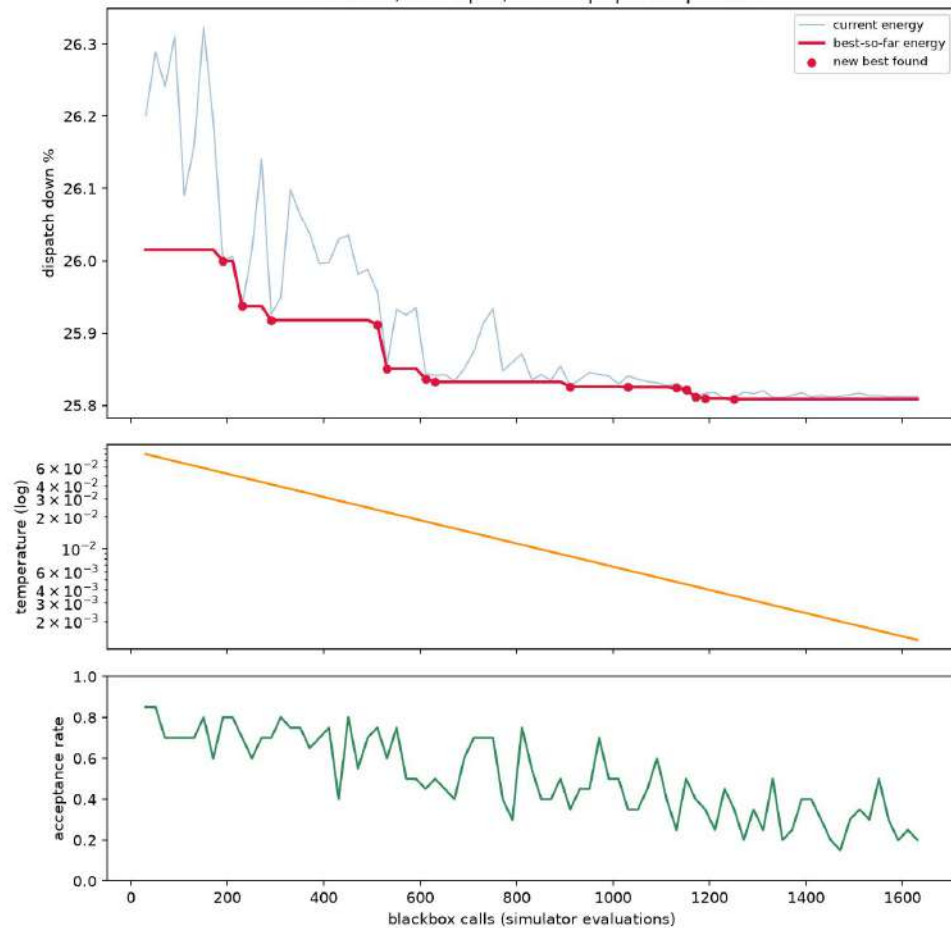
6 counties.

6-county renewable nodes

66 plotted nodes • marker size = renewable capacity • colour = primary/exclusive group



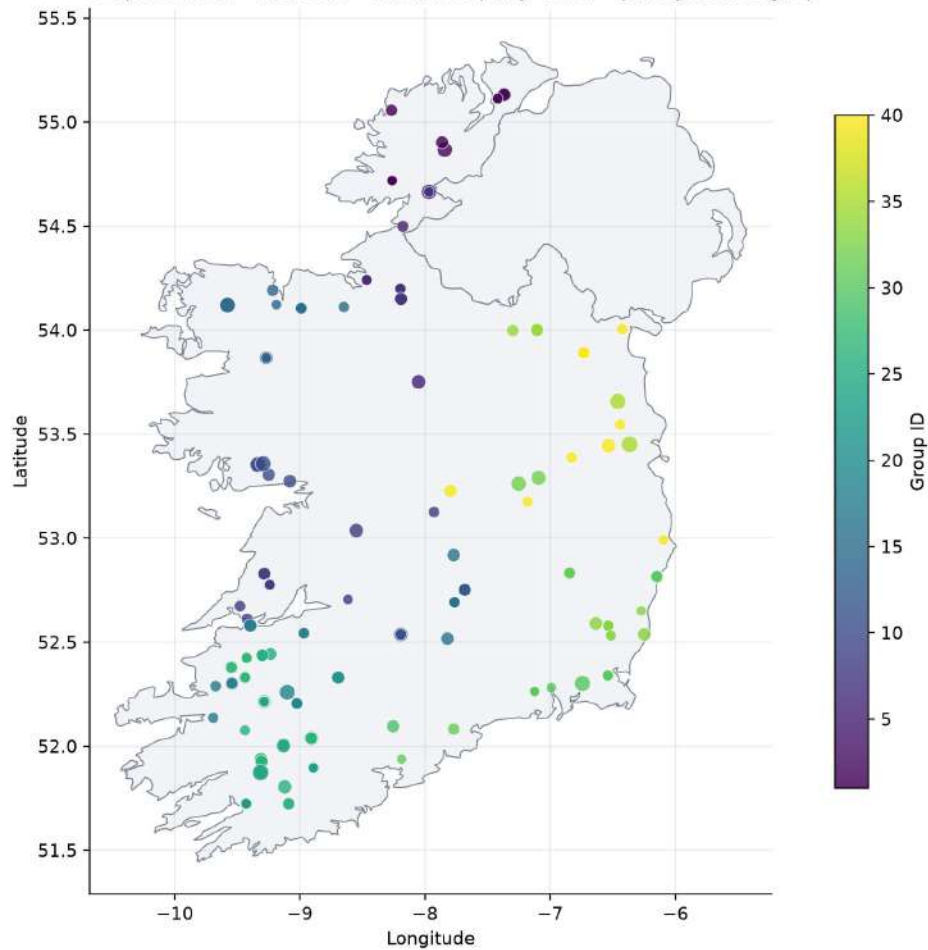
6 counties, 0.95 alpha, 20 sweeps per temperature



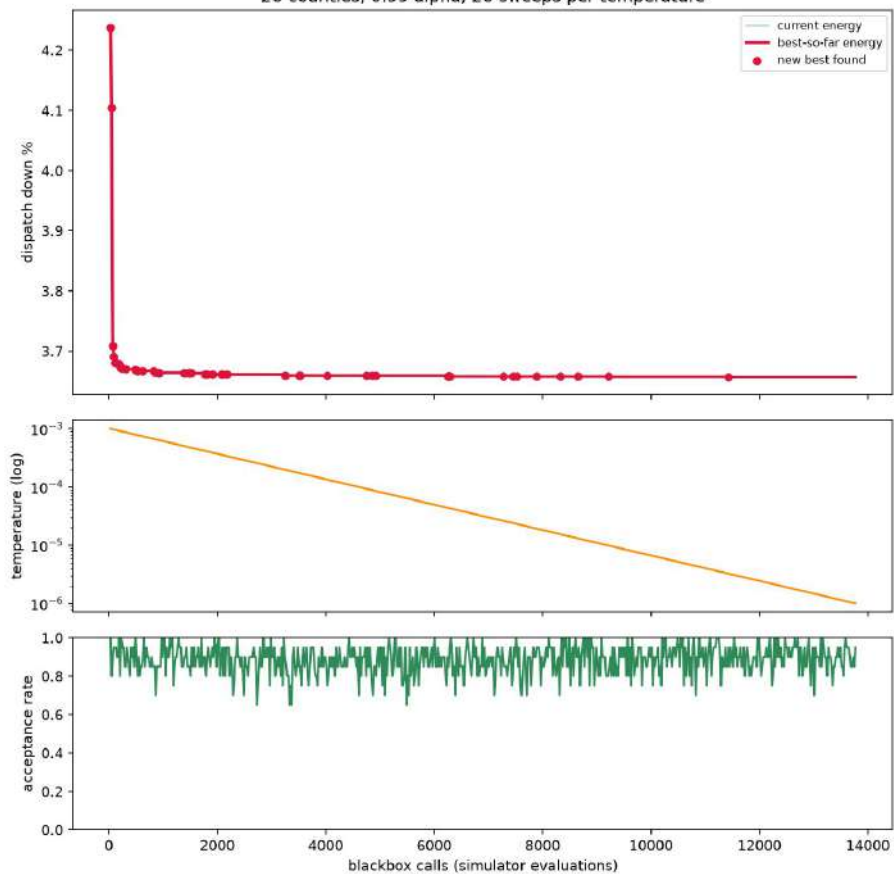
26 counties.

26-county renewable nodes

174 plotted nodes • marker size = renewable capacity • colour = primary/exclusive group



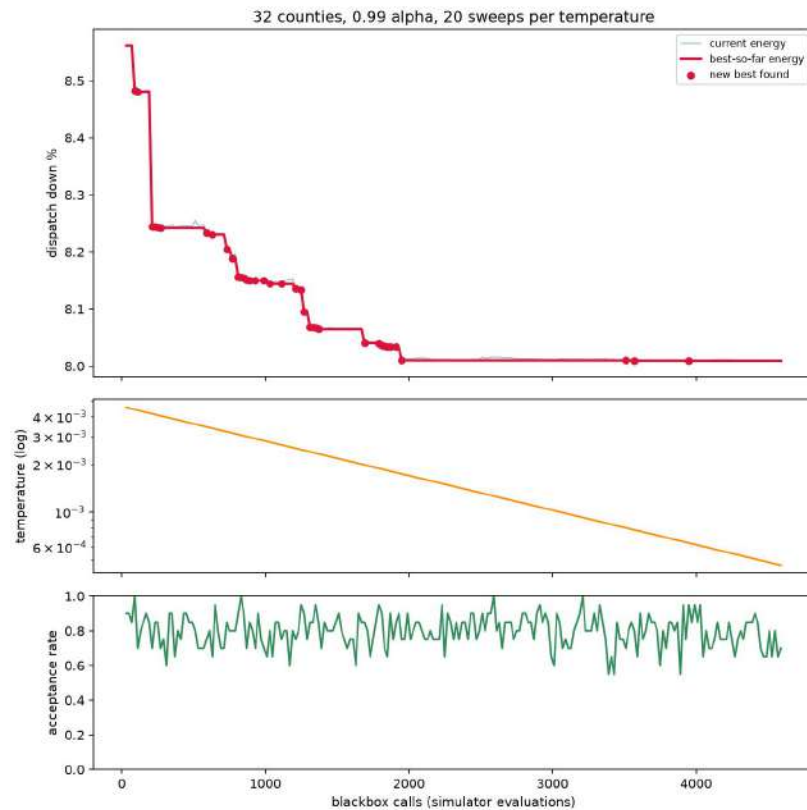
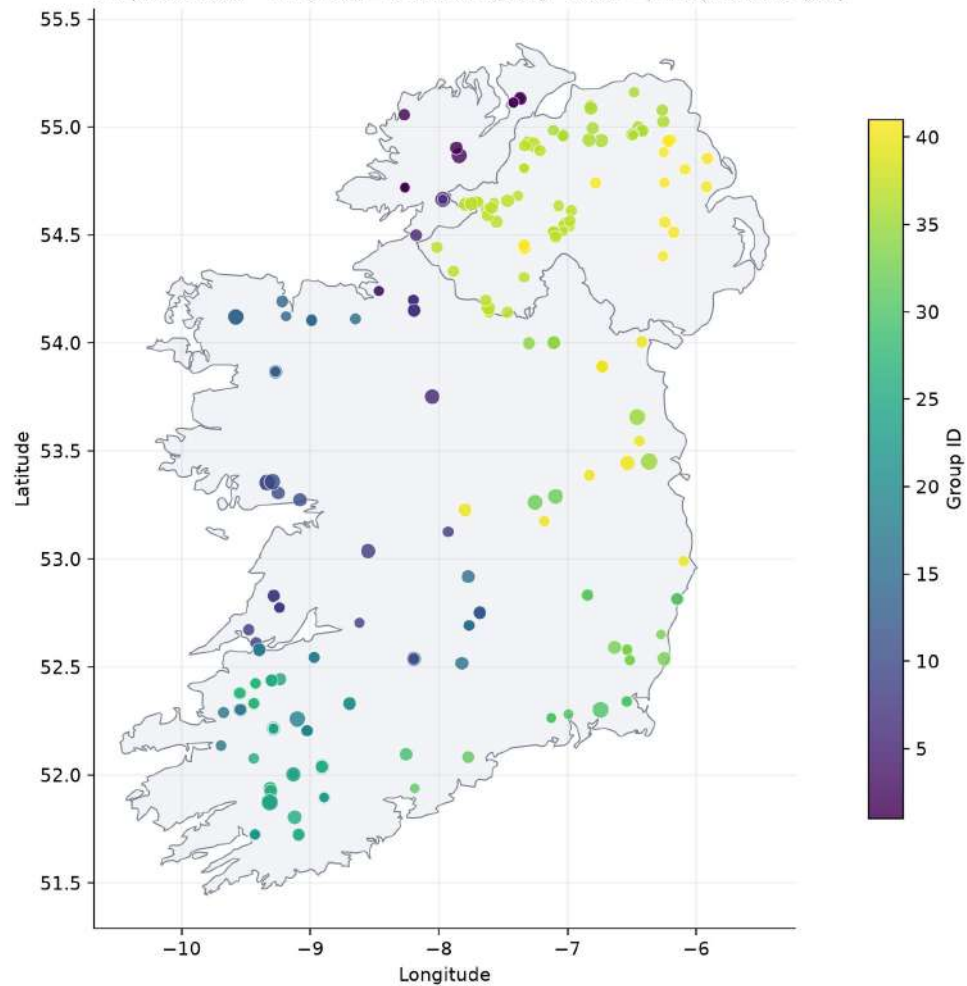
26 counties, 0.99 alpha, 20 sweeps per temperature



And we unite
Ireland.

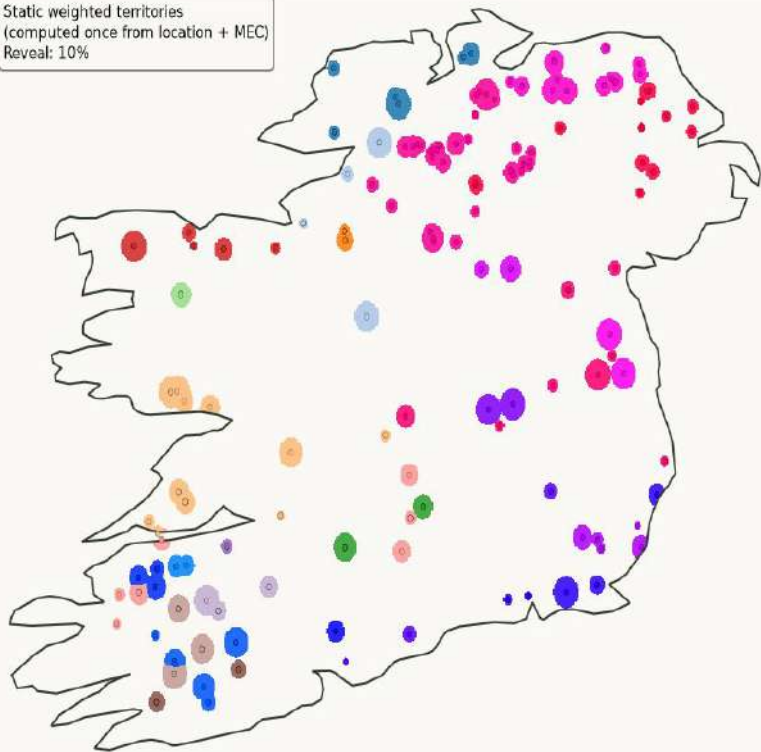
32-county renewable nodes

240 plotted nodes • marker size = renewable capacity • colour = primary/exclusive group



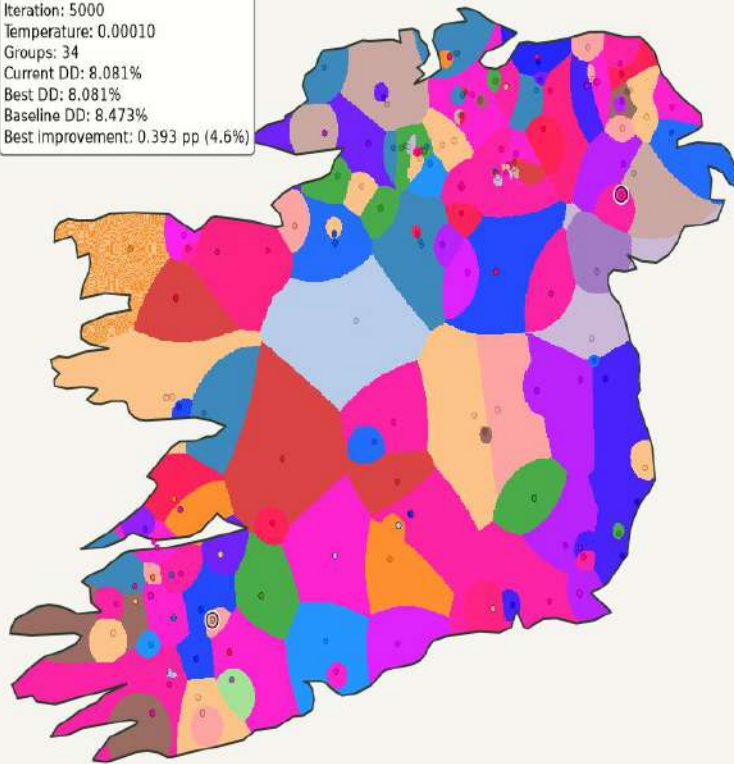
32-county annealing territories

Static weighted territories
(computed once from location + MEC)
Reveal: 10%



32-county annealing territories (final)

Iteration: 5000
Temperature: 0.00010
Groups: 34
Current DD: 8.081%
Best DD: 8.081%
Baseline DD: 8.473%
Best improvement: 0.393 pp (4.6%)

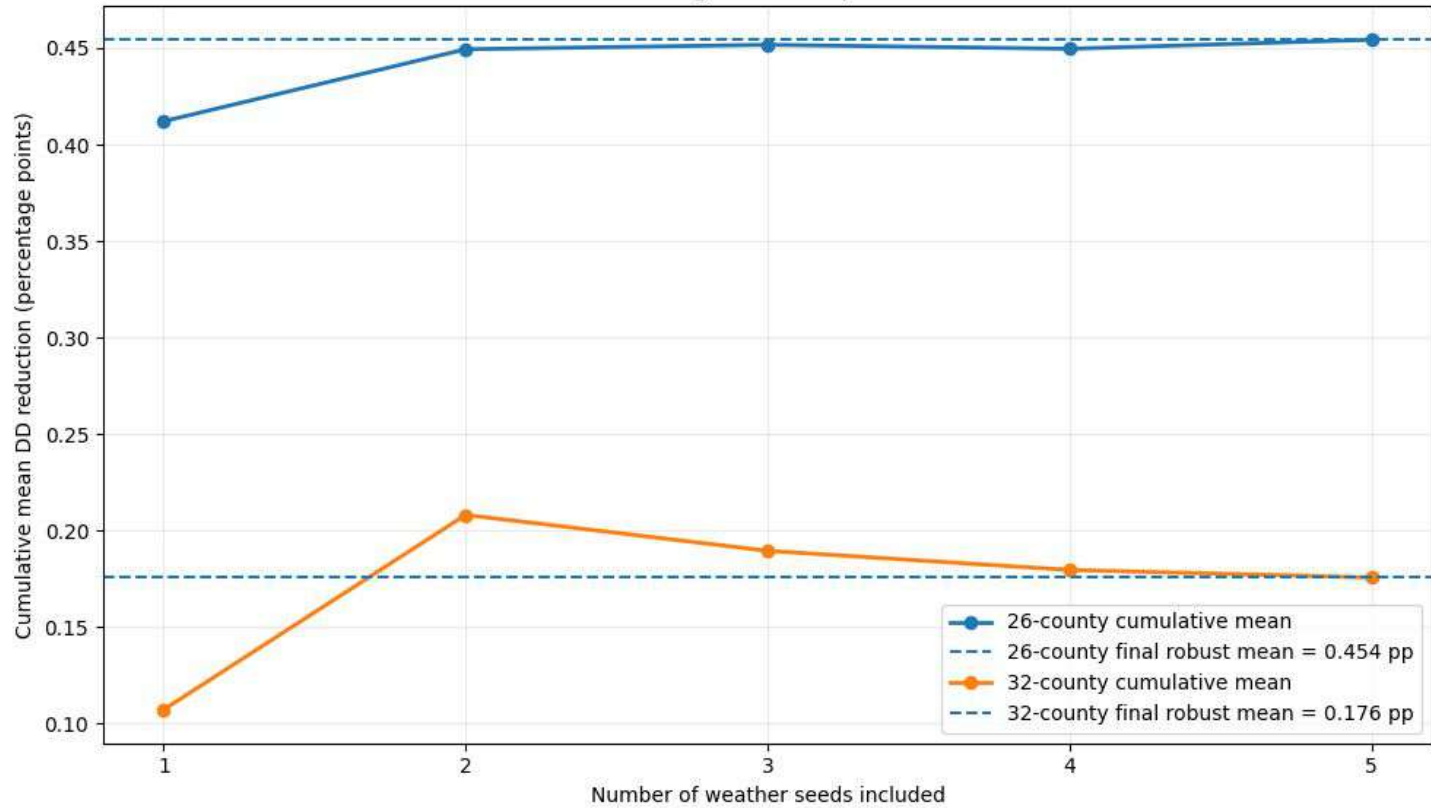


Verification.

- Multiple weather seeds are tested
- Multiple simulation seeds used
- Baseline and optimized groups are always compared under the same conditions
- N-1 contingencies are included.
- Stress scenarios are added.
- Simulator ran 5000 times (and then each sim has 1000 iterations-ish) before any decisions on the rigidity of the system are made.

- **26-county candidate:** 0.454 percentage point mean reduction, equivalent to 7.3% relative improvement, safe across all 5 nominal N-1 weather-seed checks.
- **32-county candidate:** 0.176 percentage point mean reduction, equivalent to 1.9% relative improvement, safe across all 5 nominal N-1 weather-seed checks.

Cross-weather convergence of dispatch-down reduction



Thank You

github.com/ciangregg/EIEG_Hackathon26

Questions?

Banded effectiveness ordering

Shift-factor-ordered wind redispatch under a year-to-date equity bound within existing constraint groups

Elias Mocik, Darragh Evanson, Yuhang Tong

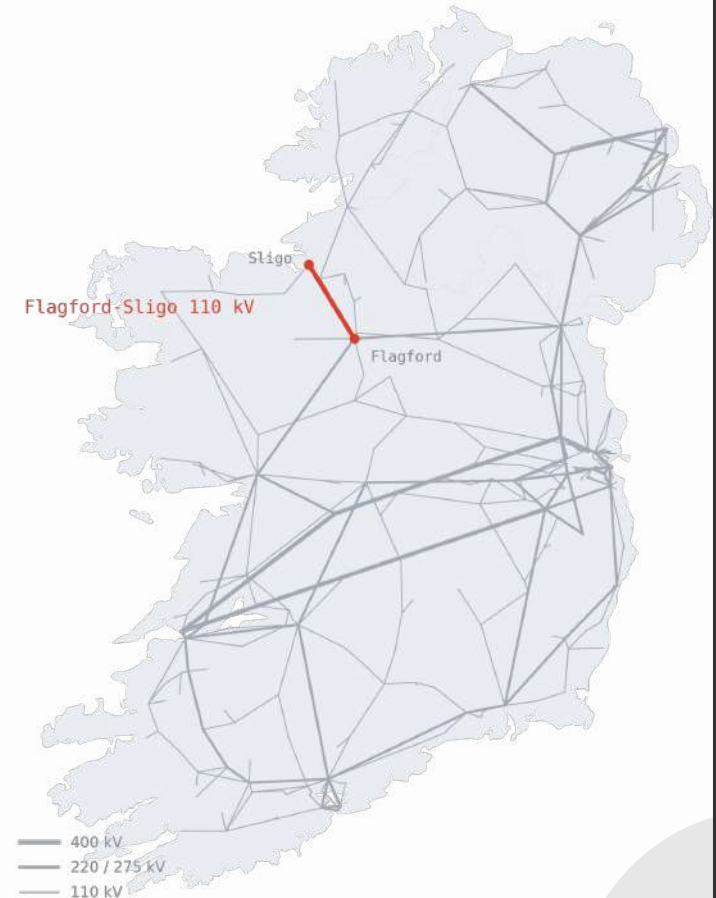
Problem introduction

Why do we need to dispatch-down?

Dispatch-down is the deliberate turn off of a wind farm.

Power generation within a region often overloads local transmission infrastructure.

Common in North-West, we will use this as a case study throughout.



The constraint group system

Used to decide which wind farms should be dispatched down for a certain **constraint**.

Eirgrid decides on which wind farm to be admitted to each group based on its **shift factor**.

Each farm within a group is **dispatched-down** by the same proportion to help address the constraint.



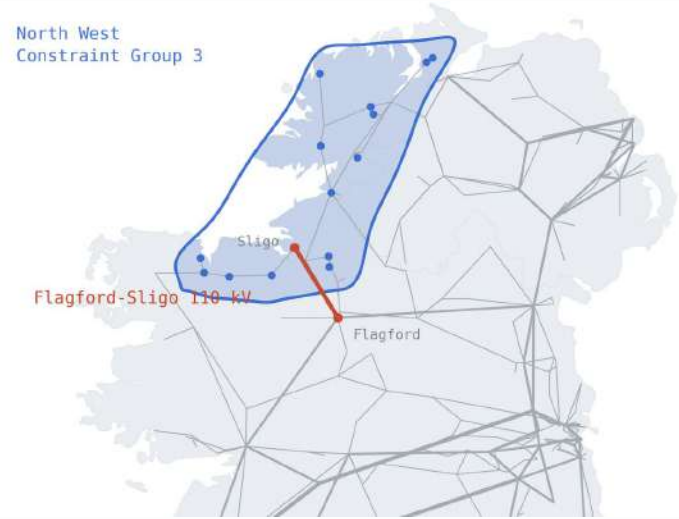
Current model

The current system: Pro rata (A)

All farms in the relevant constraint group are dispatched down by the same proportion.

This is **equitable**, but **inefficient** because certain farms relieve the constraint more than others.

The current system means you have to dispatch down more power than you would theoretically have to.



The current system: Pro rata (A)

Why don't we just make it more efficient?



Theoretically Efficient Model (B)

The optimal solution is to sequentially choose the wind farm with the highest contribution to reducing congestion (highest shift factor).

Each wind farm is considered individually.



available energy

% of available energy · one scale

NO!! Consequence: Financial Disputes

Many wind farms would want to sue the electrical grid as some farms lose much more than others without compensation. Poor future projections due to this uncertainty may even reduce the investment into wind energy in Ireland.

(1). In 2022, several wind farm operators sued the CRU over their regulations regarding compensation for dispatch down.

Current System: Pro-rata

Everyone in the same constraint group gets dispatched down the same (proportional) amount.



Inefficient

Dispatches excess power.



Mostly Equitable

Theoretically Perfect: Shift factor order

Dispatch down wind farms ordered by their shift factor alone.



Efficient



Non-equitable

Would get sued and blocked.

Proposed model

Proposed Model: "Banded effectiveness ordering" (C)

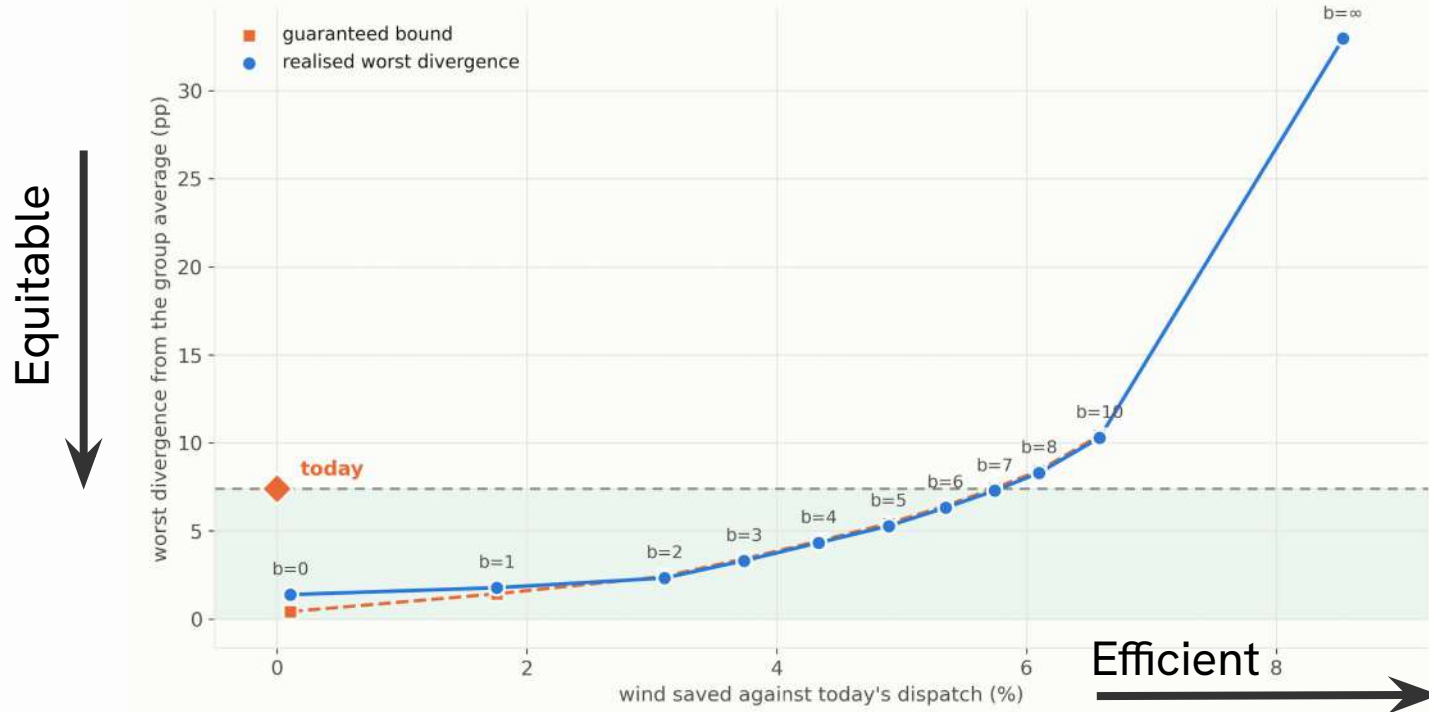
Shift-factor-ordered wind redispatch under a year-to-date equity bound within existing constraint groups



Banded effectiveness ordering (C)

✓ Equity is maintained over the time span of a year

✓ Efficiency is maximised by using the shift factor to dispatch farms with the biggest contribution



How the Algorithm works

- When a constraint occurs, power reduction needed in that section is calculated (same as pro rata)

$$R = \max(0, |F| - F_{\max})$$

- Each farm has a ledger of energy cut

$$r_i = \frac{\text{cut_energy}_i}{\text{available_energy}_i} = \frac{C_i}{V_i}$$

- The dispatch-down % of whole group sets the average

$$\bar{r} = \frac{\sum_i C_i}{\sum_i V_i}$$

- Any farm more than b points above that average is protected from dispatch-down

$$\text{Eligible} = \{i : r_i \leq \bar{r} + b\}$$

- Eligible farms are cut in shift-factor order (most effective first)

$$c_i = \alpha a_i,$$

$$\alpha = \min\left(1, \frac{R_{\text{remaining}}}{\sum_i \text{SF}_i a_i}\right)$$

- The ledgers update, so the next cut lands on a different farm

$$C_i \leftarrow C_i + c_i \Delta t, \quad V_i \leftarrow V_i + a_i \Delta t$$

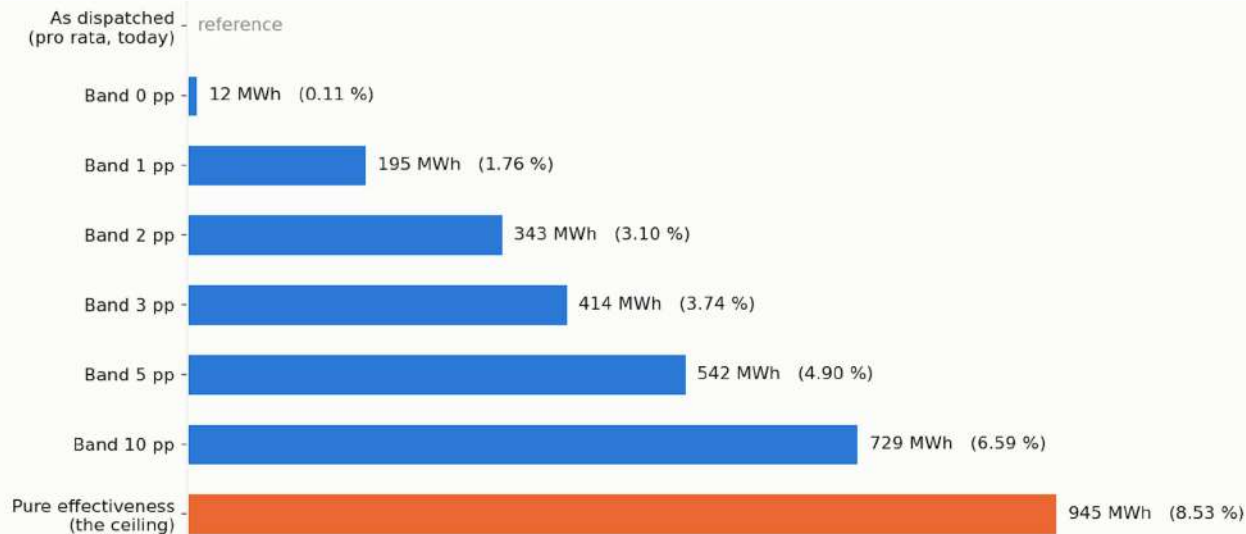
- Security check: if eligible farms can't provide relief, protected farms are added back

Effect of widening the band

If $b = 0$, the system behaves like the current system (A)

As $b \rightarrow \infty$, the system becomes pure shift-factor order (B)

Our proposition explores bands between $b=0$ and $b=10$ (C)



The Optimal band: $b=5$

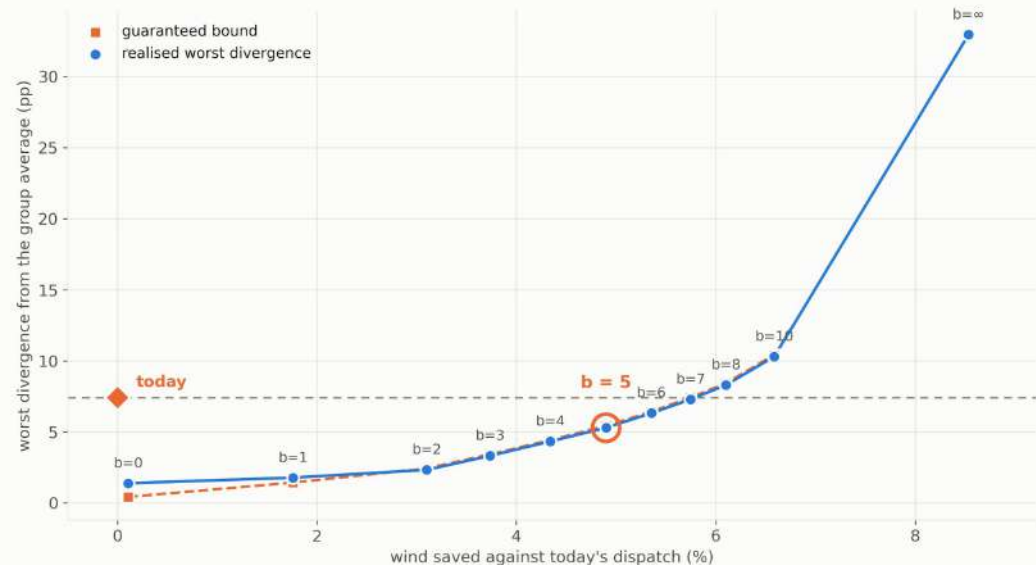
$b=5$ saves **4.90%** of the wind power.

This is **57%** of the theoretical maximum.

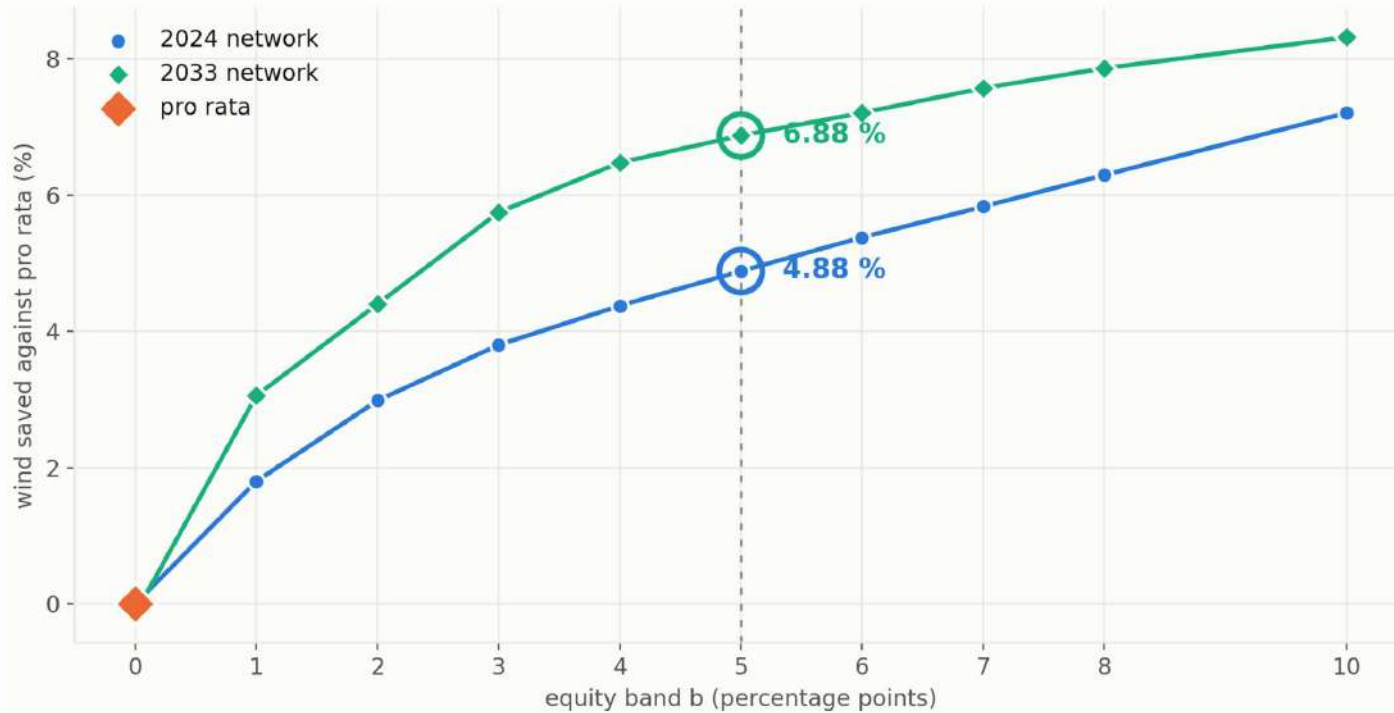
Holds every farm within 5.4 points of the group average vs 7.4 points with pro rata (A)

$b=5$ is the largest band that still does so when the data is split in half

Banded effectiveness ordering (C), summer 2026, real data, North-West Constraint Group 3



Future projection of proposed model on synthetic data



Comparison of Pro rata (A) vs Banded effectiveness ordering (C), 1 full year, synthetic data, North-West Constraint Group 3

How does this impacts key players?

Generators

4.90% less spilled at $b = 5$

Improved bound on equity

Today $\rightarrow$ 7.41 pp

Our System $\rightarrow$ 5.43 pp

No costs lost $\rightarrow$ ledger is MWh

The Grid

An algorithm optimises different dispatch levels within the constraint groups.

Same flow relief and stability.

One instruction per event.

Control groups unchanged.

Consumers

Less energy cut $\rightarrow$ lower cost placed on the consumer.

Also saves energy that would require burning gas instead.

€790m imperfections costs imparted onto customers in 2025/26

Considerations

- N-1 Security: The system must be resilient to survive the loss of any single element.
- 7-Generator Rule: around all of Ireland there must be 7 inertial power plants active at all times.
- DLR: Our system can incorporate a dynamic line rating for each power line and change the threshold for when dispatch down is triggered.

Limitations

1. We believe the proposed model can be feasibly implemented though it would require some regulation changes.
2. Equity in the ledgers is tracked in MWh rather than in euros.
3. This does not replace the requirement to invest in new transmission infrastructure.
The UK National Grid report: 4x increase in imperfection cost by 2030 without transmission upgrades.

Summary

Our method achieves **4.9%** reduction in energy dispatched down while maintaining equity among wind farms. *(from 2026 real data simulation)*

We disregard day-to-day equity but ensure equity is maintained year-to-year.

 Improve Efficiency

 Same constraint groups

 Maintain Equity

 Same security

Thank You

Brendan Tuohy - Chairperson @ EirGrid

Jack Farrell - Energy Trader @ Erova

Deirdre Merriman - Senior Grid Control Engineer @ EirGrid

Dave Linehan - Director of Policy and Research @ Wind Energy Ireland

Any Questions?

Where Ireland's Wind Hits a Wall

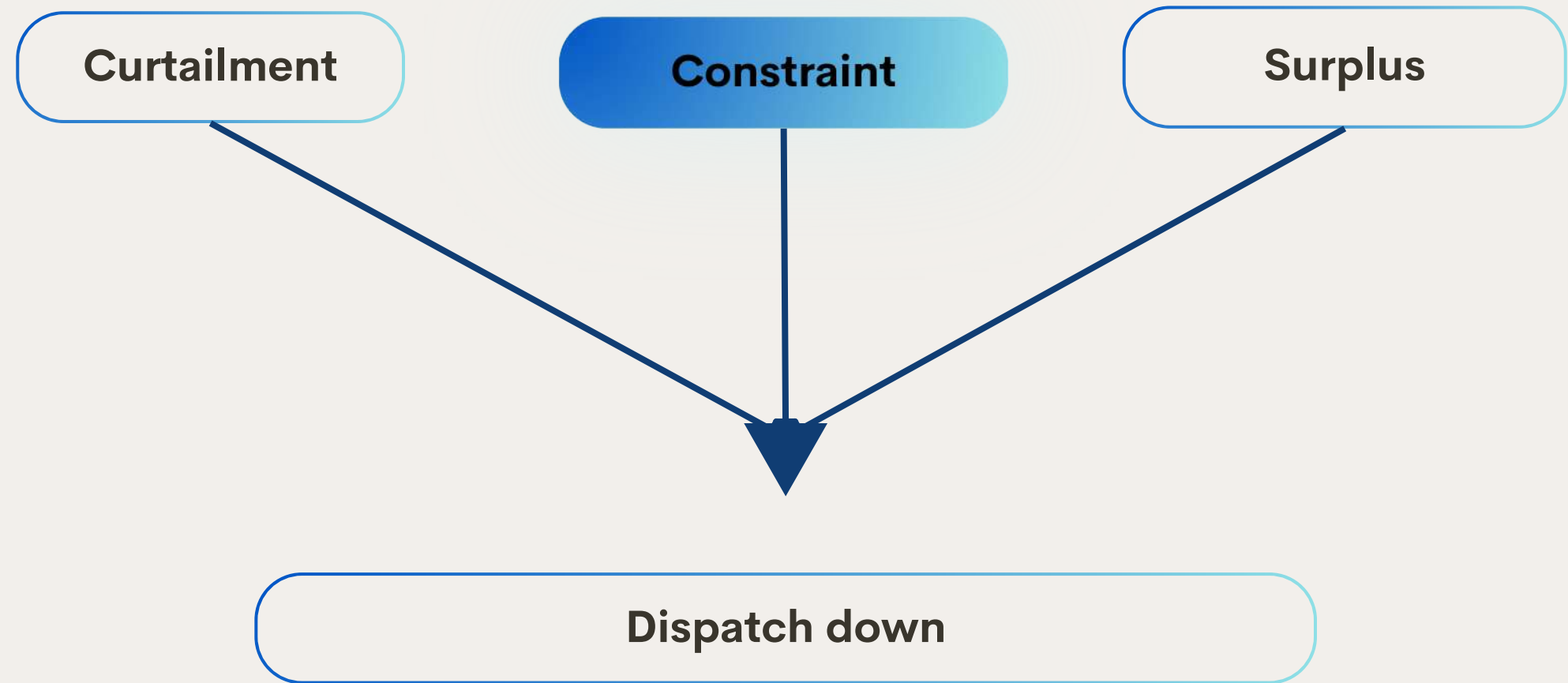
A network-first approach to grid constraint

The Problem

Ireland sometimes has renewable energy available but cannot safely transport all of it to where it is needed.



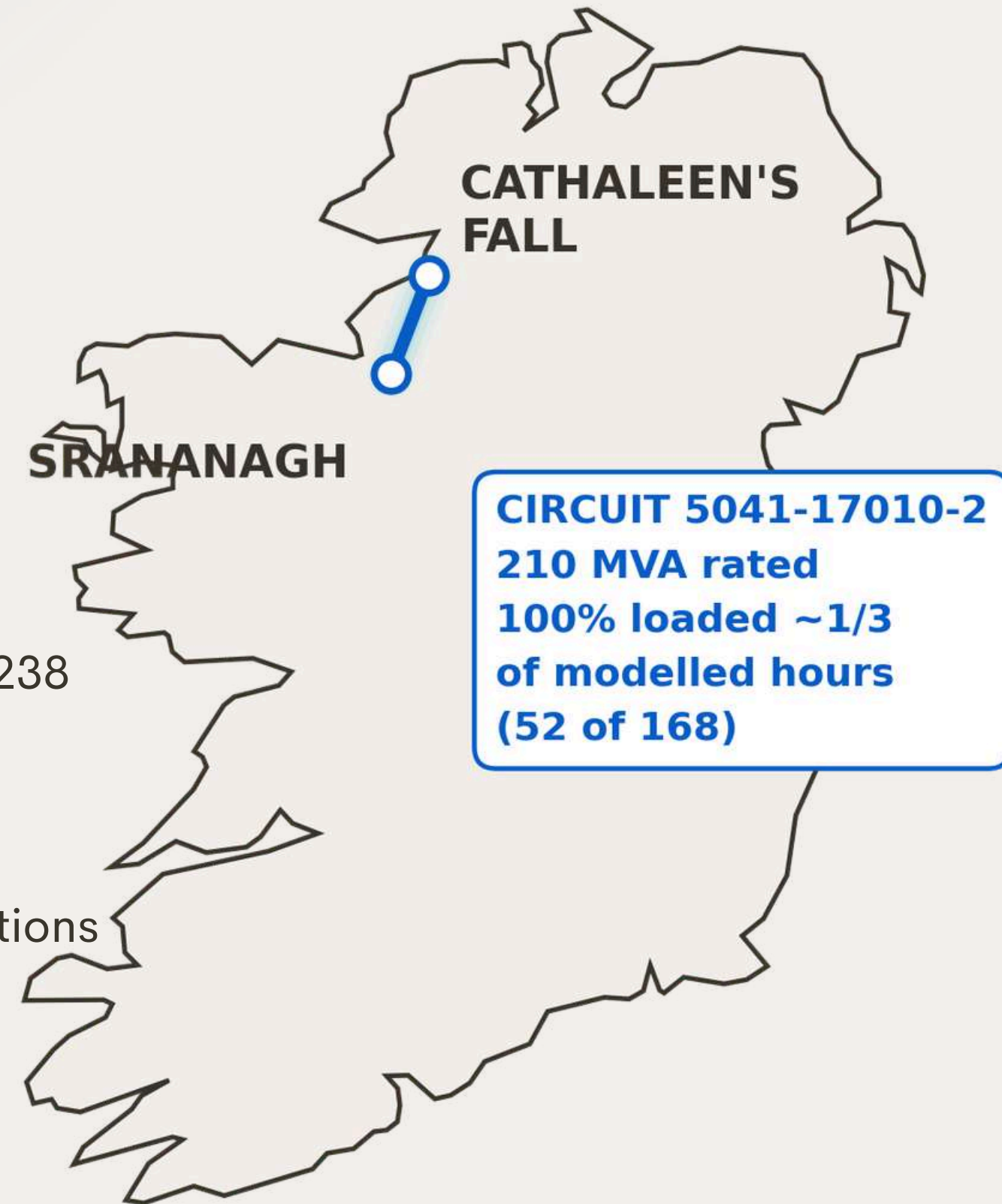
Why Renewable Output Is Reduced



Real Bottleneck: Srananagh → Cathaleen's Fall

Case Study

When wind output in the North-West is high, this circuit fills up, forcing some available wind generation to be reduced.



Regions Clear Bottleneck

At 100% loading $\frac{1}{3}$ of the time (52 of 168)

Shift Factors

10 farms reduce the bottleneck with a factor of -0.238

Constraint Group Relief

718 MW across the 10 relieving farms

Robustness

Rank correlation 1.0000 across 3 reference conventions

CIRCUIT 5041-17010-2
210 MVA rated
100% loaded ~1/3
of modelled hours
(52 of 168)

Our Solution

Instead of fixing one bottleneck at a time, we test interventions such as battery siting and dynamic line rating across the constrained network, because a fix that helps one line can quietly worsen another.

HOW WE TESTED THE IDEA

1. LOCAL FIXES

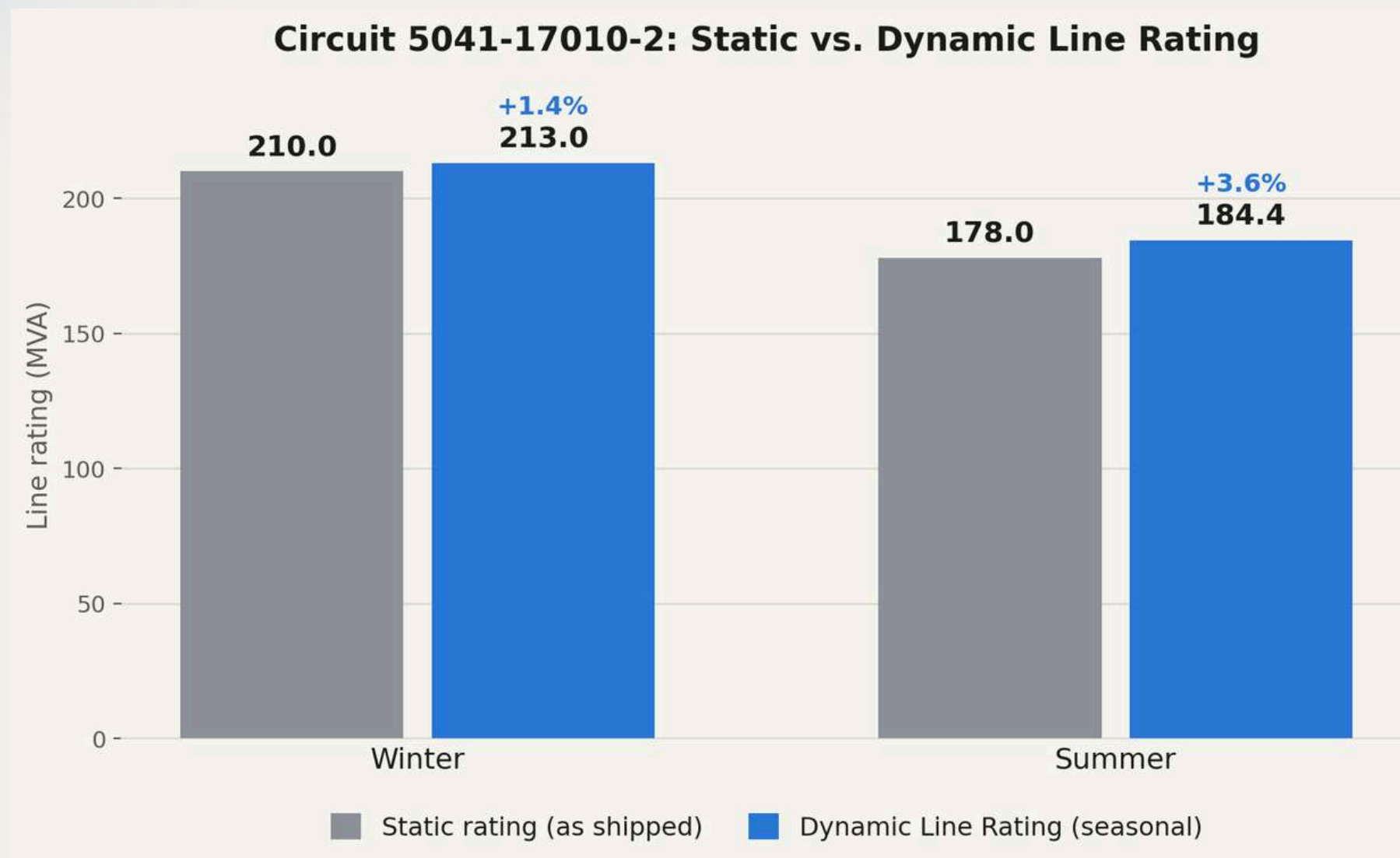
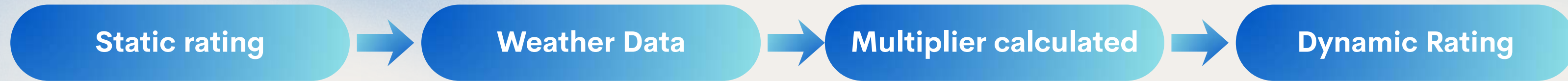
- Target circuit: Srananagh → Cathaleen's Fall
- Battery siting and sizing
- Dynamic Line Rating scenarios

2. NETWORK-WIDE CHECK

- Trade-off identification
- Constraint-group validation
- Multi-line shift-factor map

Dynamic Line Rating

How could DLR reduce wind constraint?



Winter

+1.4% capacity increase (210 → 213.0 MVA)

3,660.6 → 3,579.4 MWh/week dispatch down

Summer

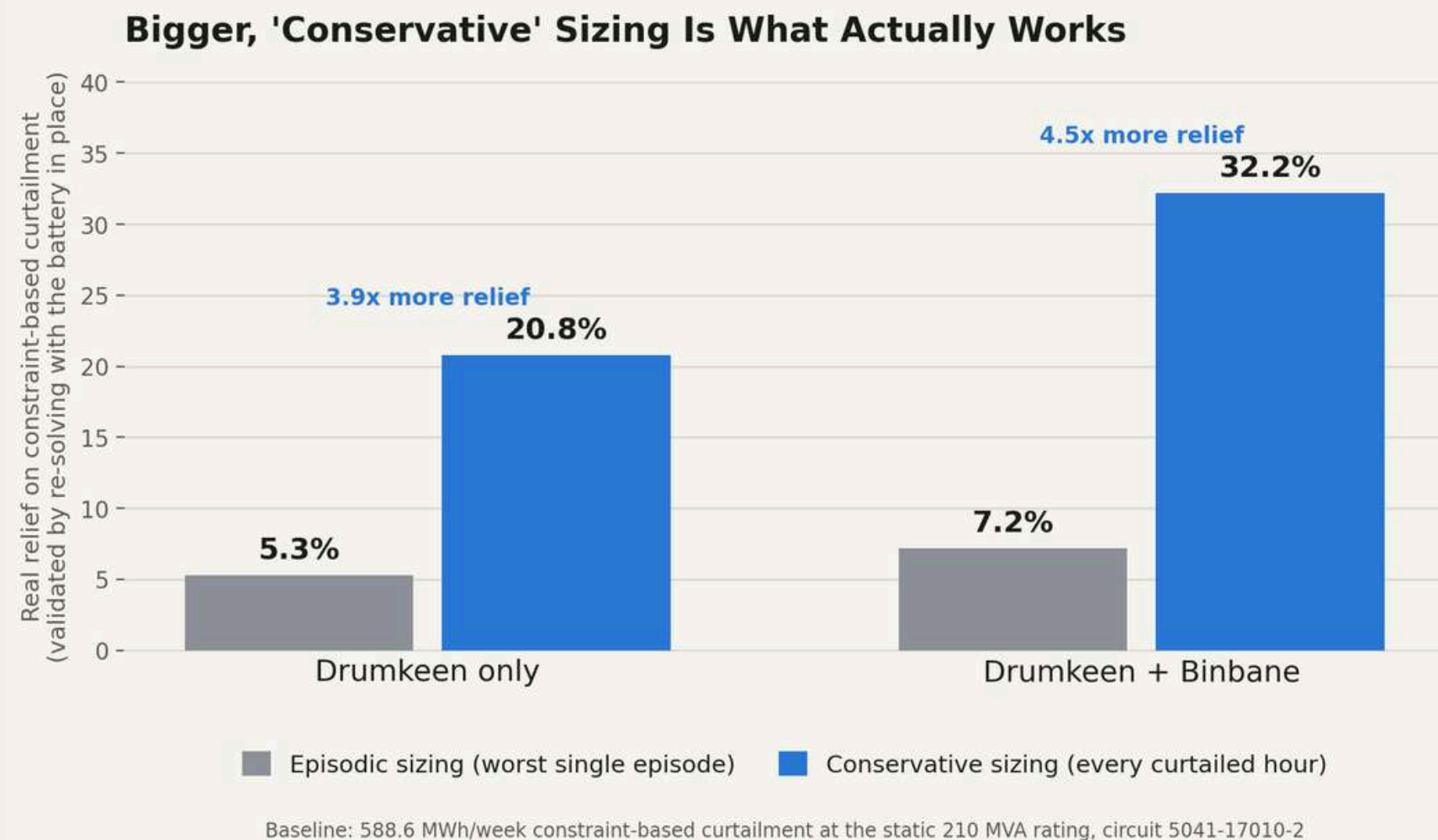
+3.6% capacity increase (178 → 184.4 MVA)

21,813.6 → 21,406.0 MWh/week dispatch down

Planning caveat: DLR is currently used as a short-term mitigation while permanent reinforcement is delivered. We propose pairing it with battery storage for a longer-term solution, DLR providing continuous weather-driven headroom, the battery covering the residual constraint.

Bigger Battery, Real Relief

- Two sites concentrate almost all of it — Drumkeen + Binbane make up 92.5% of where dispatch down happens, out of 10 candidate sites (concentration, not relief — see below).
- Baseline: 588.6 MWh/week constraint-based dispatch down (static 210 MVA rating)



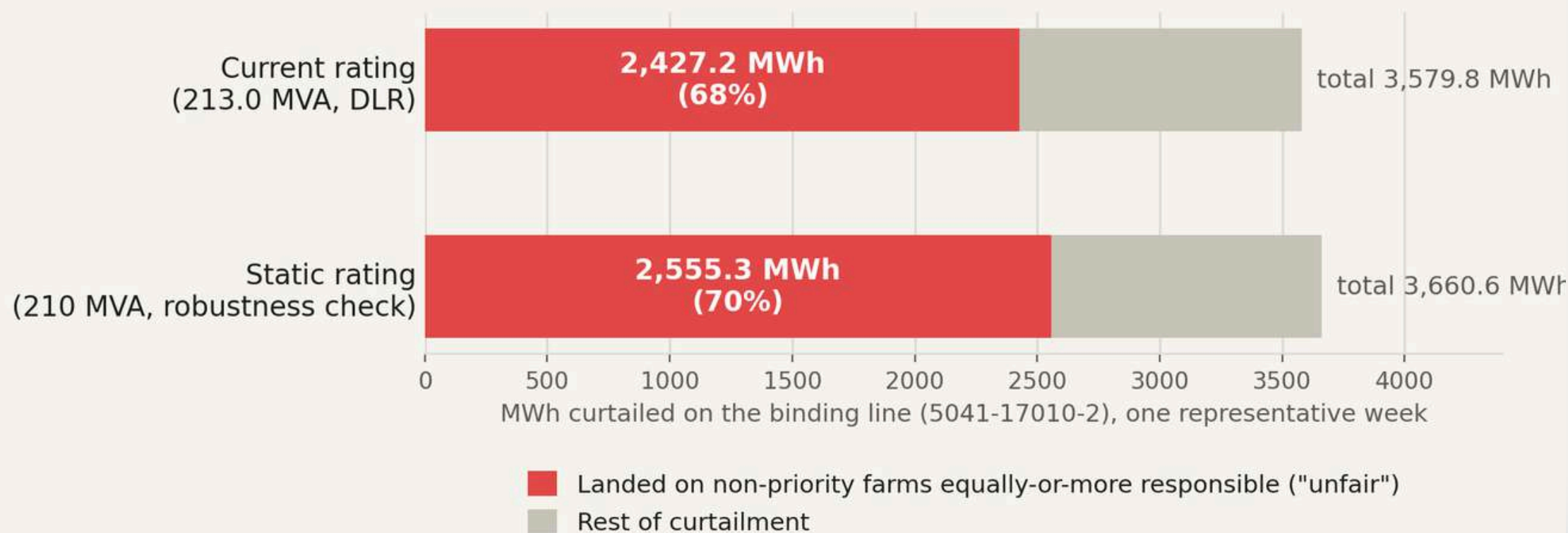
Sizing a battery to its worst episode looks efficient on paper but delivers the least relief once re-solved. Sizing conservatively — for every curtailed hour in the week — delivers 3.9–4.5x more: 42.9 MW / 346.5 MWh at Drumkeen alone reaches 20.8%; adding Binbane (91.3 MW / 544.4 MWh) reaches 32.2%

Priority Status Can Override Actual Responsibility

2,427.2 MWh

Curtailment that landed on non-priority farms which were equally-or-more responsible for the congestion than a protected farm that was spared.

Same unfair pattern holds under both ratings



Out of 3,579.8 MWh total curtailed on the binding line (using the corrected 213.0 MVA DLR rating), 2,427.2 MWh, 68%, fits this 'unfair' pattern, landing on three farms: Cathaleen's Fall, Sorne Hill, and Trillick. The same pattern holds under the older static 210 MVA rating too (2,555.3 of 3,660.6 MWh, 70%)

SECTION 3.2 · STEP 3A · MULTI-LINE CONFLICT MAP

Where a *clean fix* for one line breaks another

Click any line on the map (or in the list) to see which wind and solar nodes would need turning down to relieve it (green), which ones must stay online because turning them down would make it worse (red), and which are below the materiality threshold for that line (yellow).

WP2033 · all-island materiality: top 10% |shift factor|, per line 6 monitored · 755 lines in the all-island network 218 candidate nodes

Monitored lines (click one)

- LETTERKENNY → STRA**
3581-89516-1 · 123 binding h · 123 MW
- GALWAY → SALTHILL**
2781-4951-1 · 42 binding h · 186 MW
- ARKLOW → GLENART**
1122-11260-1 · 29 binding h · 513 MW
- ARKLOW → CARRICKMINES**
1122-1742-1 · 10 binding h · 513 MW
- INCHICORE → IRISHTOWN**
3882-3122-1 · 9 binding h · 274 MW



Selected line

LAGHTANVACK ↔ MOY

#6 rank, binding hours	1 binding hours
192 MW line rating	3691-4041-1 line id

Binds flowing LAGHTANVACK → MOY. A node's sign below is signed relative to that direction — positive (green) means more output there pushes flow the same way and worsens this line.

Shift factors 22 of 218 **Material** All 218

Candidate Group Constraint Methods

Constraint-Aware Curtailment Grouping

Which Wind and Solar sites to curtail - Without creating new problems elsewhere

We start from every bus in the network that hosts a wind /solar generator — 218 candidates. Each, screened on two conditions: is the coupling to the target line strong enough (shift factor $\geq 5\%$), and does curtailing that node actually relieve the line — checked against the line's own real binding-hour flow direction, not assumed.

That directional screen is what takes 218 down to 179, and it's the same check that catches the sign-flip trap seen elsewhere in this project (a farm with a big shift factor that's actually on the wrong side of the bottleneck)

6

Binding Lines

218

Nodes Checked

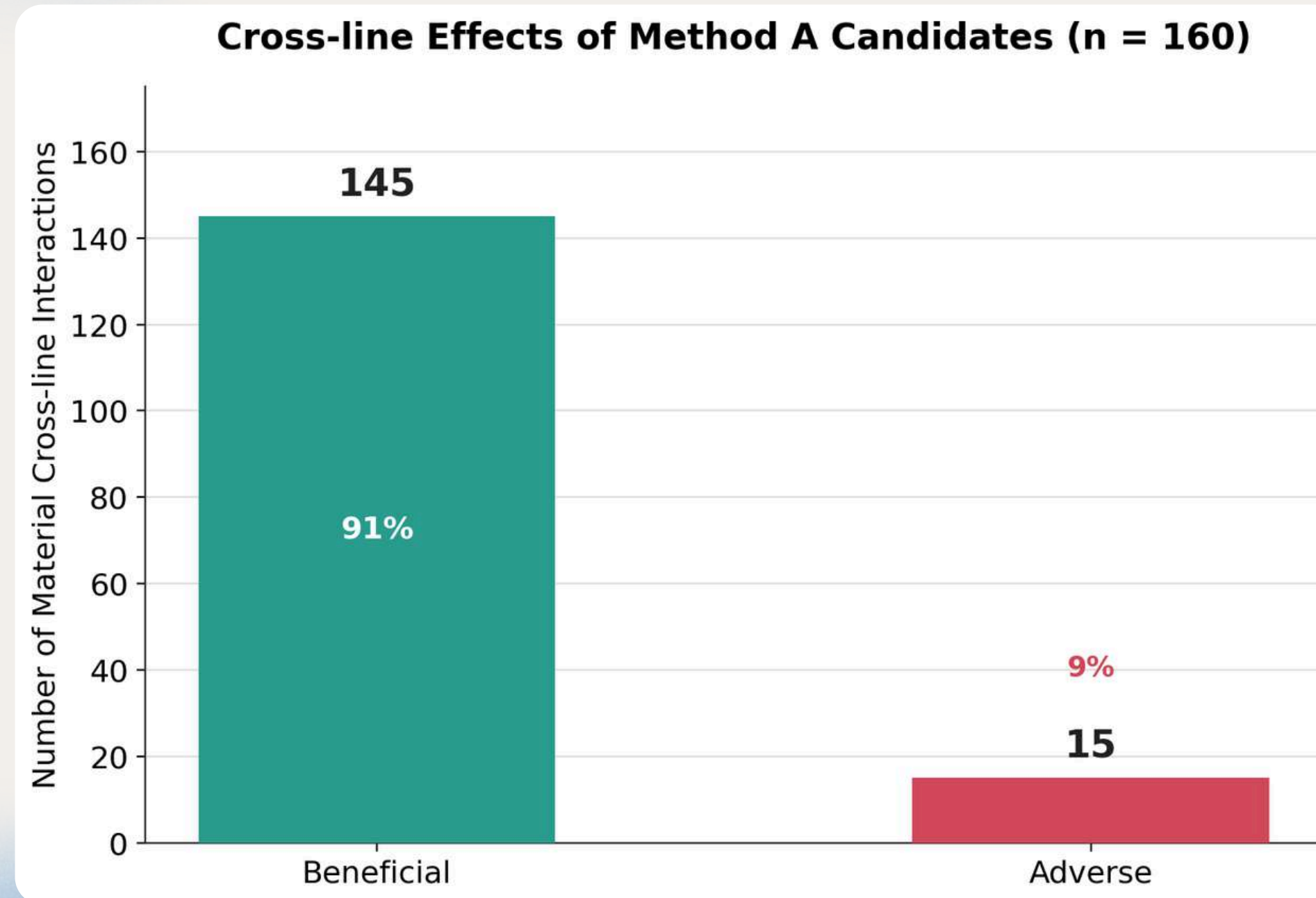
179

Candidates Shortlisted

Y— screened on both magnitude and direction, not magnitude alone

145 of 160 are Free Wins

Only 15 Candidates carry a genuine cross-line trade off



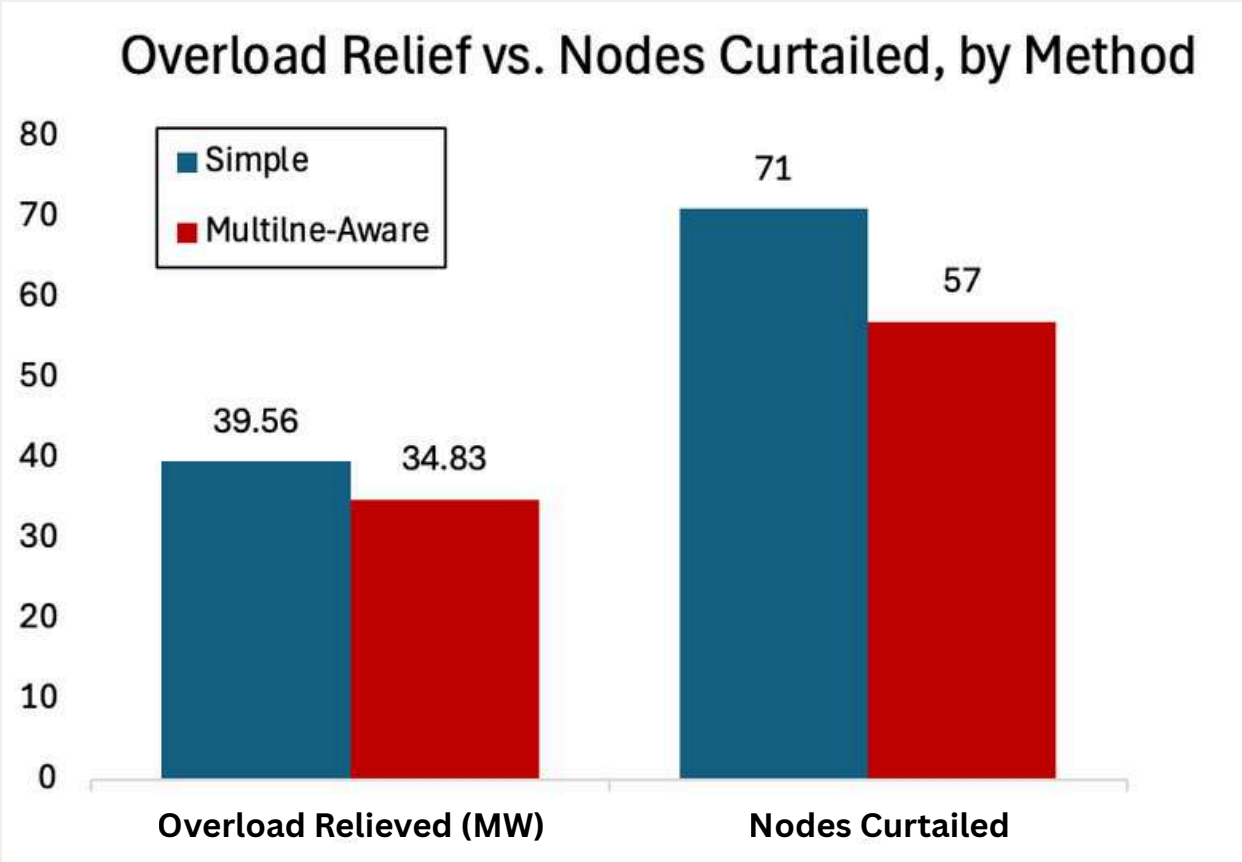
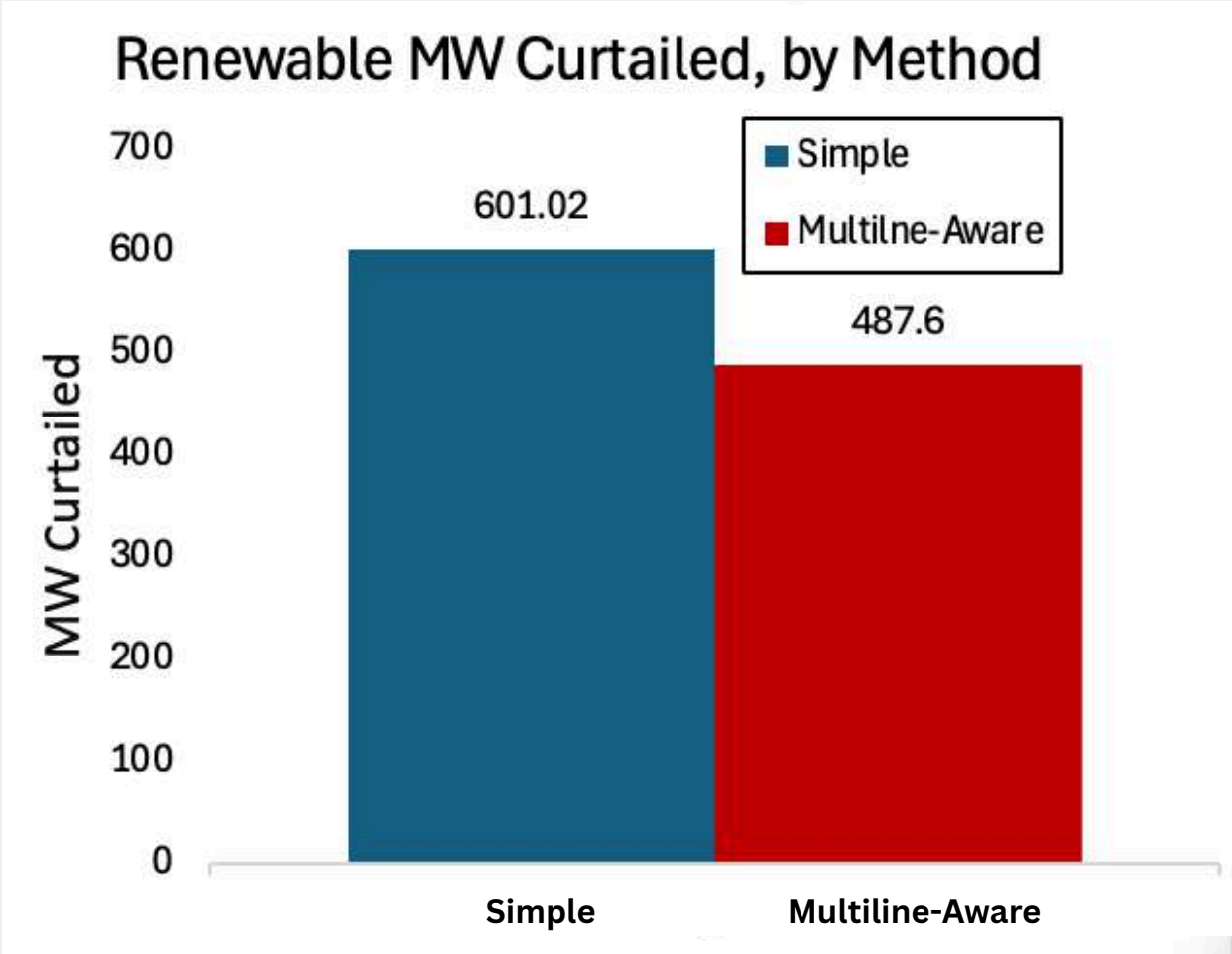
The Result: A validated , ranked shortlist ready for curtailment decisions

Does multi-line-awareness actually cost anything?

Step 4 Validation - Headline Comparison

Metric	Simple	Multiline-aware
Overload relieved	39.56 MW	34.83 MW
Renewable curtailed	601.02 MW	487.60 MW
Nodes curtailed	71	57
New overloads created	0	0
Nodes excluded (flagged)	0	15

Multi-line-awareness costs ~5 MW of relief, but saves ~113 MW of curtailment. With similar overload relief and meaningfully less curtailment paired with zero new overloads, multi-line-awareness emerges as the more efficient option.



Limitations

- Simplified models - DC power flow; folder network off by -25 MW on the target line
- Sythethic Data - Hourly genrator ouput is syntthetic
- Scope and assumptions- single-line focus; all Ireland check found zero multi-line free wins. 5% threshold is a choice, not a standard
- Optimisation behaviour- naive battery sizing implied 92.5% relief; real re-solve gave 32.2%

Conclusion

Our model takes a focused network problem and turns it into a repeatable decision-making approach for the wider grid.

Screen

Identify the binding constraints

Map

Understand how the network interacts

Test

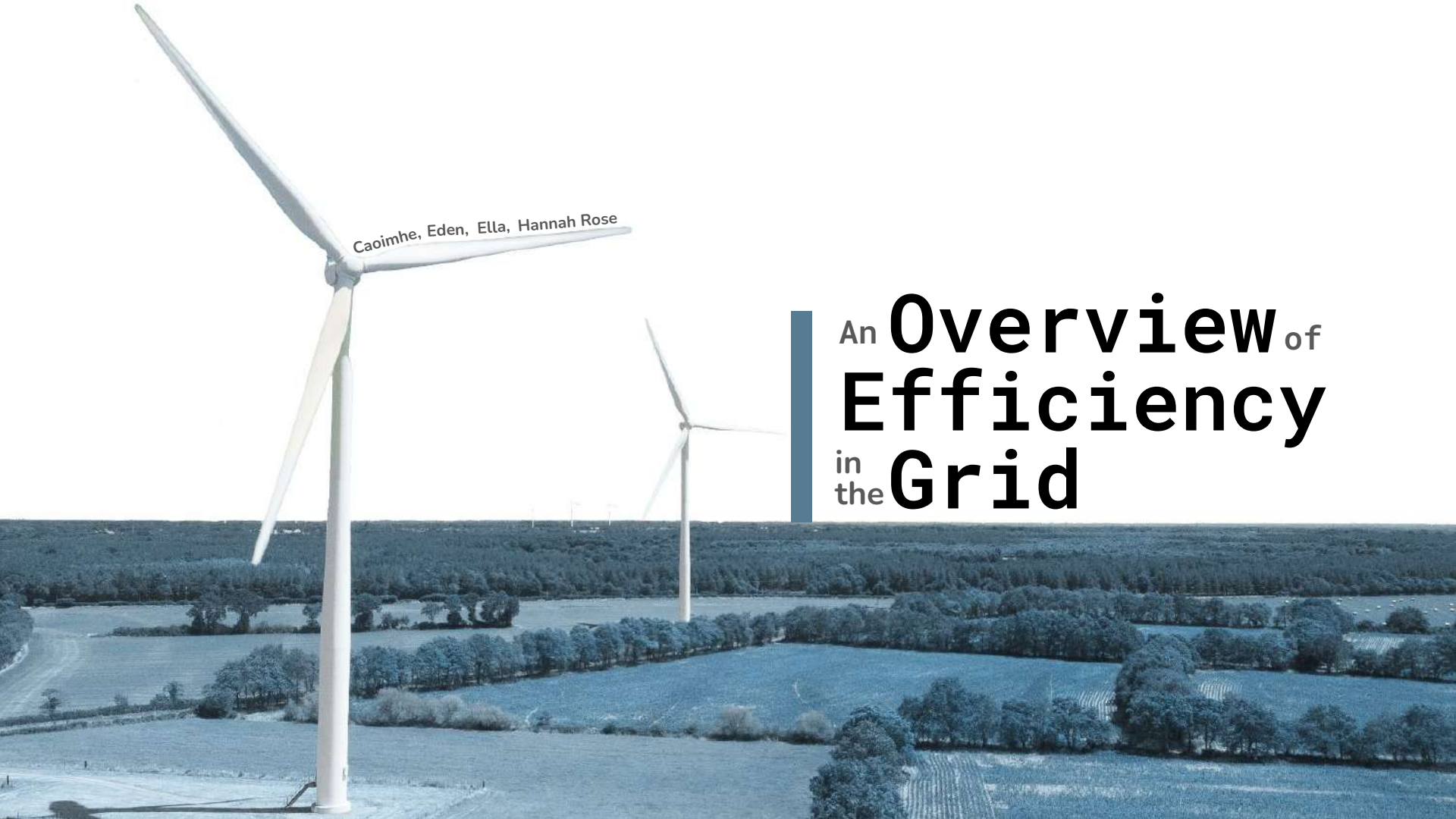
Evaluate interventions against the whole network, not one line

Validate

Re-solve the network to test whether the intervention actually delivers relief

Scale

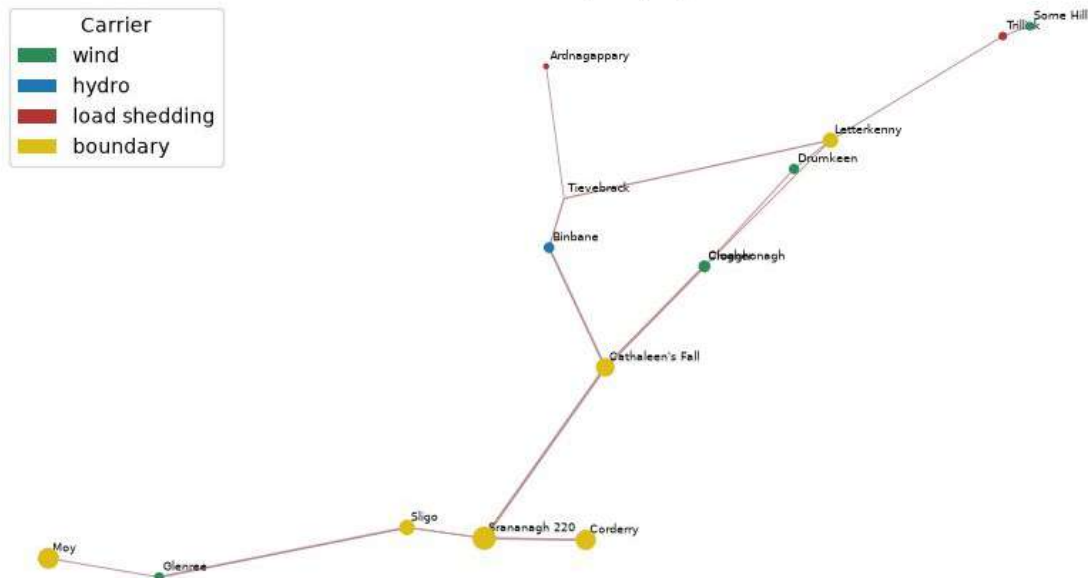
Apply the same framework across more lines, regions and the grid as a whole



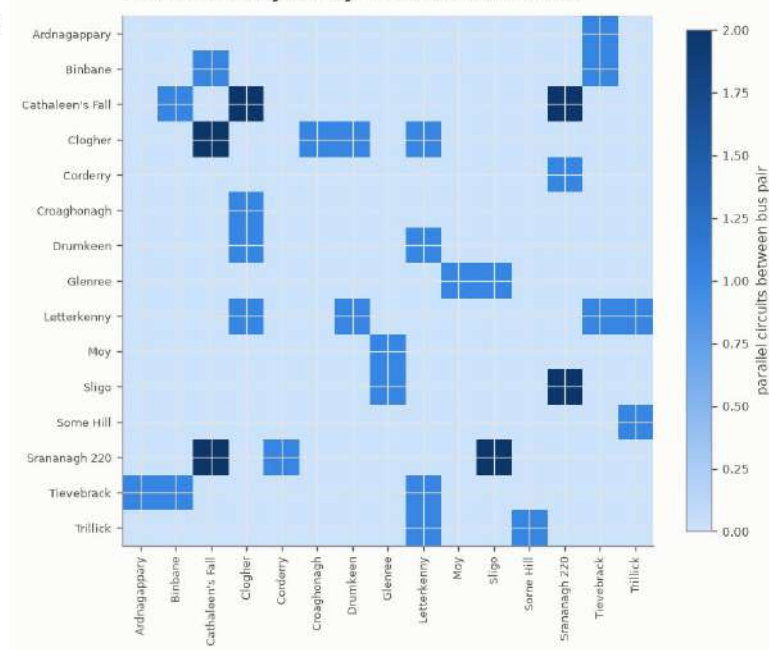
Caoimhe, Eden, Ella, Hannah Rose

An Overview of Efficiency in the Grid

North-West network: capacity by carrier



bus-to-bus adjacency - SV2024 north-west



Adjacency, degree, simple Laplacian

$$A_{ij} = \begin{cases} 1 & \text{buses } i, j \text{ directly connected} \\ 0 & \text{otherwise} \end{cases}$$

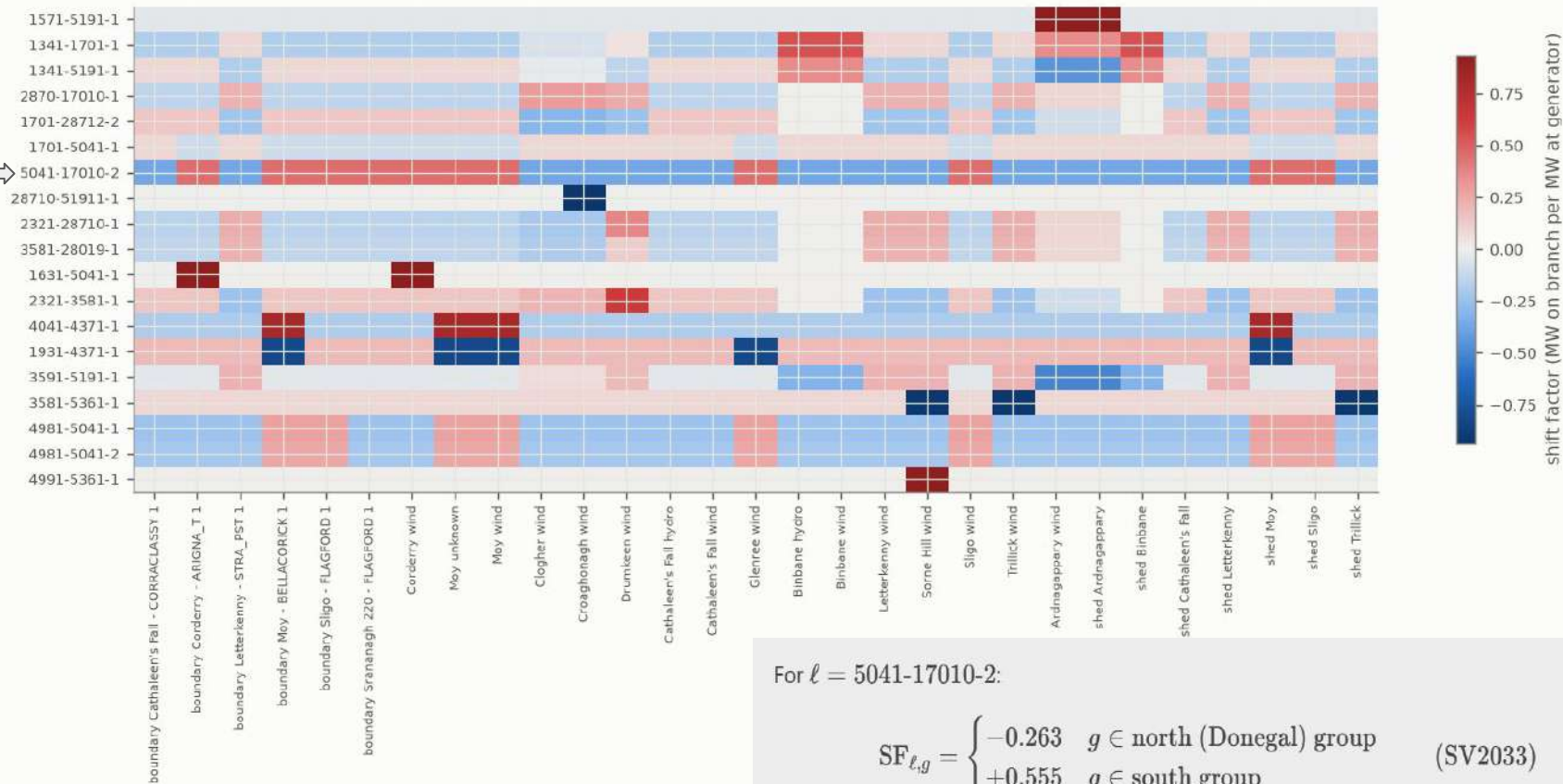
$$D_{ii} = \sum_j A_{ij} \quad L = D - A$$

Fiedler value and Fiedler vector

$$\lambda_2 = \min_{\substack{v \perp \mathbf{1} \\ \|v\|=1}} v^\top L v = \min_{v \perp \mathbf{1}} \frac{\sum_{(i,j) \in E} b_{ij} (v_i - v_j)^2}{\sum_i v_i^2}$$

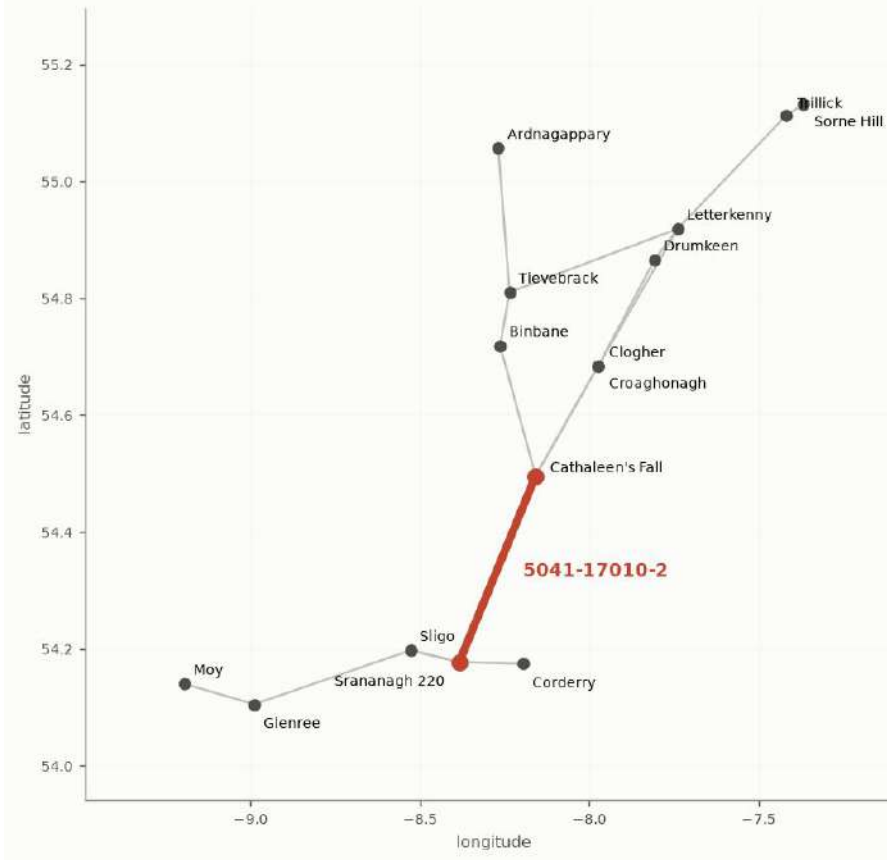
$$v_2 = \arg \min_{\substack{v \perp \mathbf{1} \\ \|v\|=1}} v^\top L v \quad \implies \quad \text{group}(i) = \text{sign}(v_{2,i})$$

shift-factor matrix, SV2024 north-west



The North-West constraint is one circuit

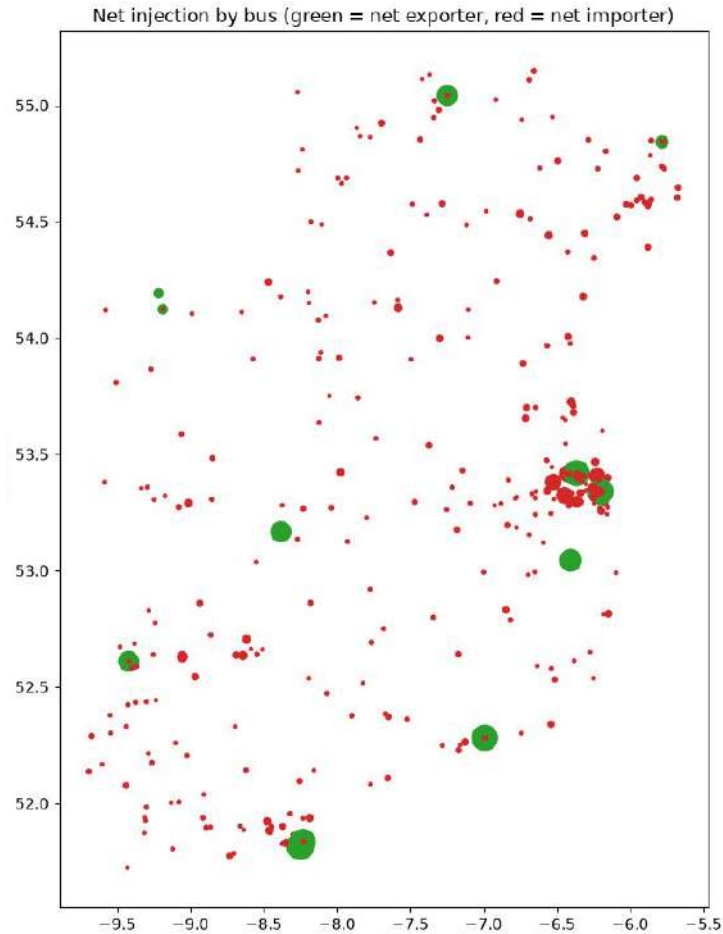
1 of 19 circuits ever binds: 5041-17010-2, at its 178 MVA limit 17 h of 168



Optimising the Grid



Where Power Goes



East-West Flow



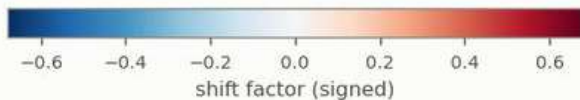
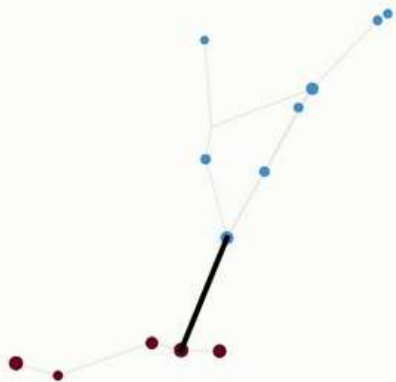
- Red - 400 kV
- Green - 220 kV
- Blue - 110 kV

Violation

WP2024 north-west: constraint groups ($\tau = 15\%$)

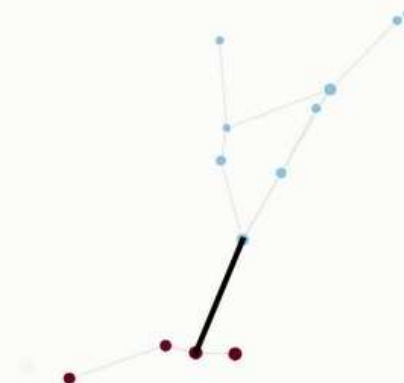
5041-17010-2
14 farms in group

4041-4371-1
1 farms in group



WP2033 north-west: constraint groups ($\tau = 15\%$)

5041-17010-2
14 farms in group

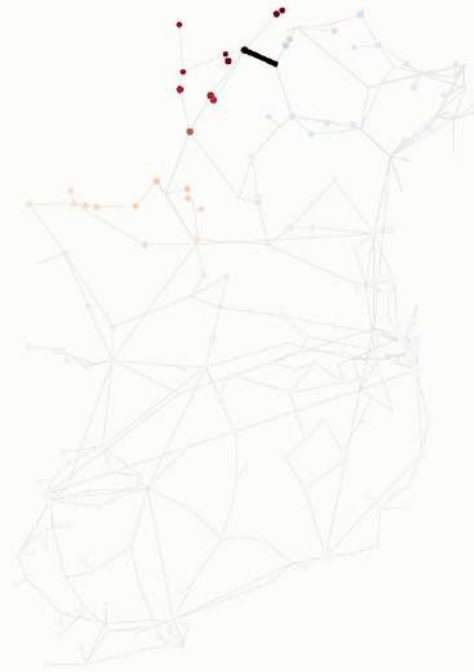


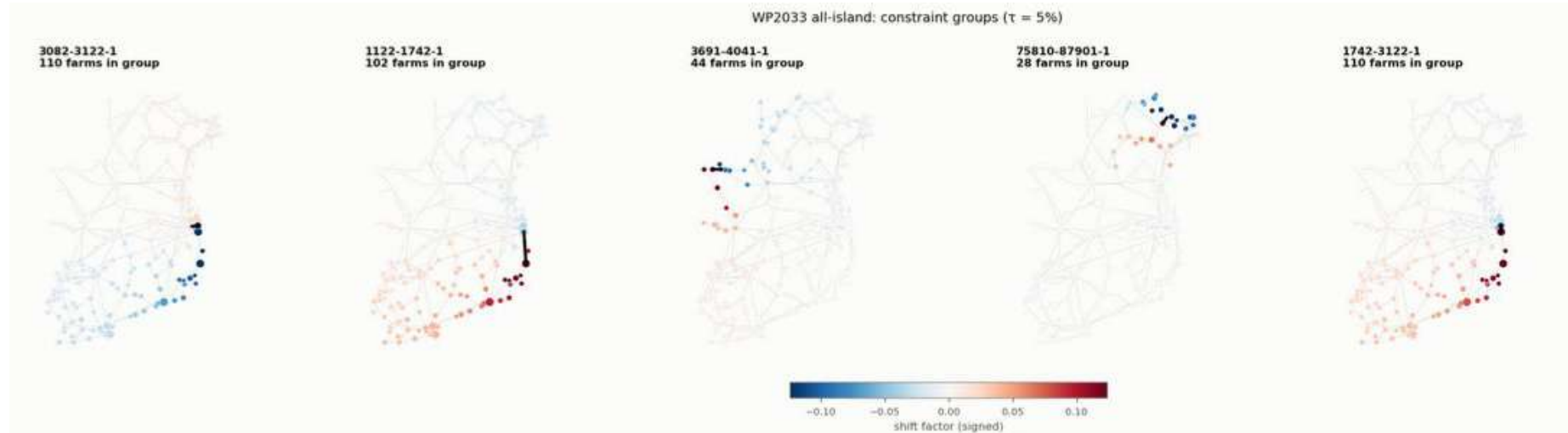
Violation

WP2024 all-island
monitored: 3581-89516-1 — VIOLATED

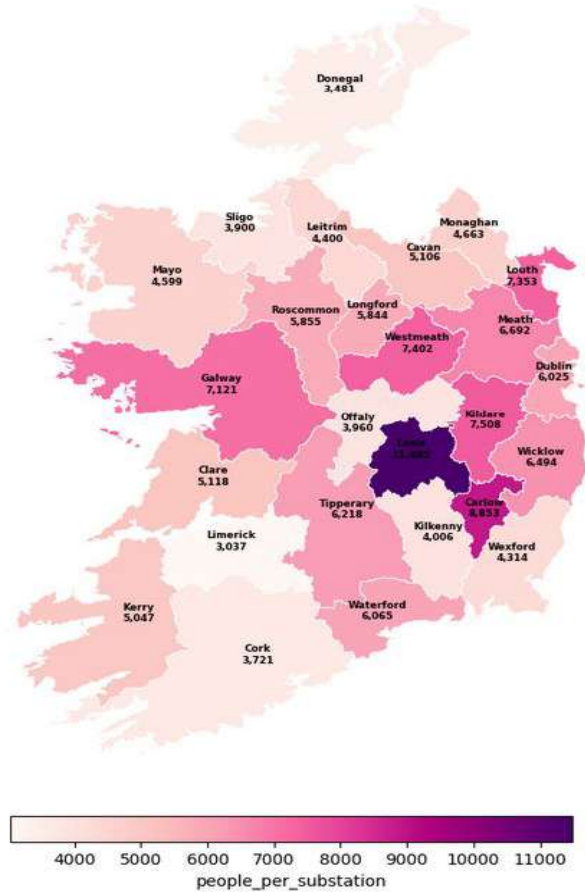


WP2033 all-island
monitored: 3581-89516-1 — VIOLATED

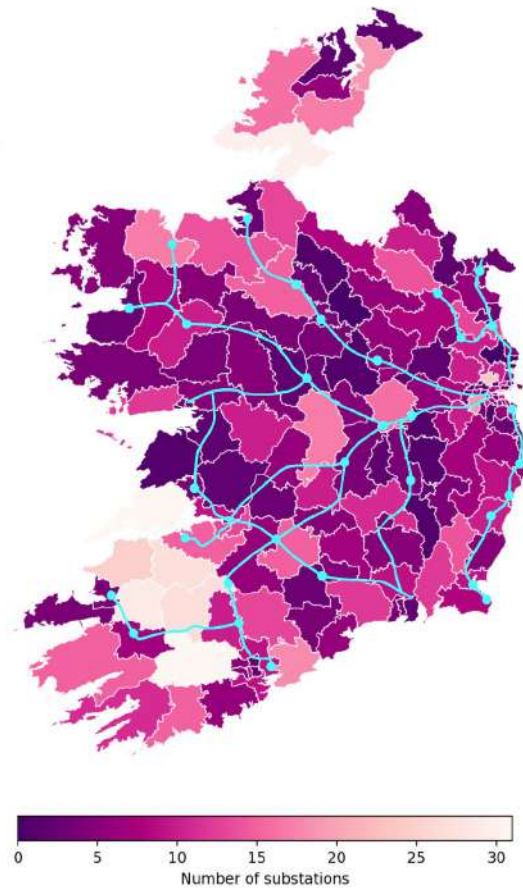




People per substation (higher = less coverage)

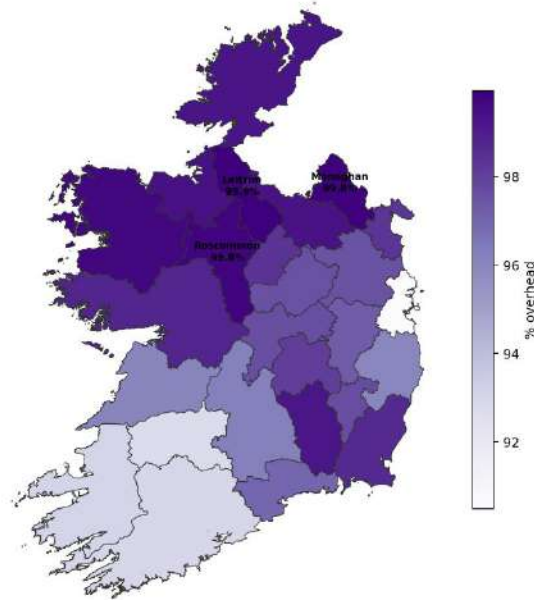


Substations per Local Electoral Area

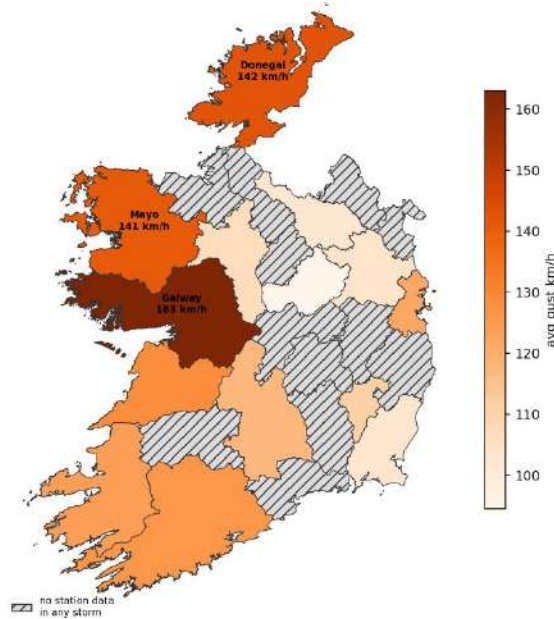


Ireland's Distribution Network: Overhead Exposure x Storm Wind Vulnerability

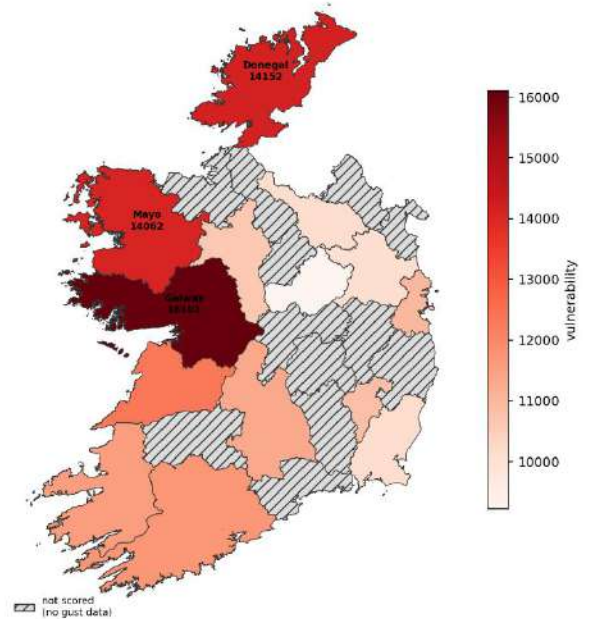
**Lack of underground protection
(% of network still overhead)**



**Average peak wind gust
(across 2 storm(s) with station data)**



**Vulnerability
(avg peak gust x overhead %)**



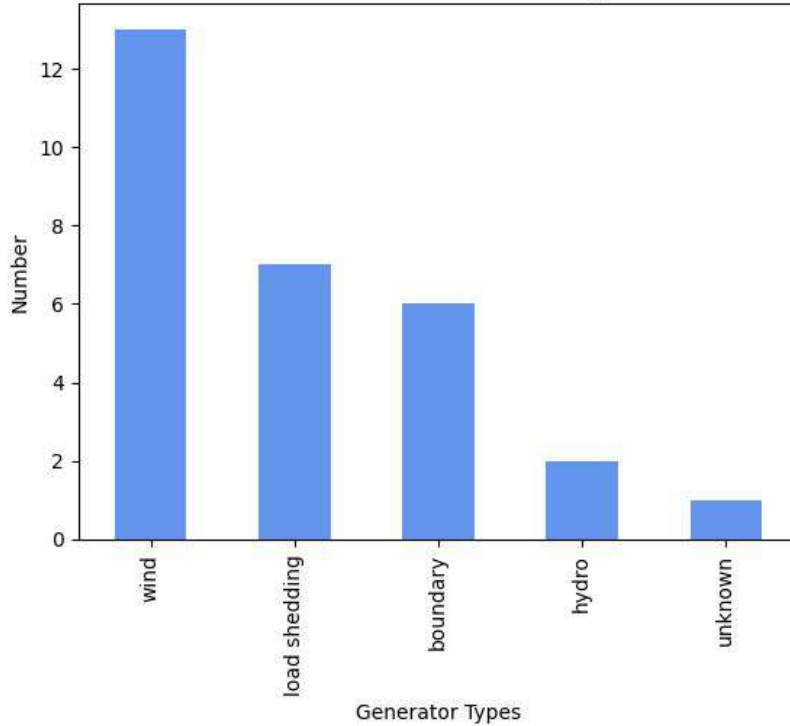
Overhead % = $100 - (\text{cable km} / \text{total distribution line km})$, from the county grid sweep.
 Average peak gust = mean of each storm's peak 3-second gust (km/h) recorded at Met Éireann synoptic stations in that county, averaged across the 2 storm(s) with a full station table available (Eowyn_2025-01-24, Daragh_2024-12-06).
 Vulnerability = avg_peak_gust_kmh x overhead_pct (direct multiplication, not normalised).
 Hatched counties have no station data in ANY of the storms above — not zero exposure, just unmeasured here.

A photograph of a large solar panel array installed in a field. The panels are dark blue and arranged in rows, tilted towards the sun. In the background, there is a dark barn and a line of trees. The sky is clear and blue. The text "Thank You" is overlaid on the right side of the image in a large, bold, black font, with a blue vertical bar to its left.

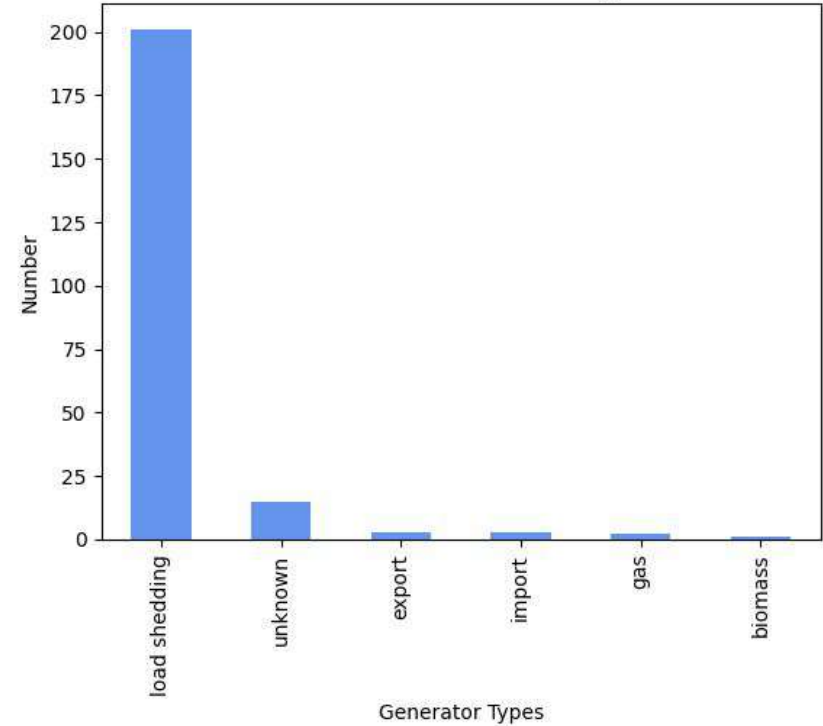
Thank You

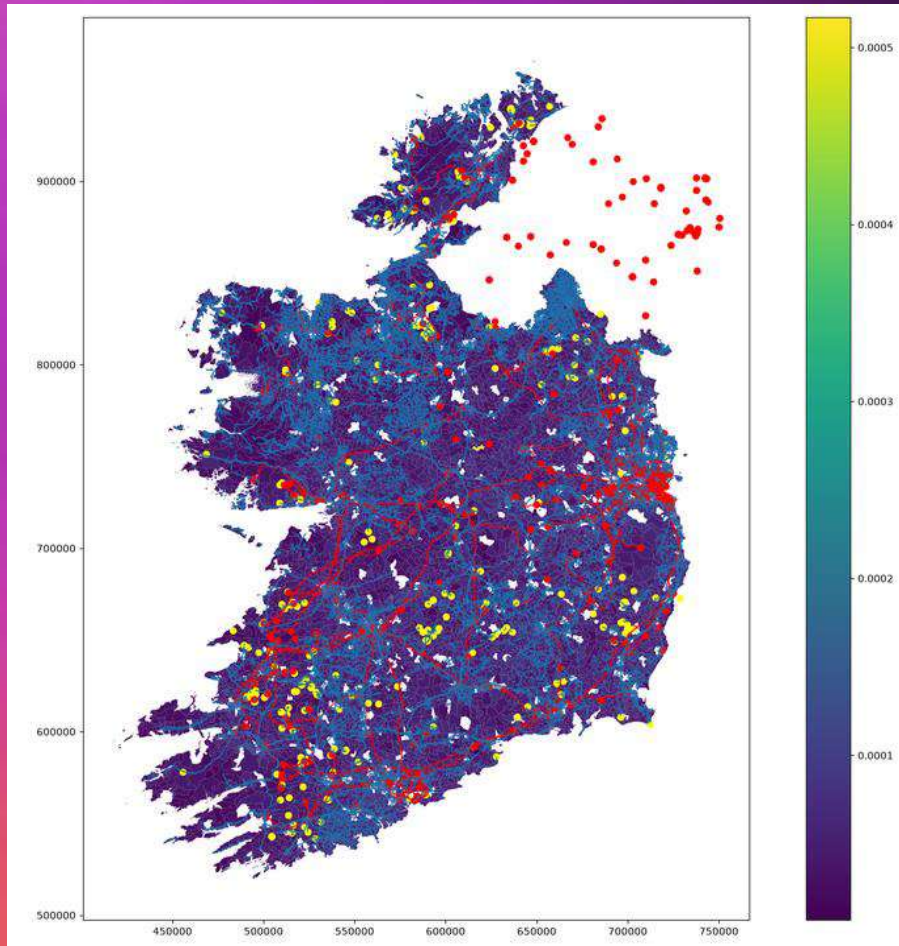
Generator Types

2024 North-West Generator Types



2024 All-Island Generator Types





Optimizing the Electric Grid – Battery Sizing & Placement

Group 4 - Alan Antony, Ishna Tripathi

Intro to battery storage and aims

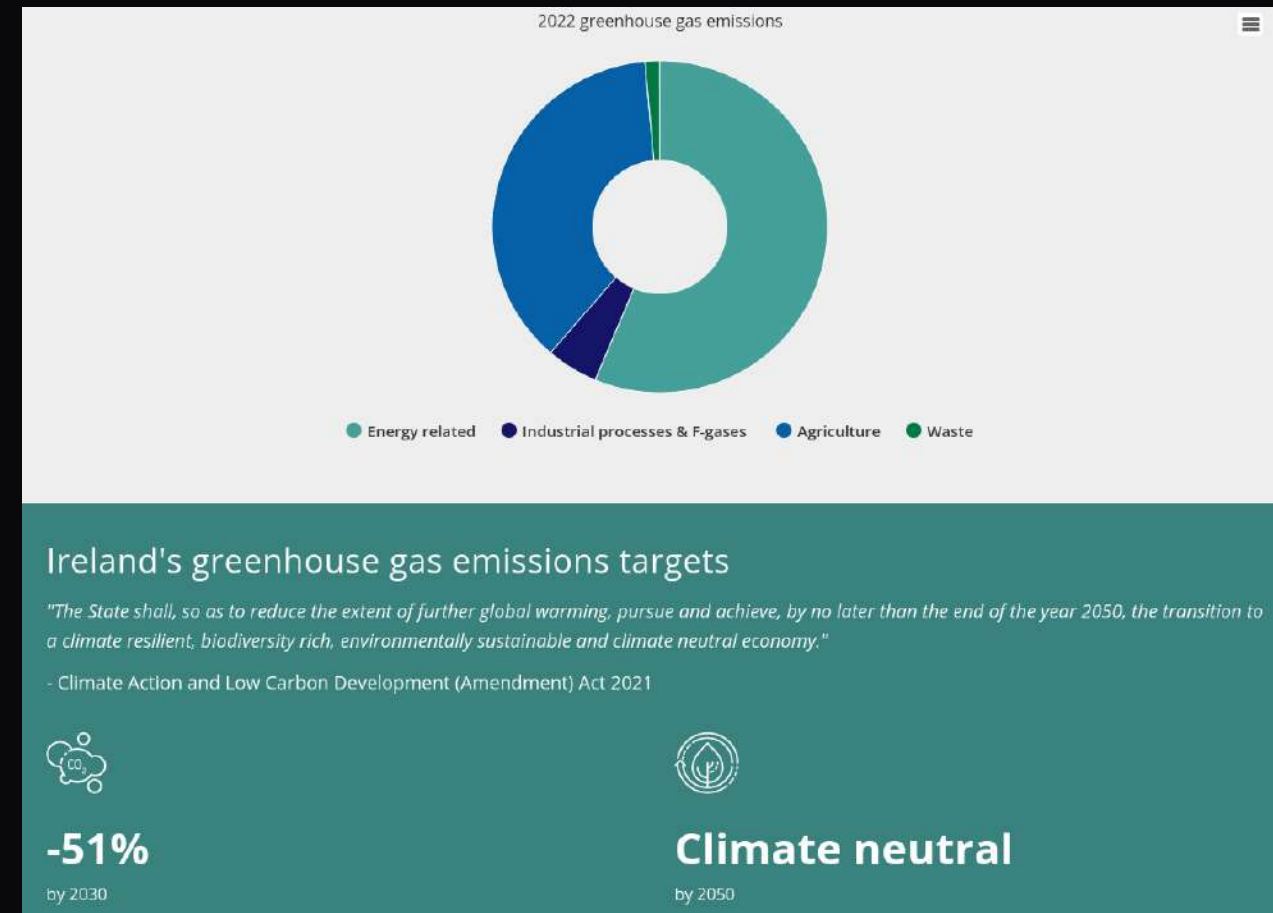
Renewable energy sources such as wind and solar are important for Ireland's development towards the goal of net zero emissions by 2050. However these sources are often restricted by temporal variables, i.e. sunlight and wind speed/direction. [1]

We can avoid this by storing surplus generated electricity through renewables into Battery Energy Storage Systems (BESS).

A report from energy storage Ireland claims nearly 10 GW of battery storage is in the development pipeline, an upgrade from the current ~1 GW of battery storage around the country. [2]

BESS are usually located close to generators, i.e. on the same bus as the windfarm itself, but is this truly the most optimal solution?

Our aim is to consider battery placement with a simple model to provide initial insights into the different site placement options.



[1] <https://www.sea.ie/about/irelands-energy-targets>

[2] <https://www.idaireland.com/latest-news/insights/energy-storage-systems>

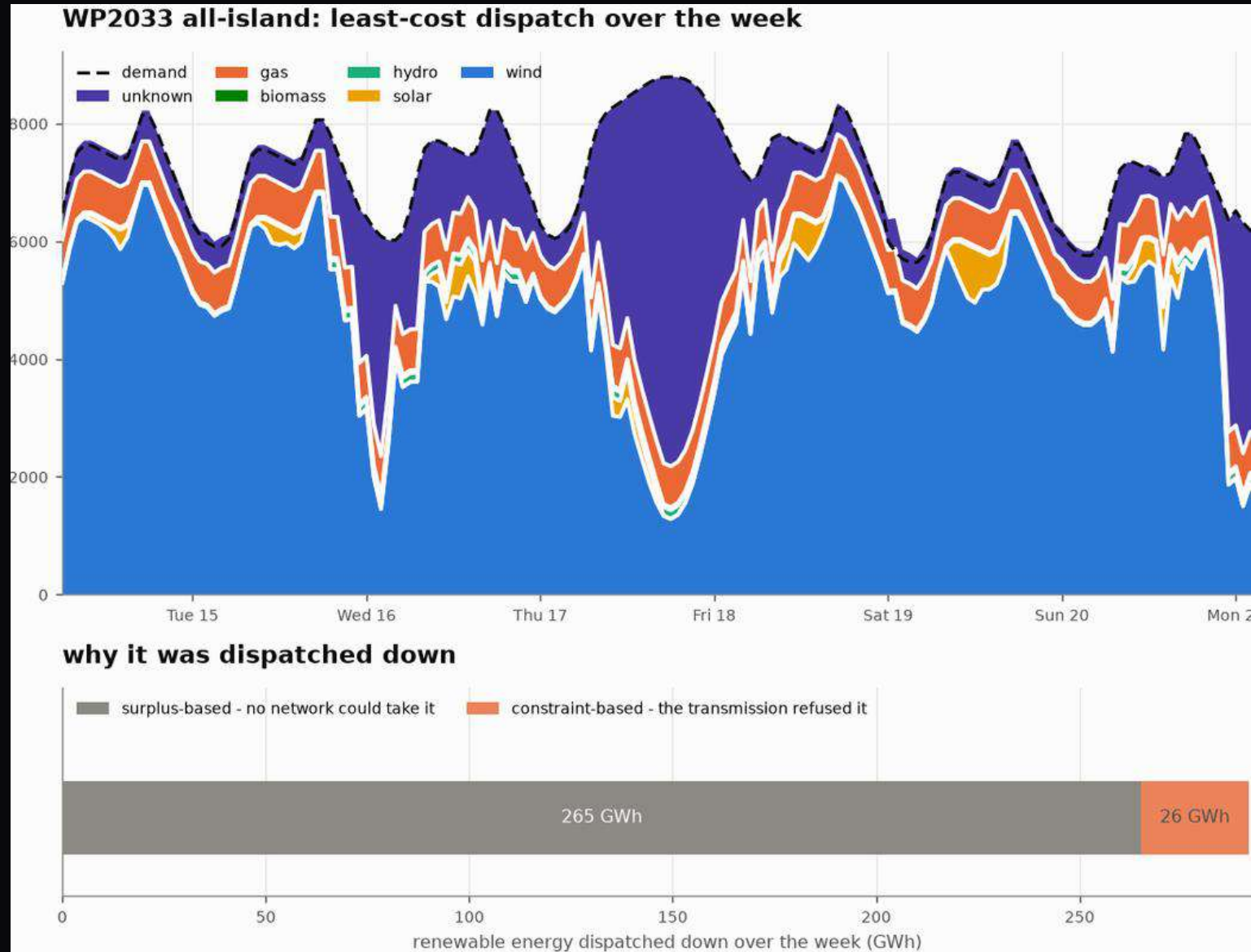
Initial Analysis

Analysis of the winter 2033 predicted dataset implies a ~ 265 GWh surplus of energy could be dispatch-downed due to low grid demand. (i.e. Surplus)

We need energy storage to prevent this produced energy from going to waste.

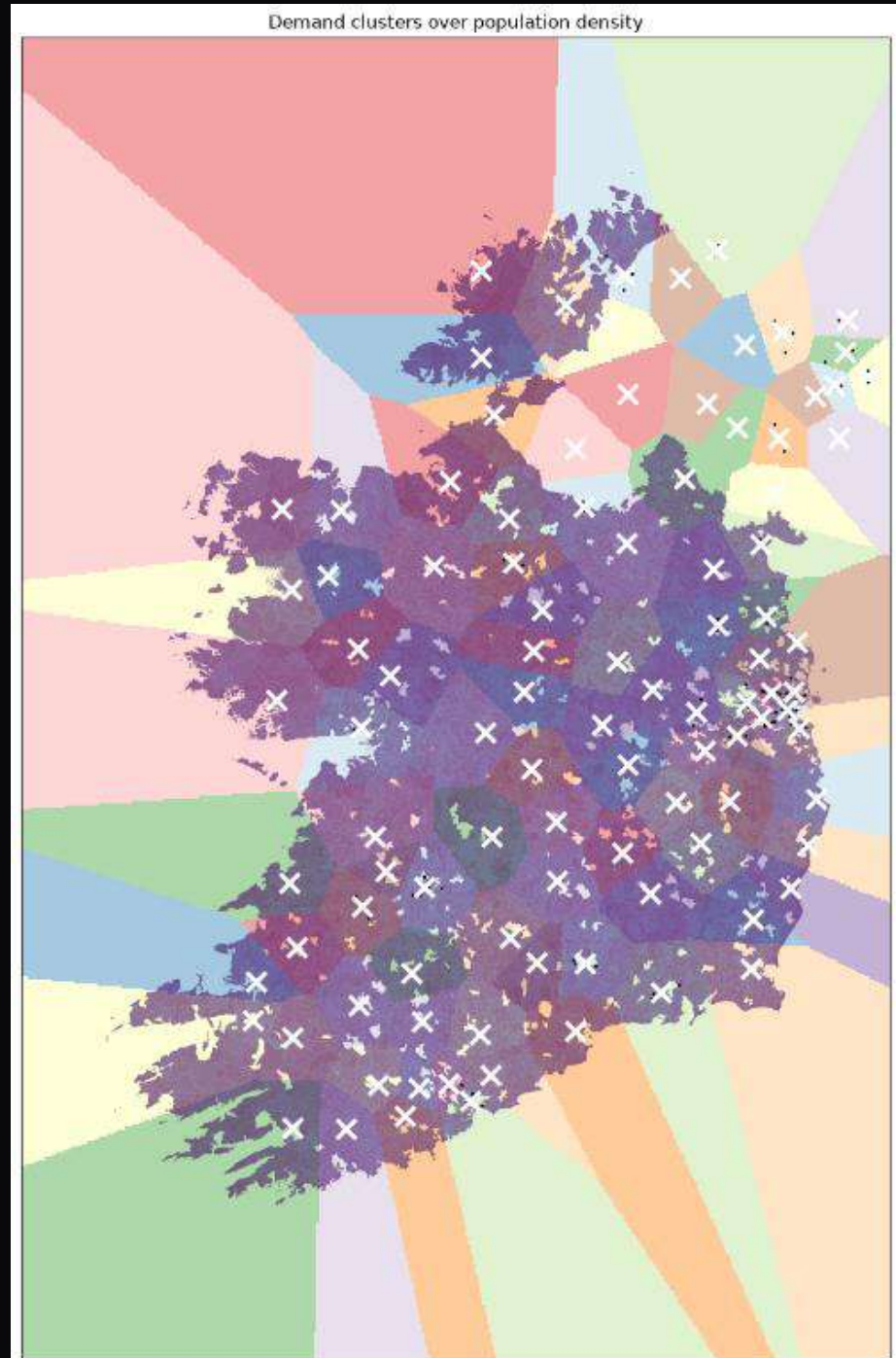
The stored energy can then be used at peak or when wind speeds are low.

BESS act as pseudo-inertia generators which means curtailment due to unstable inertia/frequency on the grid can be reduced, allowing for a larger share of renewables to be used at any time.



Methodology

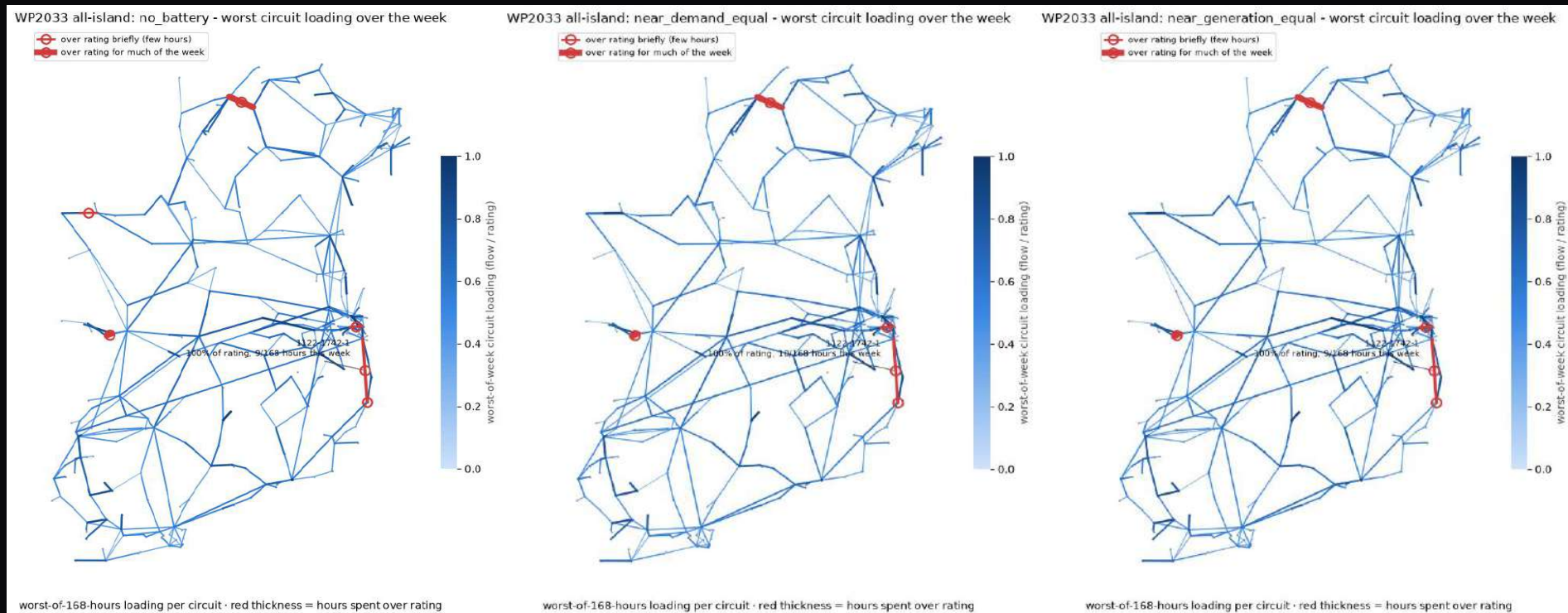
1. Cluster regions together by demand. (K-means)
2. Place battery at most demanding bus in each cluster.
3. Weight battery capacity by demand.
4. Run winter dataset and measure dispatch-down's.
5. Repeat for the naïve case of batteries at the generation buses.



Results I - Five 200 MWh batteries – 1 GWh Total

	strategy	total_GWh	surplus_GWh	constraint_GWh
	no_battery	291.457345	265.170409	26.286936
	near_demand_equal	288.202707	260.107554	28.095153
	near_demand_proportional	286.226902	260.106697	26.120204
	near_generation_equal	289.286757	262.115743	27.171014
	near_generation_proportional	288.932299	262.130468	26.801831

Lowest



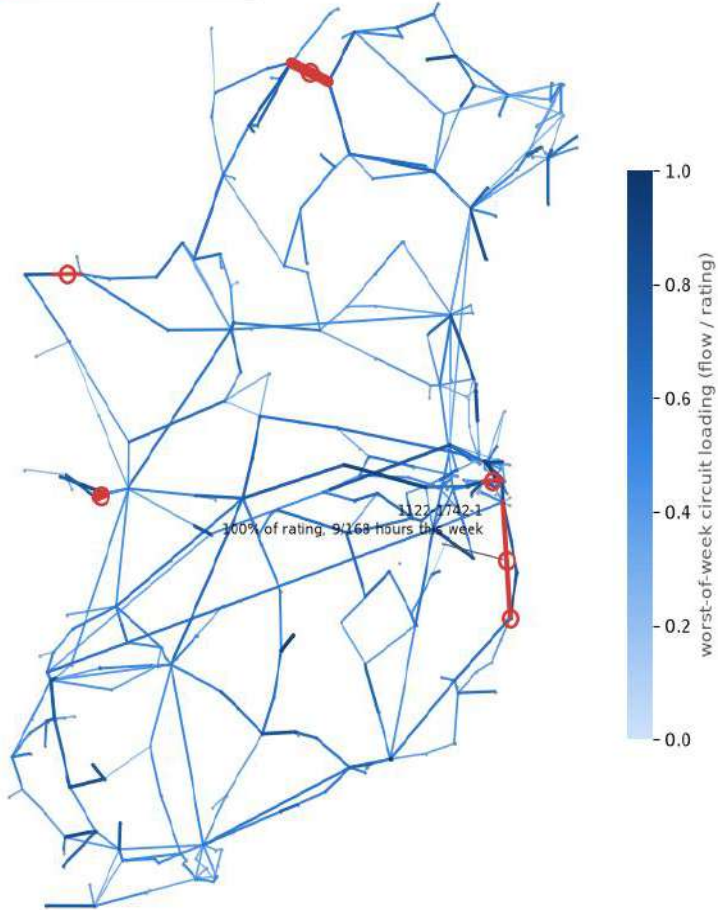
Results II - One Hundred 100 MWh batteries – 10 GWh total

scenario	strategy	total_GWh	surplus_GWh	constraint_GWh
WP2033	no_battery	291.457345	265.170409	26.286936
WP2033	near_demand_equal	263.390395	226.866395	36.523999
WP2033	near_demand_proportional	264.269498	226.692461	37.577037
WP2033	near_generation_equal	262.135818	226.044790	36.091028
WP2033	near_generation_proportional	262.196086	225.492219	36.703868

Lowest

WP2033 all-island: no_battery - worst circuit loading over the week

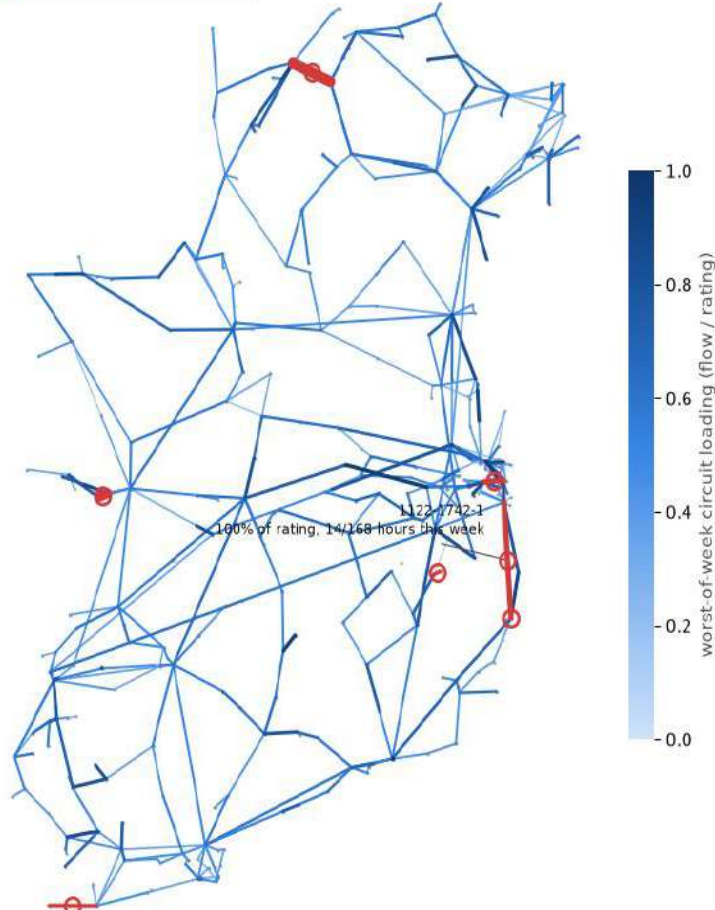
○ over rating briefly (few hours)
● over rating for much of the week



worst-of-168-hours loading per circuit · red thickness = hours spent over rating

WP2033 all-island: near_demand_equal - worst circuit loading over the week

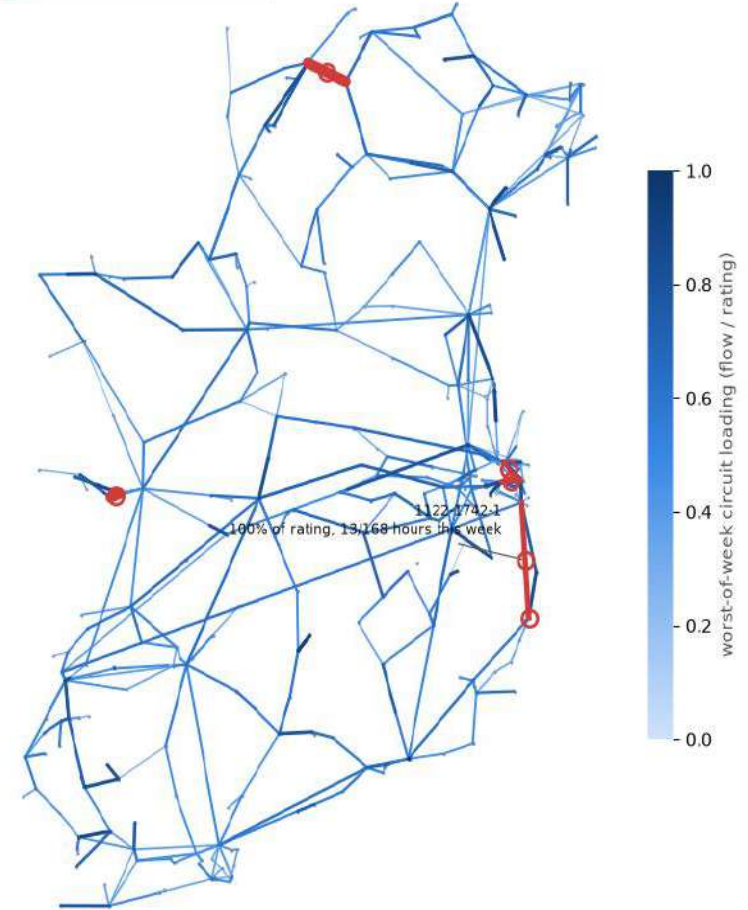
○ over rating briefly (few hours)
● over rating for much of the week



worst-of-168-hours loading per circuit · red thickness = hours spent over rating

WP2033 all-island: near_demand_proportional - worst circuit loading over the week

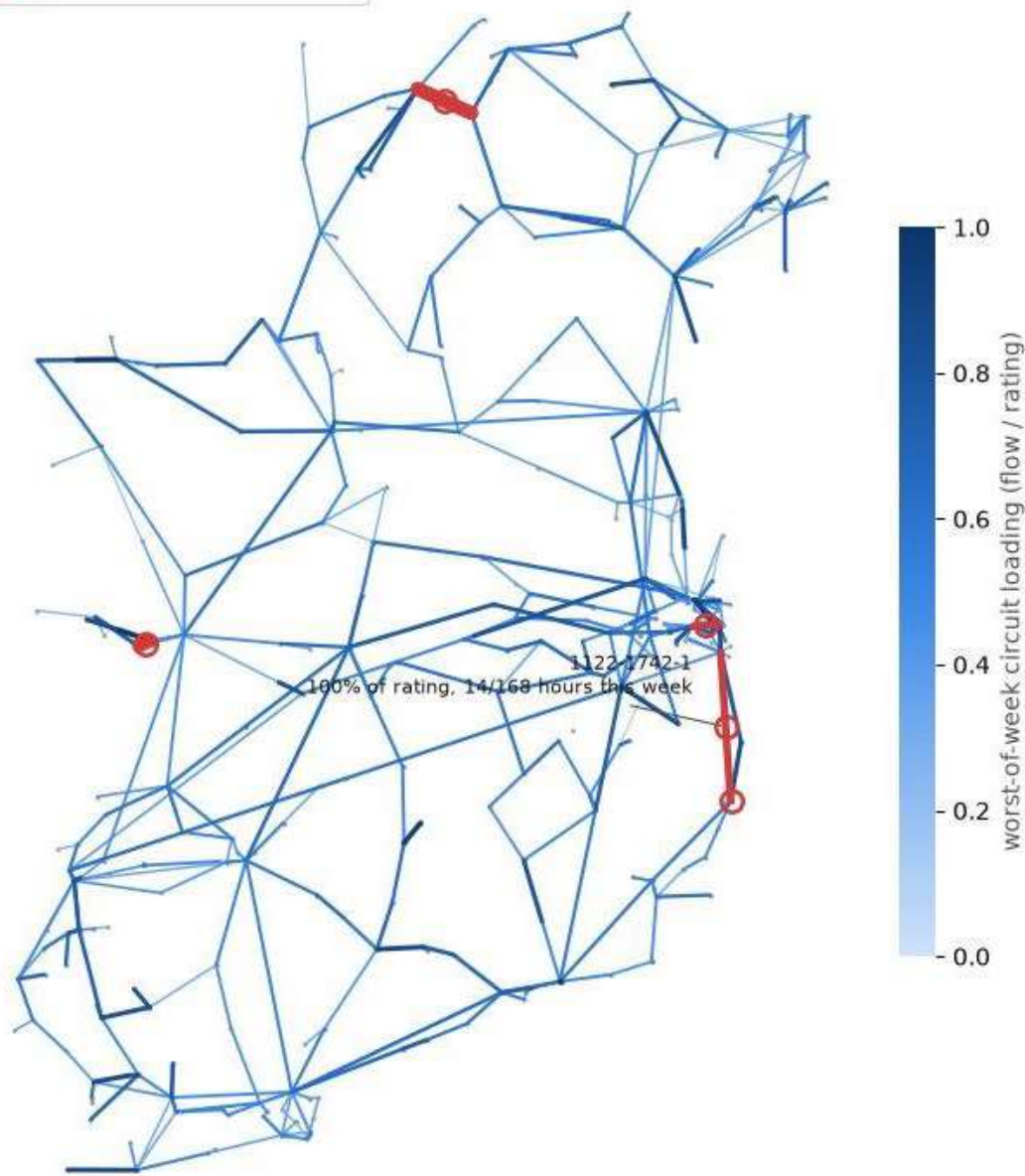
○ over rating briefly (few hours)
● over rating for much of the week



worst-of-168-hours loading per circuit · red thickness = hours spent over rating

WP2033 all-island: near_generation_proportional - worst circuit loading over the week

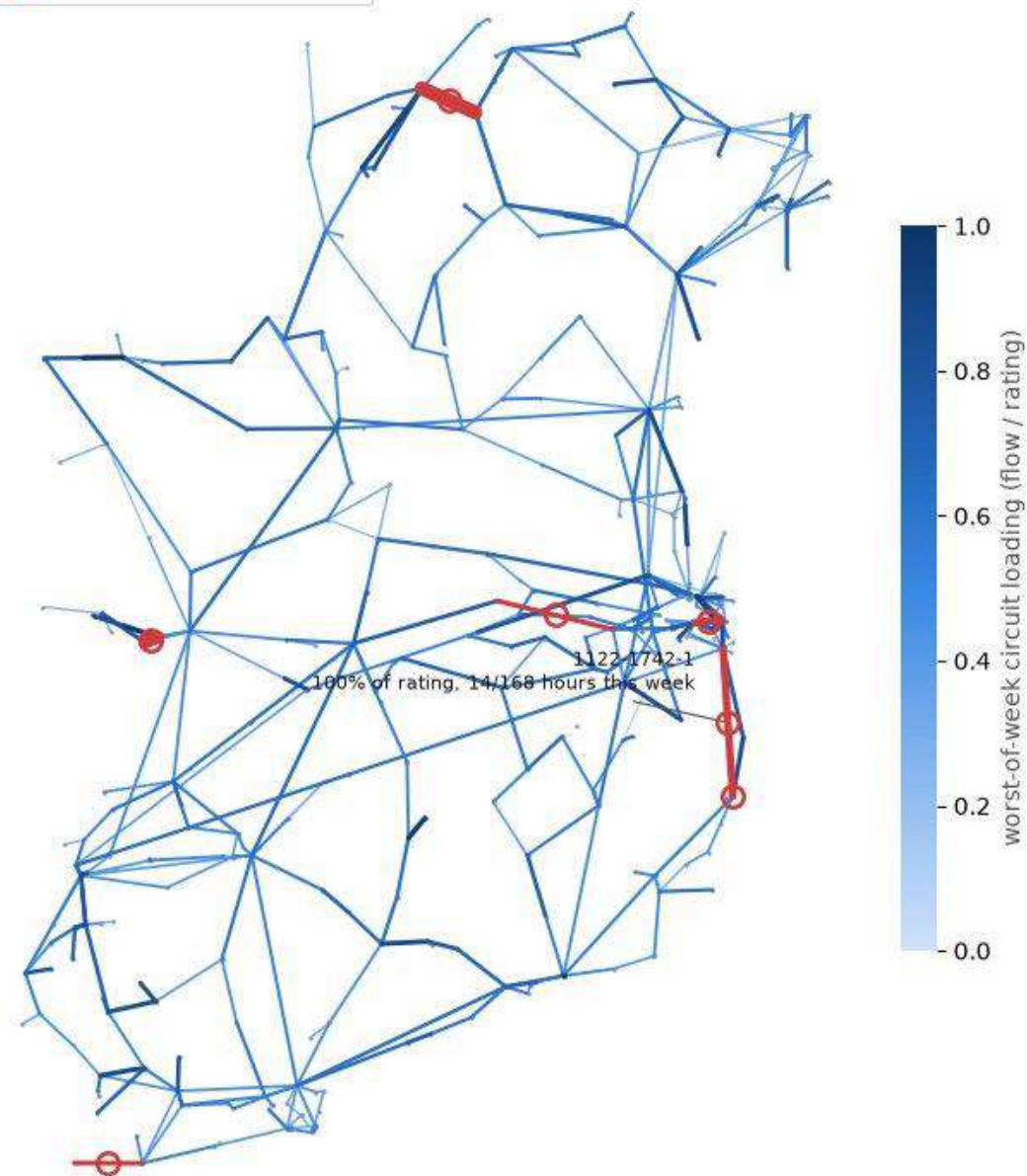
○ over rating briefly (few hours)
 ● over rating for much of the week



worst-of-168-hours loading per circuit · red thickness = hours spent over rating

WP2033 all-island: near_generation_equal - worst circuit loading over the week

○ over rating briefly (few hours)
 ● over rating for much of the week



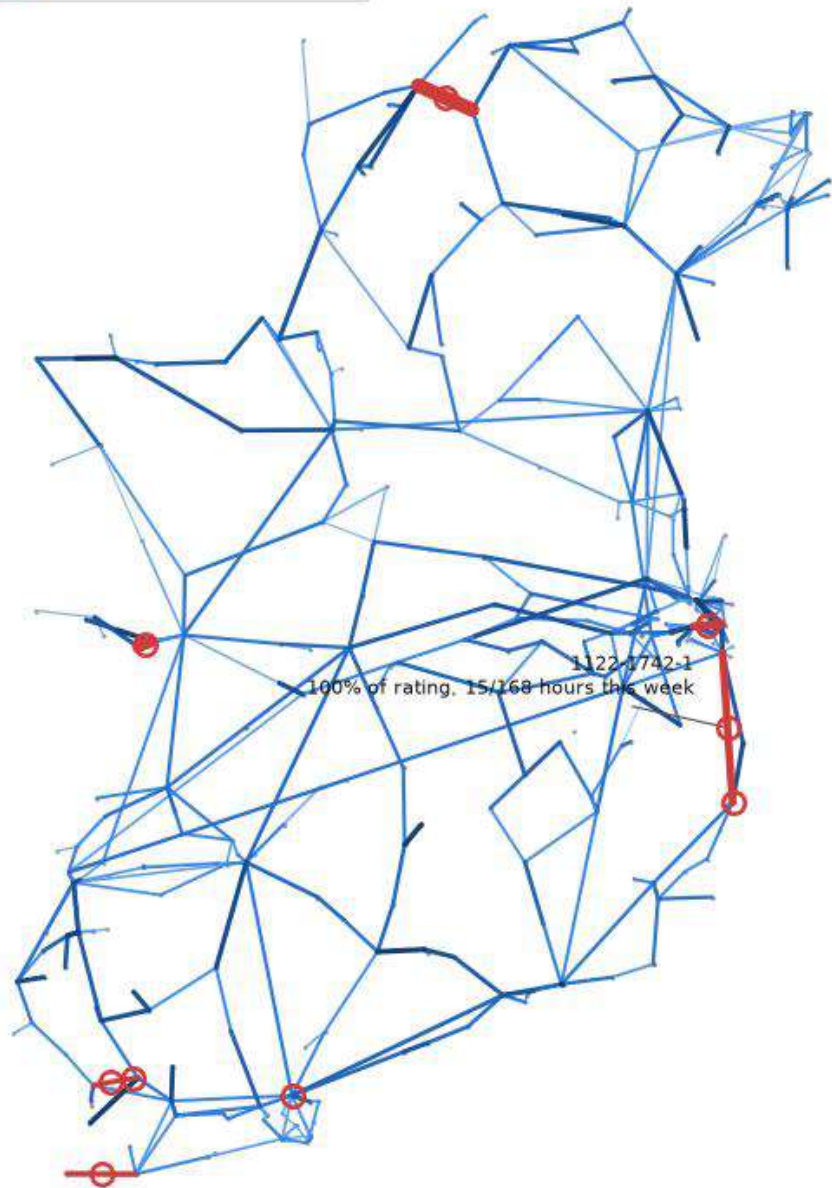
worst-of-168-hours loading per circuit · red thickness = hours spent over rating

Results III - One Hundred 300 MWh batteries – 30 GWh total

scenario	strategy	total_GWh	surplus_GWh	constraint_GWh
WP2033	no_battery	291.457345	265.170409	26.286936
WP2033	near_demand_equal	224.994400	193.832171	31.162229
WP2033	near_demand_proportional	224.391334	193.668042	30.723292
WP2033	near_generation_equal	222.958327	193.793223	29.165104
WP2033	near_generation_proportional	221.836128	193.124628	28.711499

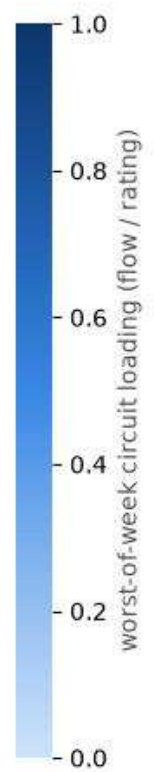
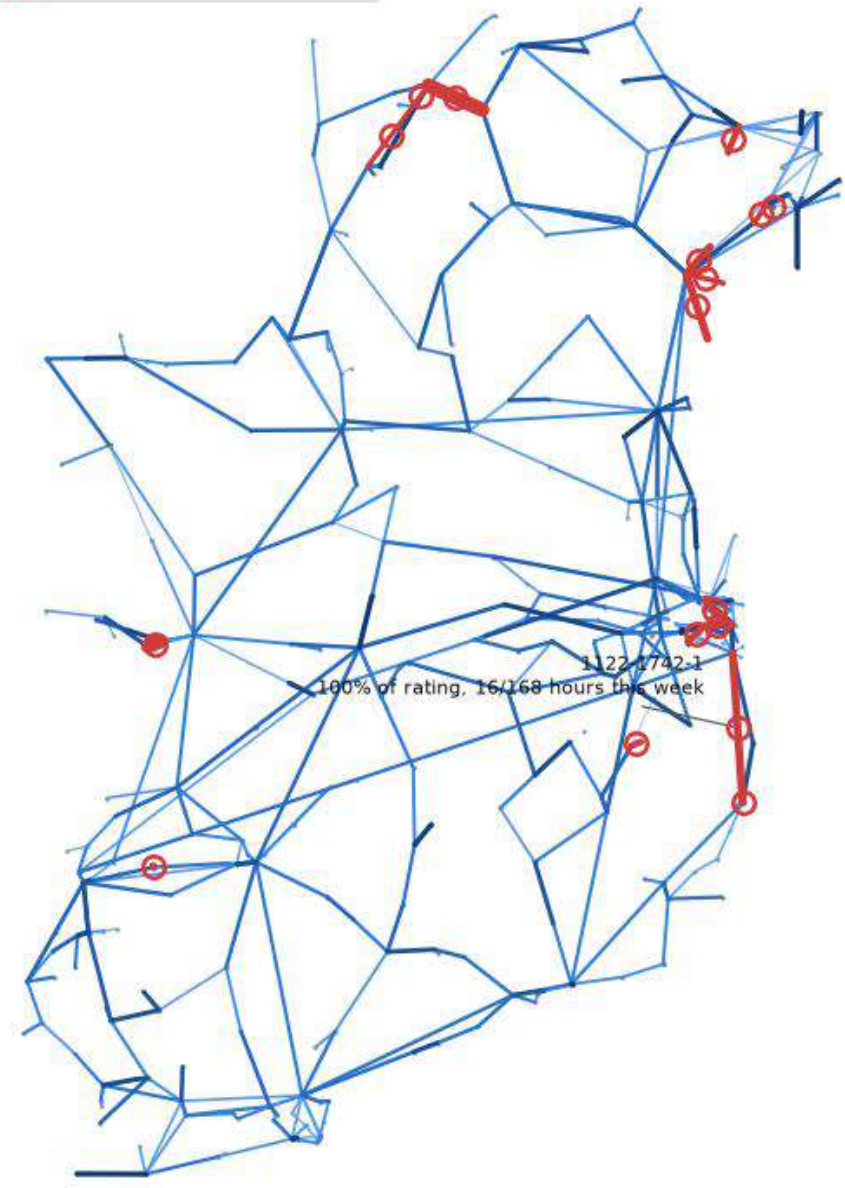
Lowest

○ over rating briefly (few hours)
 ● over rating for much of the week



worst-of-168-hours loading per circuit · red thickness = hours spent over rating

○ over rating briefly (few hours)
 ● over rating for much of the week



worst-of-168-hours loading per circuit · red thickness = hours spent over rating

Any Questions?

Email for data/scripts - antonyal@tcd.ie

EV Chargers, Adoption, and the Grid

Enda Byard Shaughnessy, Brannan Jio, Conor McCormack, Tom Ryan, Marcus
Whelan O'Connor

TPSA Hackathon 2026

Background Information

"Although Cork and Dublin together make up ~40% of the Republic's population, they account for ~50% of BEV ownership. One major question then becomes how and where might we invest in supporting infrastructure."

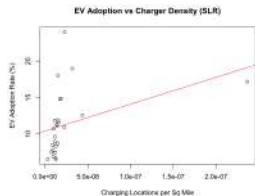
- We set out to check whether charger provision tracks EV adoption, and whether the answer changes by region.
- We then asked where would be best to implement new connections between electrical stations to efficiently power the new chargers.

Variable	Source
Registrations by engine type	SIMI Motorstats
Chargers, roads, shapefiles	OpenStreetMap
Income, population, area	CSO
Home charger grants	SEAI LA-CAP
Distance to hub city	bluejayliveability

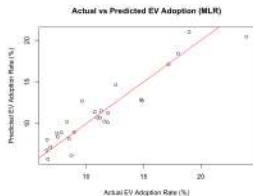
- We used data from the Republic of Ireland only, because law and policy are common across it.
- Because chargers may also follow demand, we report everything as an association rather than a proven effect.

First Regression Results

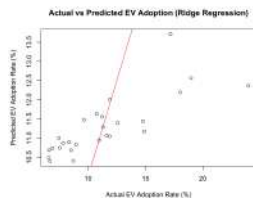
SLR



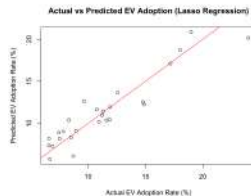
MLR



Ridge

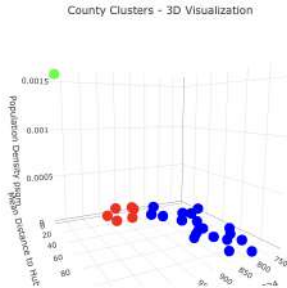


Lasso

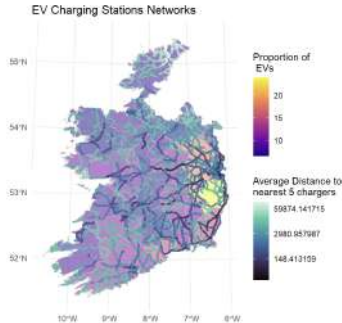


- The SLR slope is positive, but almost every county sits in one vertical column, so the line is set by a few major outliers.
- MLR and lasso track the diagonal well, while ridge shrinks every coefficient and collapses into a narrow band.
- With 26 counties we can only detect large effects, and Dublin is an extreme value on every variable we have.
- A single national slope therefore describes no specific county accurately due to the variety between counties.

Why we split Ireland into three groups



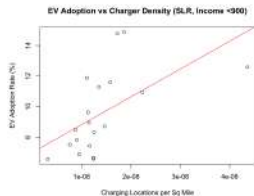
1
2
3



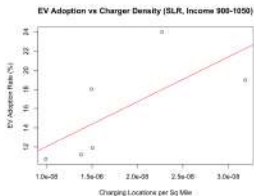
- We ran k -means over twelve variables and used the elbow method to choose the number of groups.
- Three clusters came out, matching Dublin, minor city counties and rural counties.
- The map shows why: EV proportion and charger connectivity both radiate outward from Dublin.
- Dublin ends up alone, confirming it was driving the national result.

Regression on the more rural regions

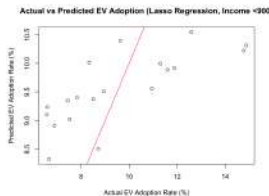
SLR, rural regions



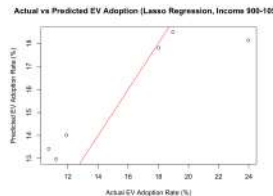
SLR, minor cities



Lasso, rural regions



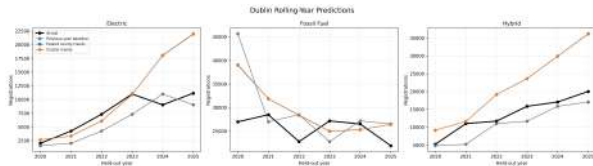
Lasso, minor cities



Rural quasi-binomial GLM	Estimate	<i>p</i>
Distance to hub city	-4.67×10^{-3}	0.055
Mean income	1.78×10^{-3}	0.237
Charger locations	-5.60×10^6	0.852

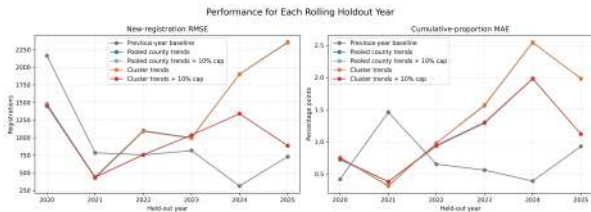
- The vertical stacking disappears once we consider the clusters separately, because within a group charger provision genuinely varies.
- In the rural GLM the only near-significant predictor is distance to the nearest hub city, and it is negative. This supports the notion that, the further one must drive, the less practical an EV becomes.
- Charger placement itself is predicted by nothing we measured.

Forecasting registrations, and why Dublin broke it



- Using ten years of data we fitted a negative binomial GLM with a log link, chosen for the multiplicative growth in EV purchases.
- Standard cross-validation failed badly on both electric and hybrid registrations.
- The reason is visible here: Dublin adopted EVs far faster than the rest of the country, so the model overfits to it.
- Our error on Dublin hybrids reached 97%.

Clustering and a growth cap both improve the forecast



Model	MAE	RMSE	R^2
Cluster trends	430	1,270	0.862
Pooled county trends	482	1,514	0.806
Capped Cluster Trends	357	1035	0.909

- We switched to rolling validation, predicting each year in turn to give a better estimation of how the model performs over time.
- Clustering measurably improved the forecast, which independently justifies the earlier decision to split.
- Capping total growth at 10% a year cut the error further, because nobody can physically buy that many new cars.

Treating the grid as a graph

- Our charger analysis showed the network radiates from Dublin, so we asked where new connections would help most.
- Substations are nodes and transmission lines are edges, weighted by one over their length.
- The effective resistance between two stations measures how easily power moves between them.
- We sum that over every pair to score the whole network in one number.

$$R_{ij} = L_{ii}^+ + L_{jj}^+ - 2L_{ij}^+, \quad R_g = \sum_{i < j} R_{ij}$$

- A lower R_g means electricity moves more easily everywhere.
- Our network has 65 buses and 82 edges, with a baseline R_g of 187,637.
- We screened 1,249 candidate lines, and our fast method matched a full recalculation to zero error.

The ten best new connections, and what they cost

#	Connection	km	ΔR_g	€m	€/unit
1	Hannahstown – Louth	85.5	−6.9%	65	5.04
2	Magherafelt – Louth	95.4	−6.8%	72	5.63
3	Tamnamore – Louth	68.5	−6.8%	52	4.12
4	Kells – Louth	108.2	−6.5%	82	6.75
5	Castlereagh – Louth	89.9	−6.4%	68	5.71
6	Ballycronan More – Louth	123.9	−6.0%	93	8.30
7	Louth – Kilroot	111.5	−5.8%	84	7.71
8	Louth – Coolkeeragh	143.2	−5.6%	108	10.17
9	Louth – Glanagow	297.7	−4.8%	222	24.60
10	Louth – Aghada	294.3	−4.8%	220	24.54

Costed with EirGrid's published Standard Transmission Charges.

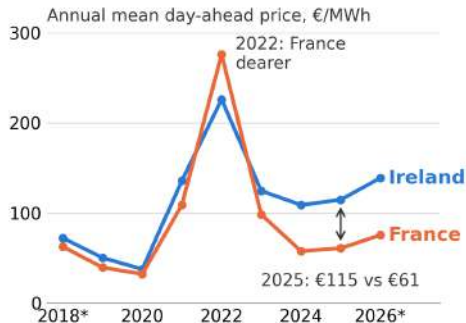
- Every one of the top ten ends at Louth, which makes it the single most valuable place to connect to.
- The winners all attach to remote, poorly connected substations such as Coolkeeragh and Srananagh.
- Tamnamore is 17 km shorter than Hannahstown but gives almost the same benefit, making it €13m cheaper.
- Ranking on benefit alone therefore picks the wrong line to build, as cost is also an essential factor.

Ireland's Electricity Grid



What a 700 MW link to France would do to Irish prices

- Celtic, 700 MW from Cork to Brittany, is due at the end of 2028.
- Replaying each hour with the cable flowing to the dearer side, it cuts the Irish price by about €9/MWh whenever it imports (2023–25).

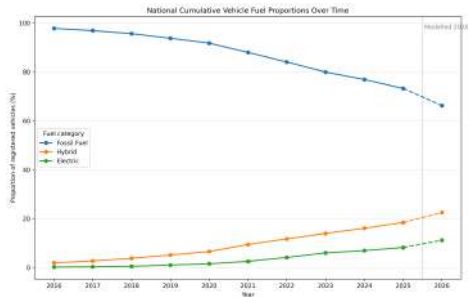


Period	Mean change (€/MWh)	Hours exporting
Apr 2023–Jan 2025	−6.3	12%
Apr–Dec 2025	−7.4	8%
Oct 2018–Sep 2026	−3.3	23%

- It runs full in about 80% of hours: 700 MW is too small to close the gap.

Fitted Irish supply curve (price against demand minus wind, scaled by gas and carbon cost); flows solved each hour until prices meet or the cable is full; French price held fixed; 3.3% losses. *Part years. Data: Ember, EirGrid, ICE.

Conclusion



- Charger implementation alone do not explain rural EV uptake; distance to a hub city also does.
- Charger placement itself is not predicted by any variable we measured, which points to an ad hoc siting strategy.
- Clustering improved both our regressions and our forecast, taking R^2 from 0.806 to 0.862.
- We forecast electric vehicles reaching roughly 11% of the national fleet* in 2026.
- On the grid, ranking on benefit per euro rather than benefit alone puts Tamnamore–Louth first: €52m for a 6.8% efficiency gain.
- Celtic, the 700 MW link to France, would cut Irish prices by about €6–7/MWh in recent conditions, but is too small to close the gap.

Optimal Battery Siting and Sizing

GROUP 7

Ulaş İçer · Kacper Kotwica · Zara Negura
Mohamed Shehata · Deea Zaharia

Problem

The Irish electricity grid with high penetration of renewable energy sources (wind) faces significant challenges:

- **Surplus:** Renewable electricity that is in excess when renewable supply exceeds the demand.
- **Curtailment:** Renewable energy generation often exceeds demand, forcing operators to reduce generator output

Therefore, our project addresses the question:

Where should a battery be placed, and how many grid connections should it have, to maximize renewable energy utilization?



Approach

Our algorithm comprises the Genetic Algorithm with a fitness function that incorporates the PyPSA optimisation function.

Our constraints:

- Only evaluate locations on land (using Cartopy/Natural Earth data)
- Focus on buses at the same voltage level as the new battery (110 kV, avoids transformer requirements)
- Maximum transmission line length (100 km)

Latter two of these constraints can be adjusted from the code.

Time Complexity

$O(G \cdot P \cdot K \cdot S)$ where:

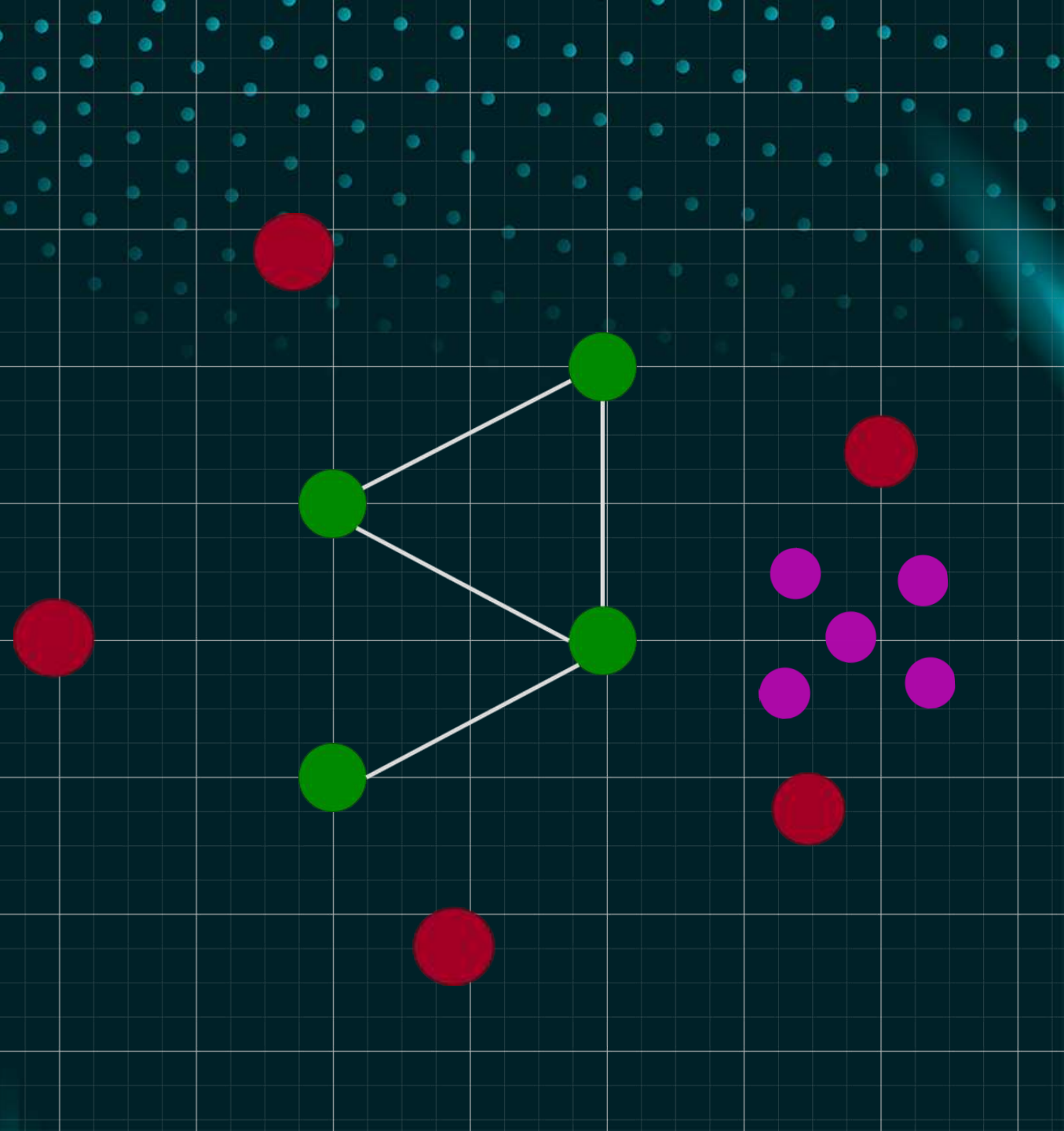
- **G**: Number of generations (default: 5)
- **P**: Population size (default: 10)
- **K**: Connection counts to test (default: [1,2,3,4,5])
- **S**: Solver time per network optimization

Typical execution: ~9 minutes for 500 total optimization runs

Genetic Algorithm

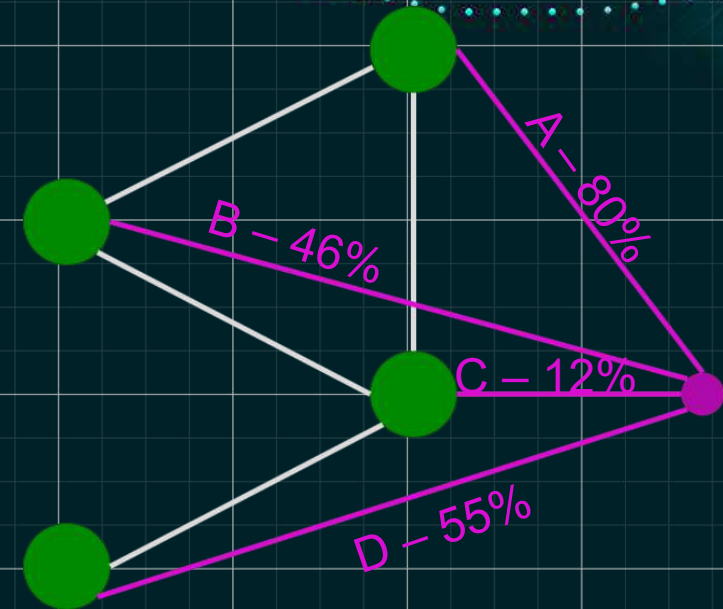
Searches for the optimal battery site.

- **Population:** Multiple candidate locations ((LON, LAT) pairs) on land
- **Fitness Function:** Percentage of curtailment & surplus waste reduction achieved
- **Natural Selection:** Elimination of the worse performers
- **Regrowth:** Grow population with inherited locations from parent nodes plus random mutation

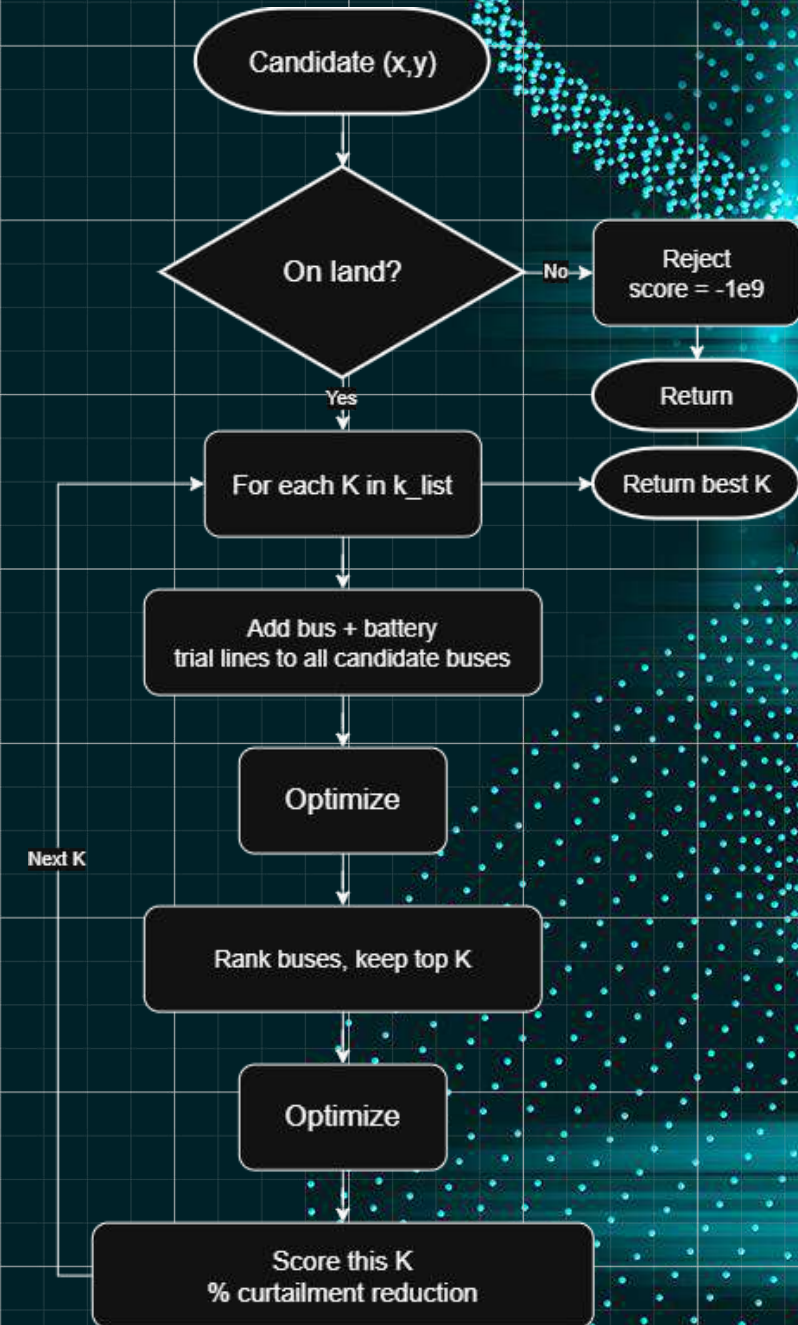
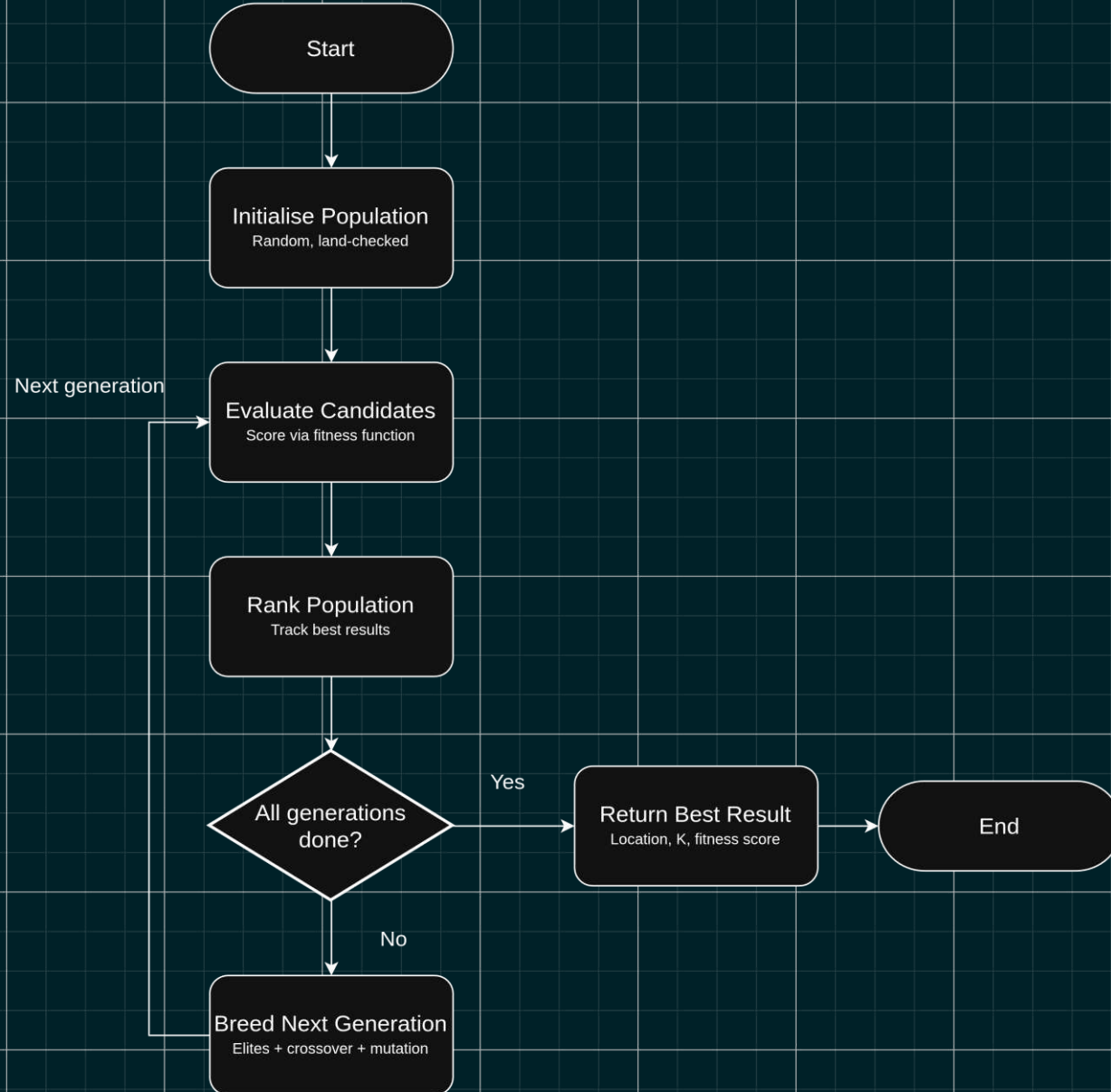


Fitness Algorithm

- **Population:** Drawing candidate, eligible lines
- **Optimize:** Using the PyPSA algorithm
- **Ranking:** What percentage of lines are used
- **Regrowth:** Only keep top K lines
- **Optimize:** Using the PyPSA algorithm
- **Output:** The surplus waste reduction percentage



Flowcharts



OUTPUT RESULTS

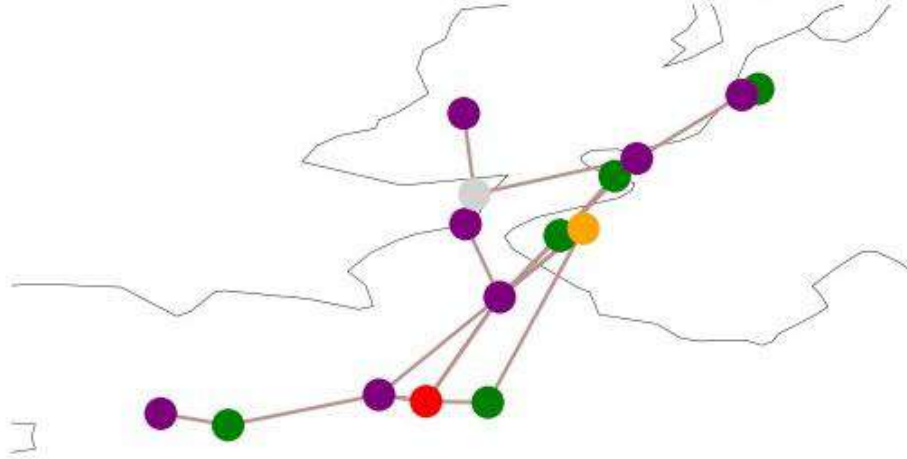
For the optimal solution, the project reports:

- **Optimal location:** Geographic coordinates (latitude, longitude)
- **Connection count (K):** Number of transmission lines to connect
- **Battery capacity:** Optimal power (MW) and energy (MWh) sizing
- **Curtailment reduction:** MWh and percentage reduction in wasted renewable energy
- **Dispatch reduction:** Percentage change in dispatch-down penalties



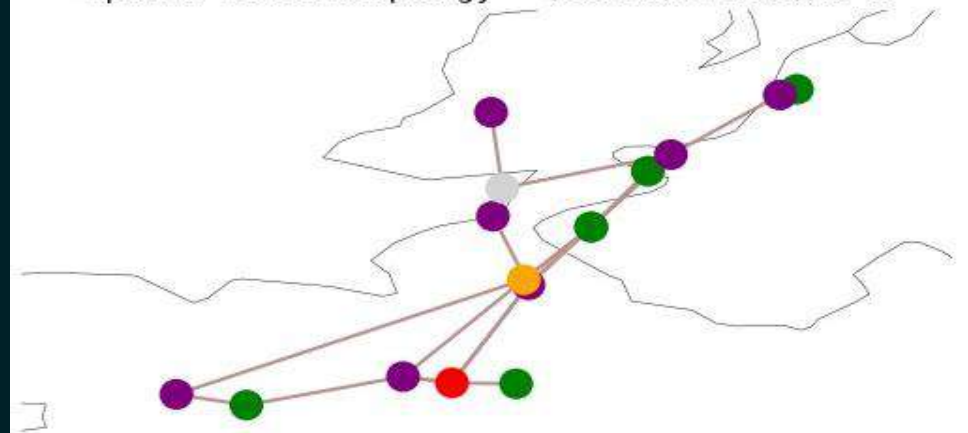
SV 2024

Optimal network topology — $(-7.903, 54.705)$, $K=4$



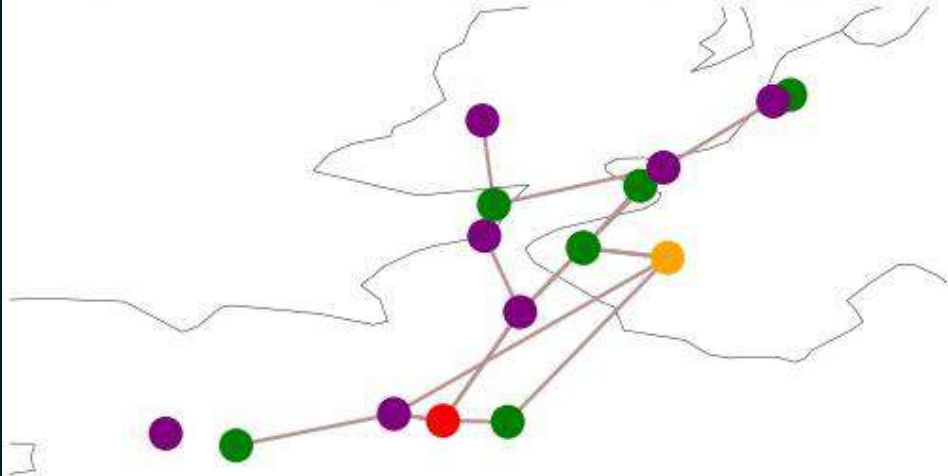
WP 2024

Optimal network topology — $(-8.173, 54.513)$, $K=5$



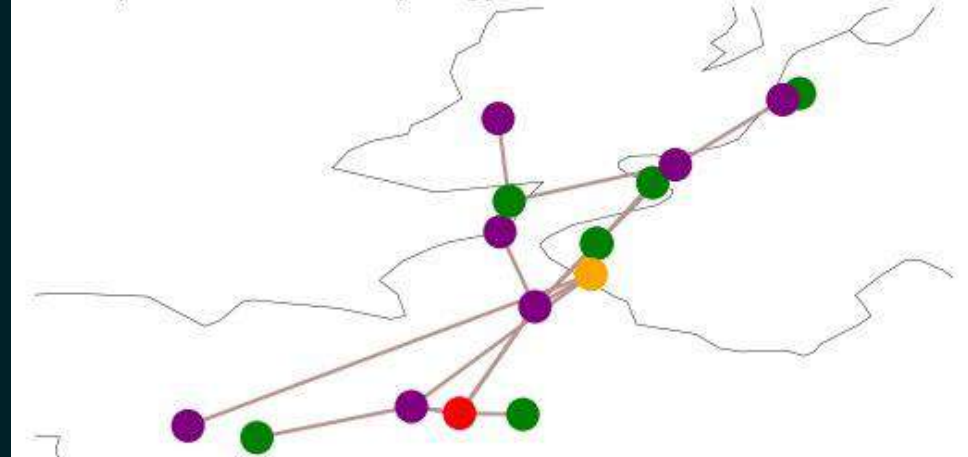
SV 2033

Optimal network topology — $(-7.727, 54.655)$, $K=4$



WP 2033

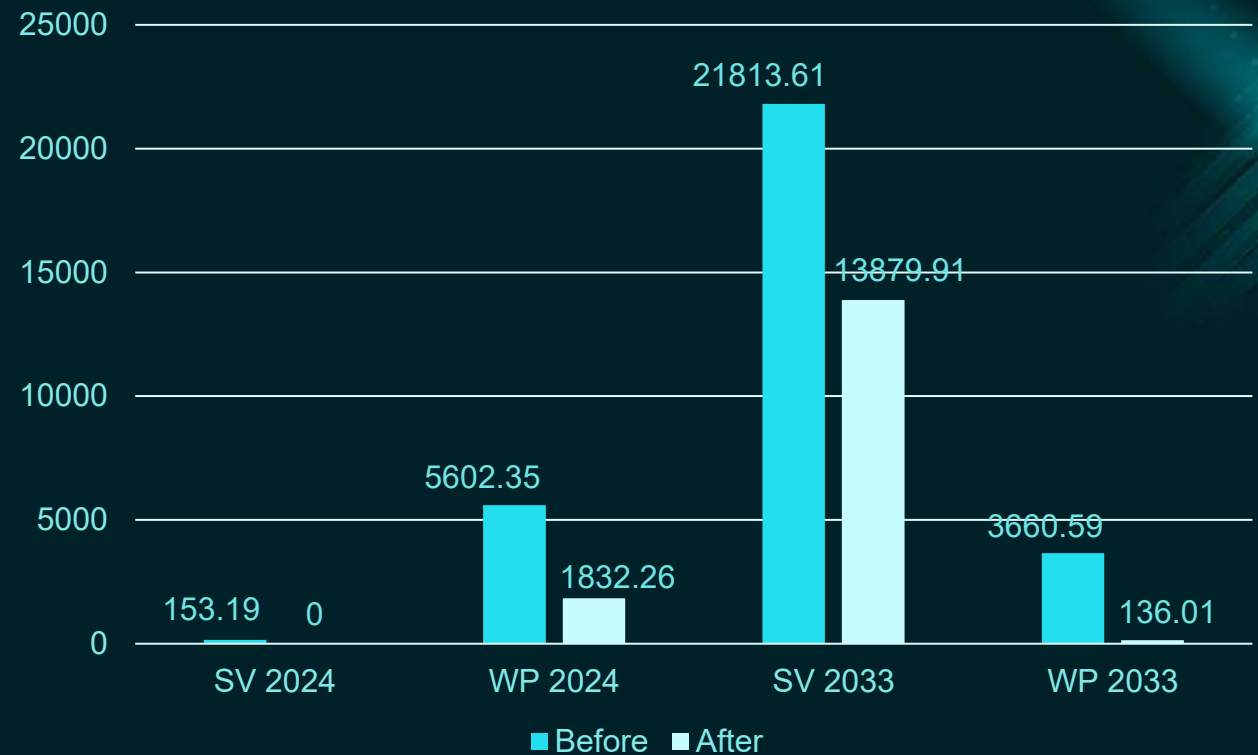
Optimal network topology — $(-7.991, 54.592)$, $K=5$



Evaluating Results

- **Storage effectiveness is highly scenario-dependent:** Dispatch-down reduction ranges from 36% (SV 2033) to 96% (WP 2033).
- **2024 baselines show minimal curtailment** (more renewable energy source will be available in the future)
- **Winter Peak scenarios respond best to battery intervention** (curtailment is cut by ~67% in WP 2024 and by ~96% in WP 2033)
- **Total curtailment reduction** across all four scenarios is substantial

Curtailment (MWh) against Scenarios

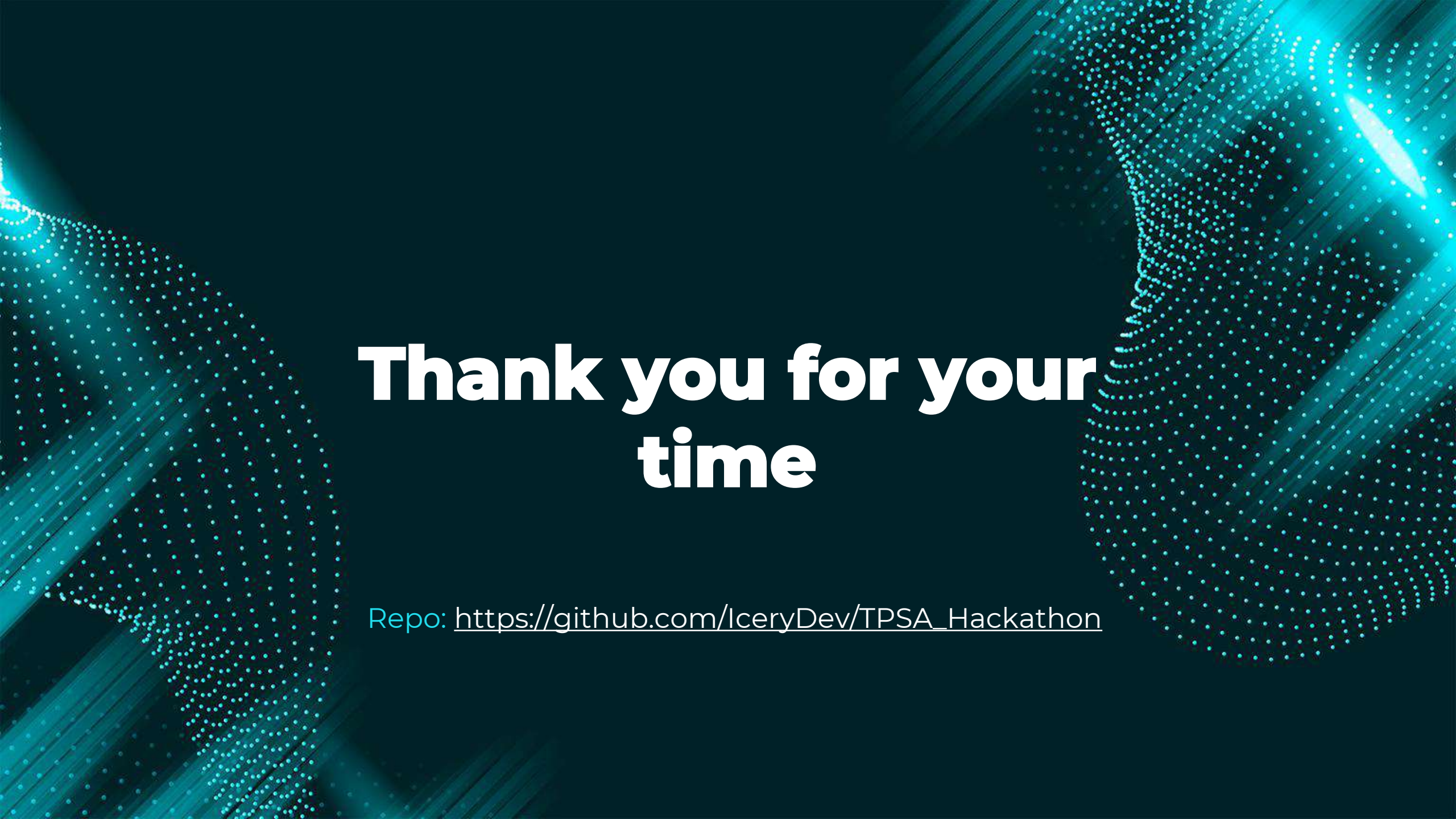


CONCLUSION

- Used a genetic algorithm paired with PyPSA network optimization to search for the best battery site and size on the Irish grid
- Placement matters as much as capacity, a battery sited where curtailment occurs delivers outstanding returns, while a poor location barely helps
- Results show 7.95% renewable waste reduction in 2024, growing to 15.13% by 2033 as curtailment volume rises
- As Ireland's grid leans further into wind, the value of smart siting and sizing doesn't stay flat; it compounds, making this kind of optimization more important, over time

REFERENCES

- **Boonluk, P., Siritaratiwat, A., Fuangfoo, P., & Khunkitti, S. (2020).** *Optimal Siting and Sizing of Battery Energy Storage Systems for Distribution Network of Distribution System Operators.*
- **NREL. (2024).** Annual Technology Baseline 2024b: Utility-Scale Battery Storage. National Renewable Energy Laboratory.
- **Calatayud, B. (2025).** *Europe's Battery Storage Edge.* Capstone DC.
- **PyPSA-Eur:** A Sector-Coupled Open Optimisation Model of the European Energy System. GitHub Repo.



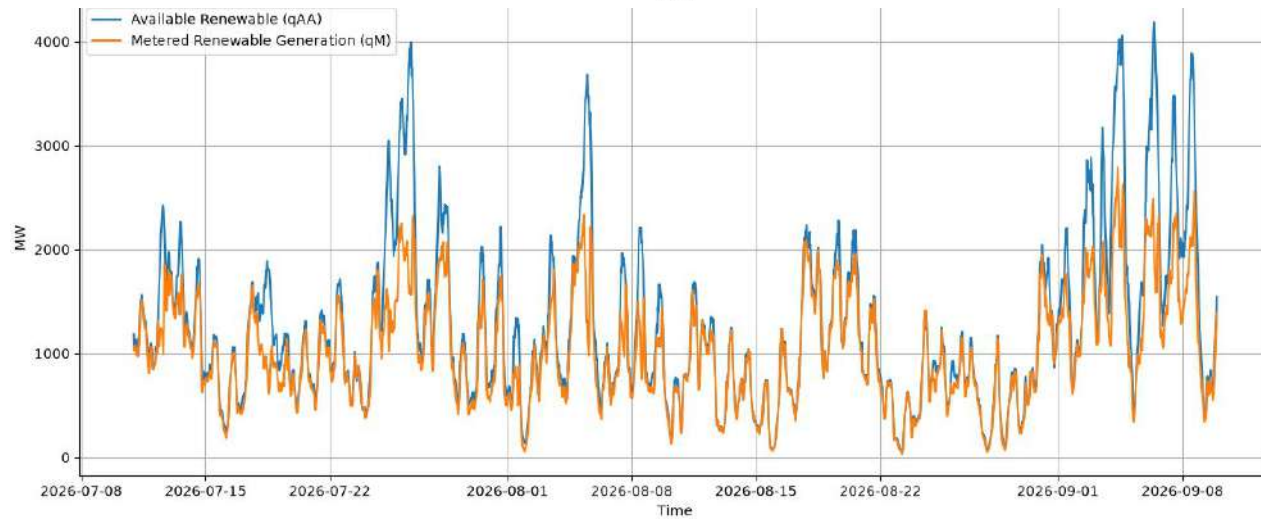
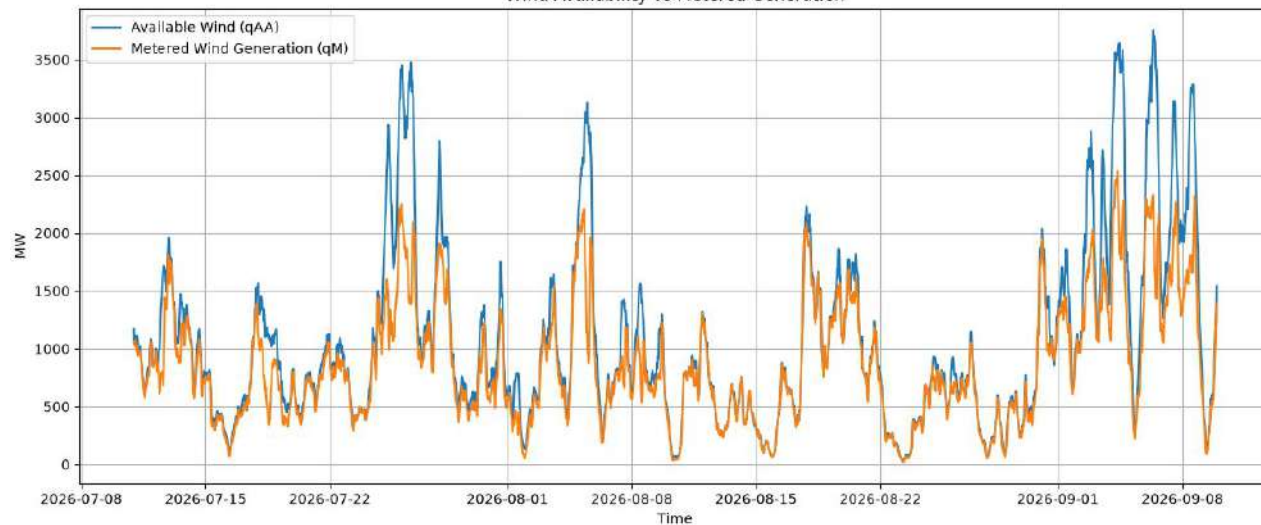
**Thank you for your
time**

Repo: https://github.com/IceryDev/TPSA_Hackathon

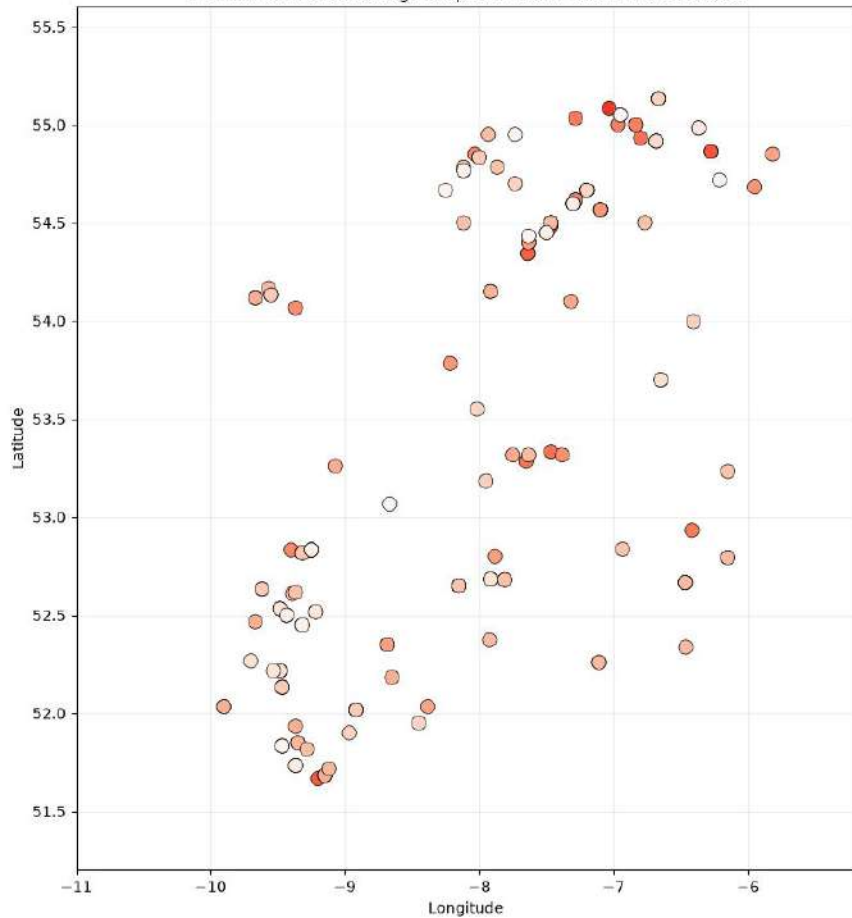
**Tackling the problem of constraint group construction
to reduce wind energy waste and defend against cascading outages**

Team 6

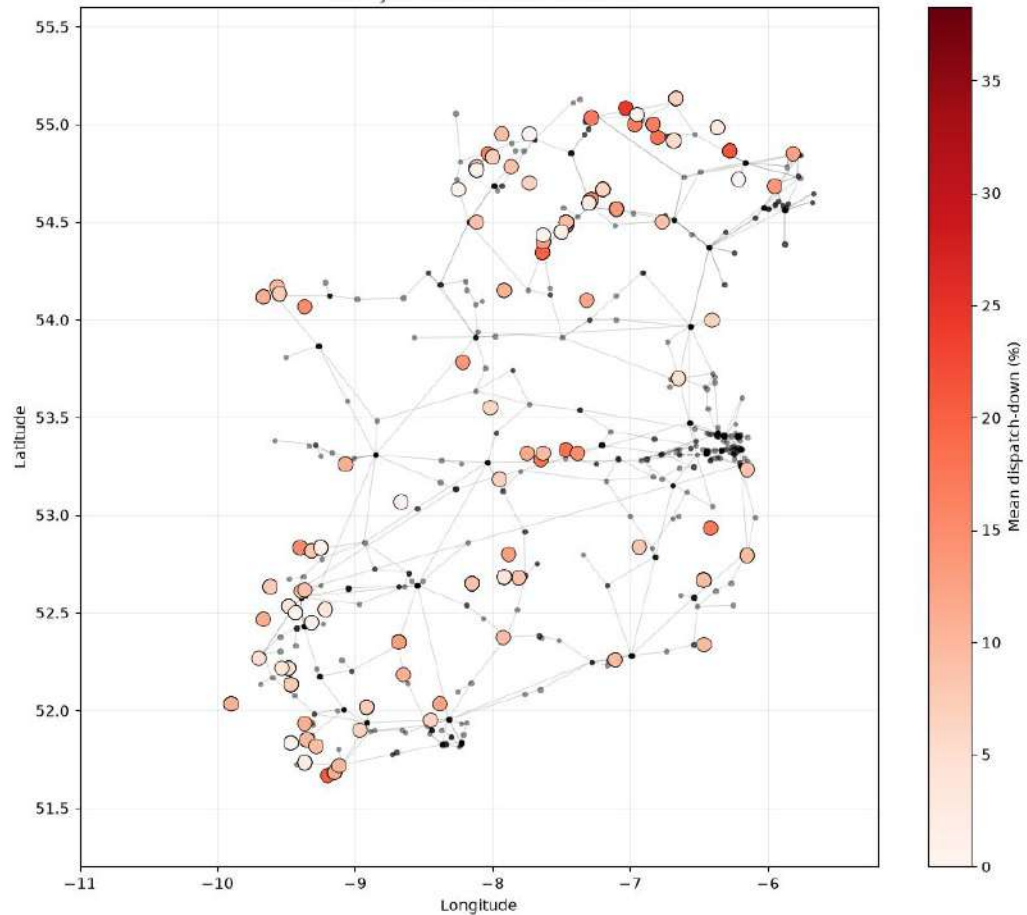
Wind Availability vs Metered Generation



Irish Wind Farms: Average Dispatch-Down Over Entire Dataset



Irish Wind Farms: Average Dispatch-Down with PyPSA Transmission Network



1 The shift factor, as it is used today

For a branch e and a generator at node n :

$$\text{SF}_{e,n} = \frac{\Delta f_e}{\Delta P_n}. \quad (1)$$

Two properties define how it is used:

It is transmission-line specific. One number describes one branch against one generator. A constraint group is built by computing a column of these for a single congested circuit.

It is empirical. [WDT] (Appendix 2, p. 71) obtains it by perturbation: move one generator by a fixed amount, re-run the load flow, read the change in branch current. One solve per number.

2 Sensitivity: a new way of looking at it

$$S_{e,n} = \text{PTDF}_{e,n} \cdot \text{sign}(f_e). \quad (2)$$

The same physical quantity as (1), with the sign re-pointed along each branch's real direction of flow, so that **positive always means relief**.

Without the correction a positive entry means “flow increases in the branch's stored **bus0**→**bus1** direction”, which is overload on one branch and relief on another. With it, one row may be ranked, and two rows may be added.

It is **dimensionless**, MW off the branch per MW curtailed at the node, so entries from different branches are directly comparable.

3 How to get to S : K , B , L

K , **incidence**, 754×980 . Topology only. Column e has exactly two non-zeros: $+1$ at the branch's **bus0**, -1 at its **bus1**.

$$K_{ie} = \begin{cases} +1 & i = \text{bus0}(e) \\ -1 & i = \text{bus1}(e) \\ 0 & \text{otherwise.} \end{cases}$$

B , **susceptance**, 980×980 **diagonal**. Electrical strength only: $B_{ee} = b_e = 1/x_e$, from the per-unit reactance.

$L = KBK^\top$, **the weighted Laplacian**, 754×754 . Topology and strength combined, and the matrix that must be inverted:

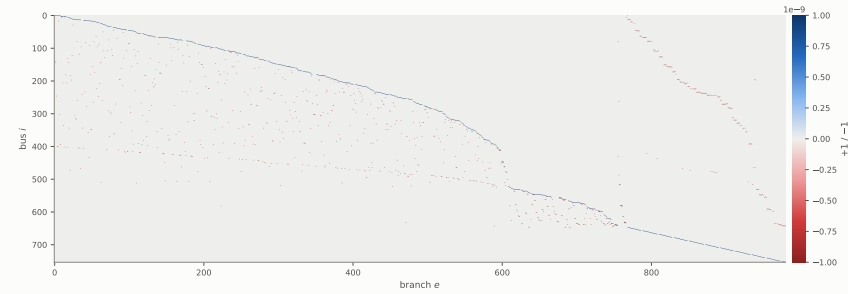
$$L\theta = p. \tag{3}$$

Why any of this is linear. The AC equations are not. The DC approximation holds bus voltages at 1 p.u., takes $r \ll x$ so losses vanish, and takes angle differences small so $\sin(\theta_i - \theta_j) \approx \theta_i - \theta_j$. Flow is then exactly linear in the angle difference, $f_e = b_e(\theta_i - \theta_j)$. That is the whole reason a closed-form sensitivity exists at all.

4 The matrices themselves

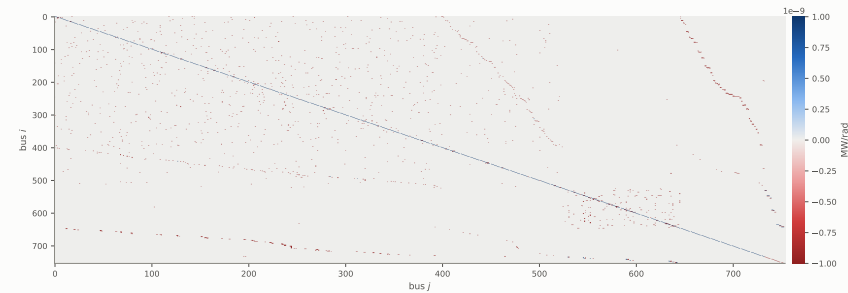
K — the full 754×980 incidence matrix

Exactly two non-zeros per column: +1 at bus0, -1 at bus1. 1,960 non-zeros, 0.27% dense.



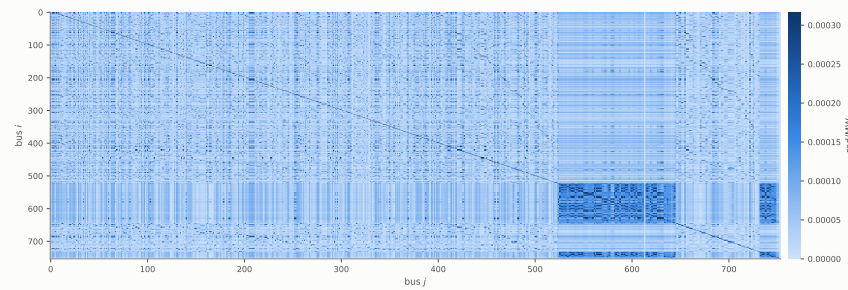
$L = KBK^T$ — the full 754×754 Laplacian

2,621 non-zeros (0.46% dense). The structure is the grid's own topology.



L^+ — the full 754×754 pseudoinverse

99.2% dense. Inversion destroys sparsity: every bus is now coupled to every other.



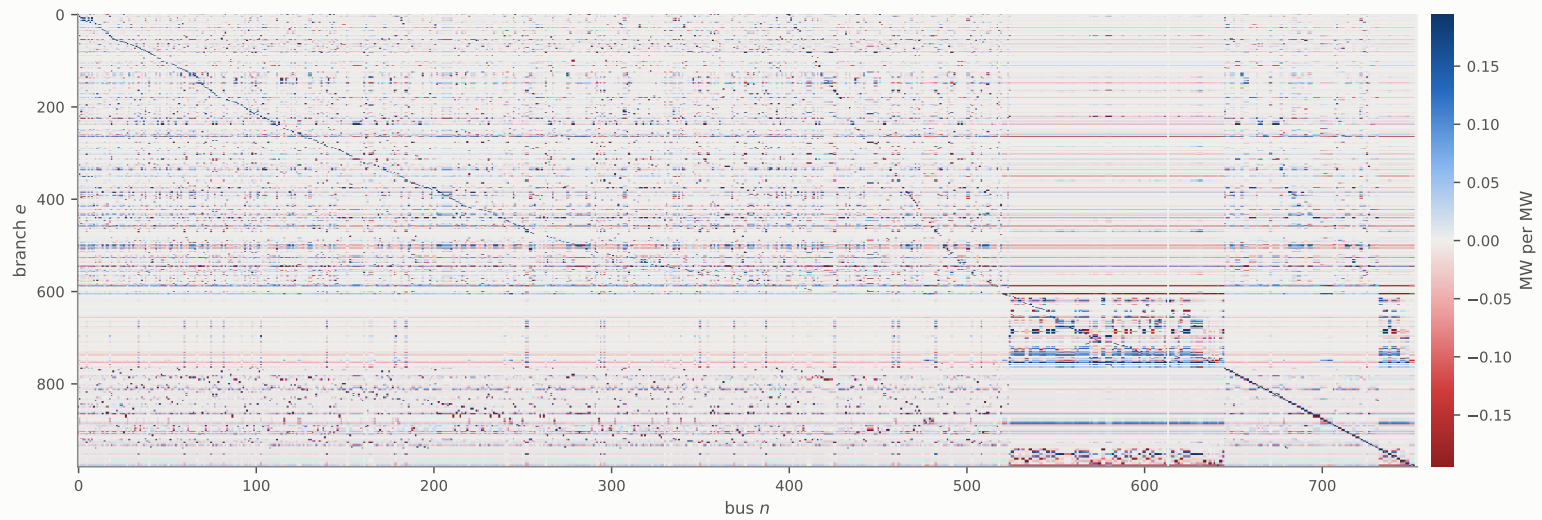
5 The PTDF

$$\text{PTDF} = BK^{\top}L^{+}, \quad 980 \times 754, \quad 738,920 \text{ entries.} \quad (4)$$

$\text{PTDF}_{e,n}$ is the flow appearing on branch e when 1 MW is injected at bus n and withdrawn across the slack.

PTDF = $BK^{\top}L^{+}$ — all $980 \times 754 = 738,920$ entries

Every branch against every bus. No numeric form of this is readable, so it is shown whole as an image.



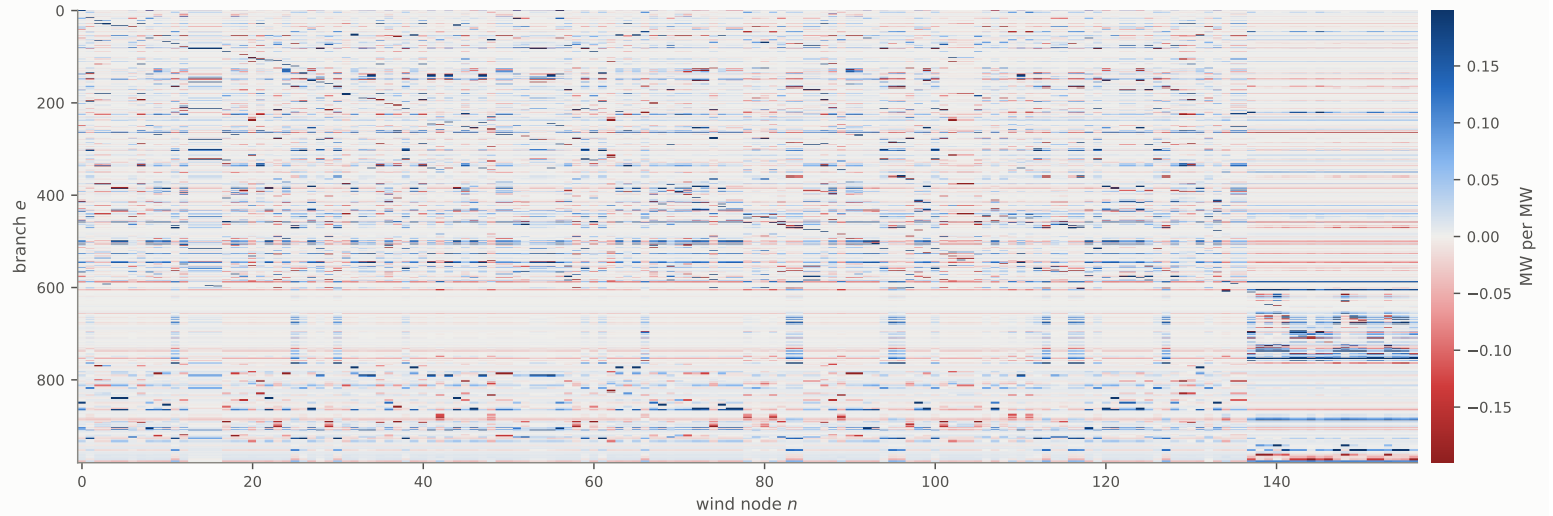
6 The sensitivity matrix

Only one quantity in the pipeline moves hour to hour: the direction of flow.

$$p_{\text{eff}} = p + KB\varphi, \quad f = \text{PTDF} \cdot p_{\text{eff}} - B\varphi, \quad S_{e,:} = \text{PTDF}_{e,:} \cdot \text{sign}(f_e). \quad (5)$$

Sensitivity — 980 × 157, signed along real flow

PTDF re-pointed by $\text{sign}(f_e)$. Positive means curtailing there unloads the branch.

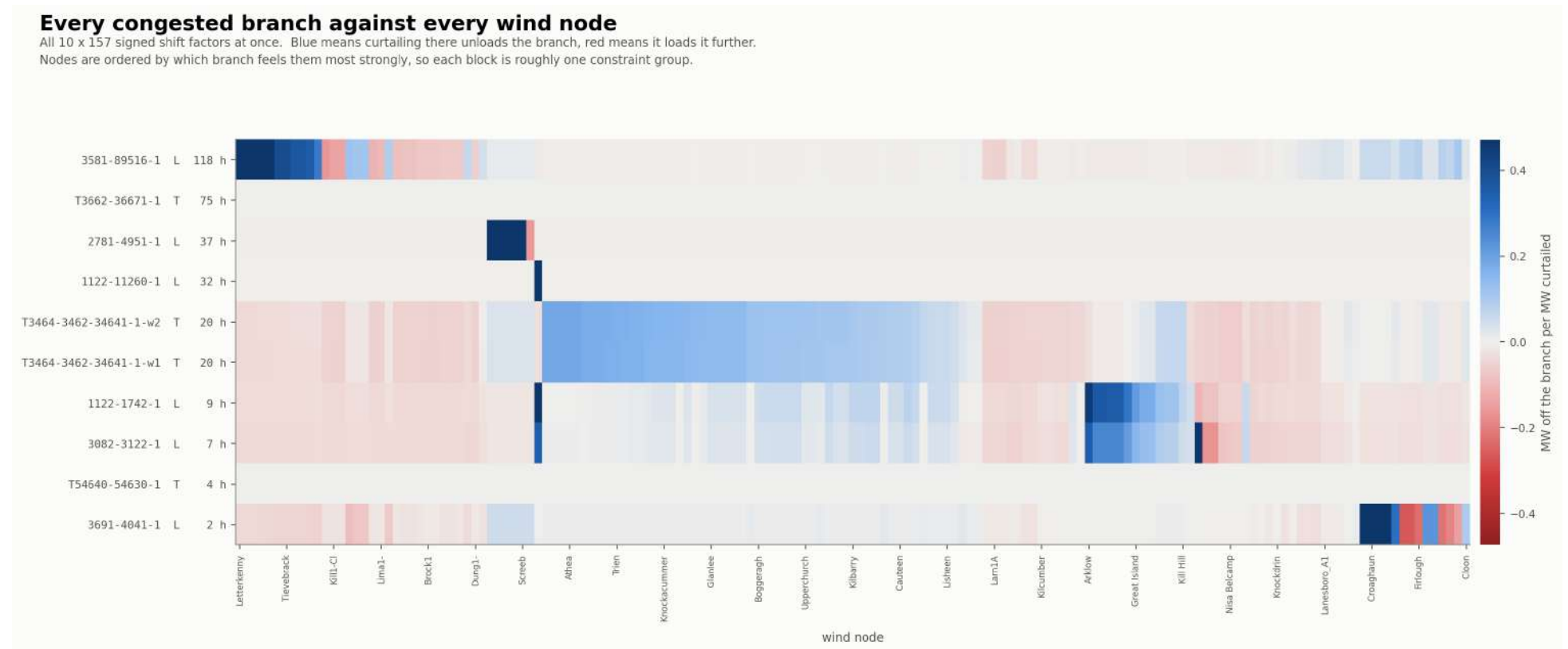


positive	curtailing at that wind node unloads the branch
negative	curtailing at that wind node loads it further

φ holds the phase shifts of the 2 phase-shifting transformers. Dropping the $-B\varphi$ term puts the flow, and therefore the sign, and therefore every entry of S , wrong on those branches.

7 In practice, only a small set of lines is congested

Of 980 branches, only a handful reach their rating at any hour of the week, and several of those are transformers, which a lines-only model would miss entirely.



Every constraint question in the rest of this document is a question about these rows. The remaining rows exist, are exact, and are never needed until the network changes, which is what an outage is.

8 Net sensitivity: a constraint group that cannot break what it did not look at

The group, as it is drawn today

$$\text{group}(e) = \{ n : S_{e,n} \geq \tau \}, \quad \tau = 0.05. \quad (6)$$

Sensitivities add. Net sensitivity is nothing more than that.

If several branches are over their rating at once, a node's total usefulness is its sensitivities to each of them, summed:

$$S_{\text{net}}(n) = \sum_{e \in \text{Over}} S_{e,n}, \quad \text{Over} = \{ e : |f_e| > s_e \}. \quad (7)$$

The critical principle

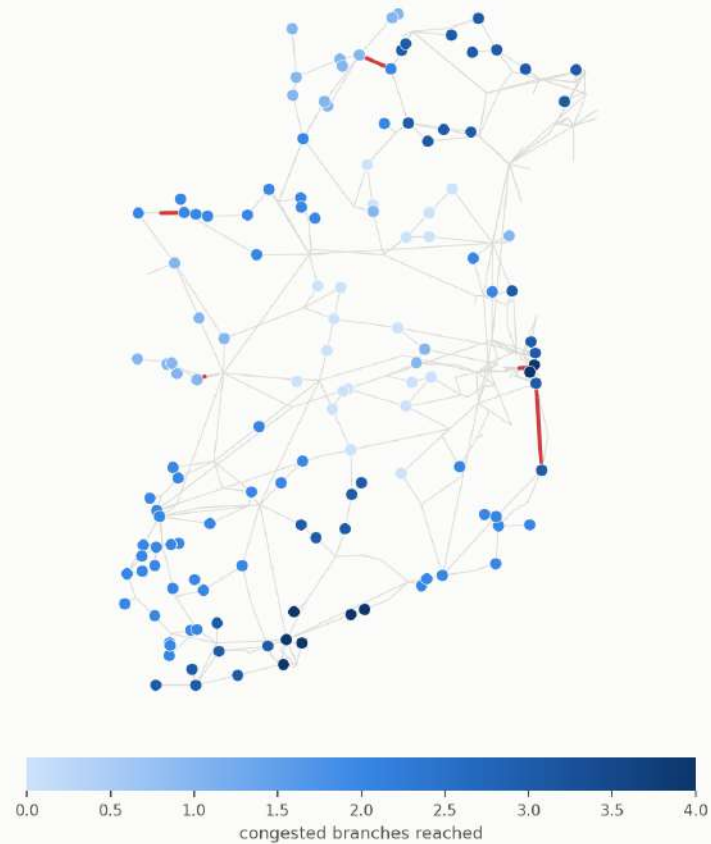
No dispatch down of a node may cause an overload on any other branch.

Reach and net relief, on the ground

Left: how many of the ten congested branches each wind node reaches. Right: its net weighted relief, which is what breadth is worth once the signs are counted. A node can be dark on the left and pale on the right — broad, but pulling in both directions at once.

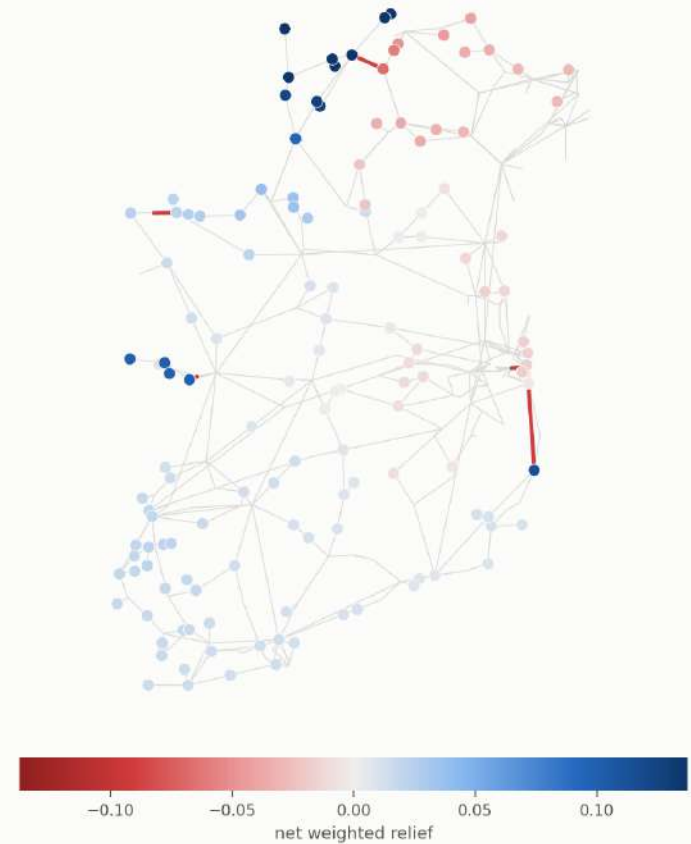
Reach

Red edges are the ten congested branches.



Net relief

Red edges are the ten congested branches.



Left: how many congested branches each wind node reaches. Right: its net weighted relief, (7). A node dark on the left and pale on the right is broad but pulling both ways at once, which counting rows cannot distinguish and summing them can.

9 Doing this for every N–1 contingency

The groups are fixed. Membership is read off S , which is topology and reactance only, so it does not move with the hour or the weather. The group for every single-branch outage is built once, offline, and looked up: 432 outages, 58 distinct circuits ever at risk, 1,308 lookup rows.

Most of the fleet is almost never in one. Across all 20,203 events, 119 of 157 nodes, 65.9% of installed capacity, are dispatched down in under 1% of them. The median node keeps 99.61% of everything it offers, and over the whole study only 0.85% of all wind offered is curtailed.

For those generators a constraint group binds on a handful of hours a year. It is not a standing limit on output, and they run at full power essentially all the time.

And a few of these outages do not stop at one circuit.

10 Another application: stopping cascading outages

Most single-branch outages are absorbed. Flows redistribute and nothing else reaches its rating. A small number are not: the first trip pushes other elements over their limits, those trip in turn, and the failure propagates.

This is where the two constraint groups separate. A group drawn from one row can relieve the congested circuit while pushing a healthy one over, which starts the next link in the chain. A group built under the critical principle cannot.

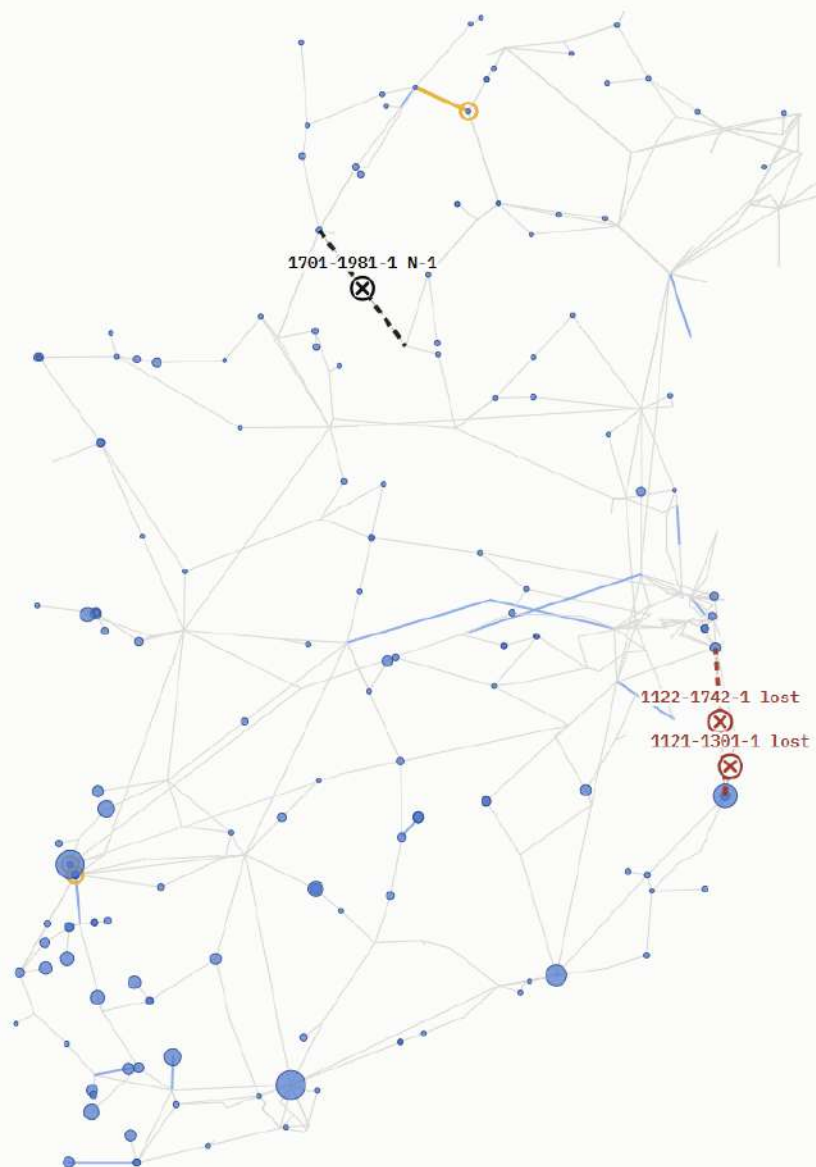
Shift factor constraint groups

LINE OUTAGES

2

WIND CUT

744 MW · 14%



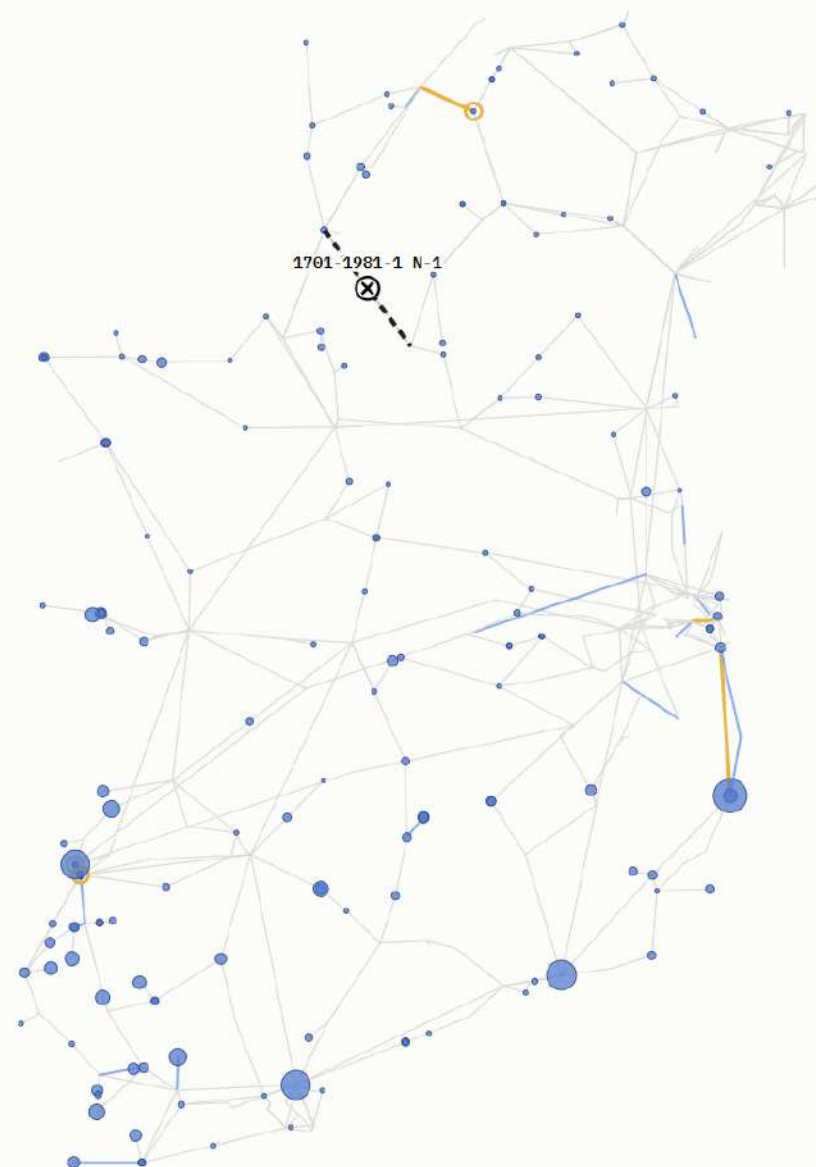
Net sensitivity constraint groups

LINE OUTAGES

0

WIND CUT

16 MW · 0%



Spectral Clustering of the Irish Transmission Grid

Where the grid comes apart

Oleksandr — Team 6, Electrical Grid Hackathon 2026

Networks: WP2024 and WP2033, all-island

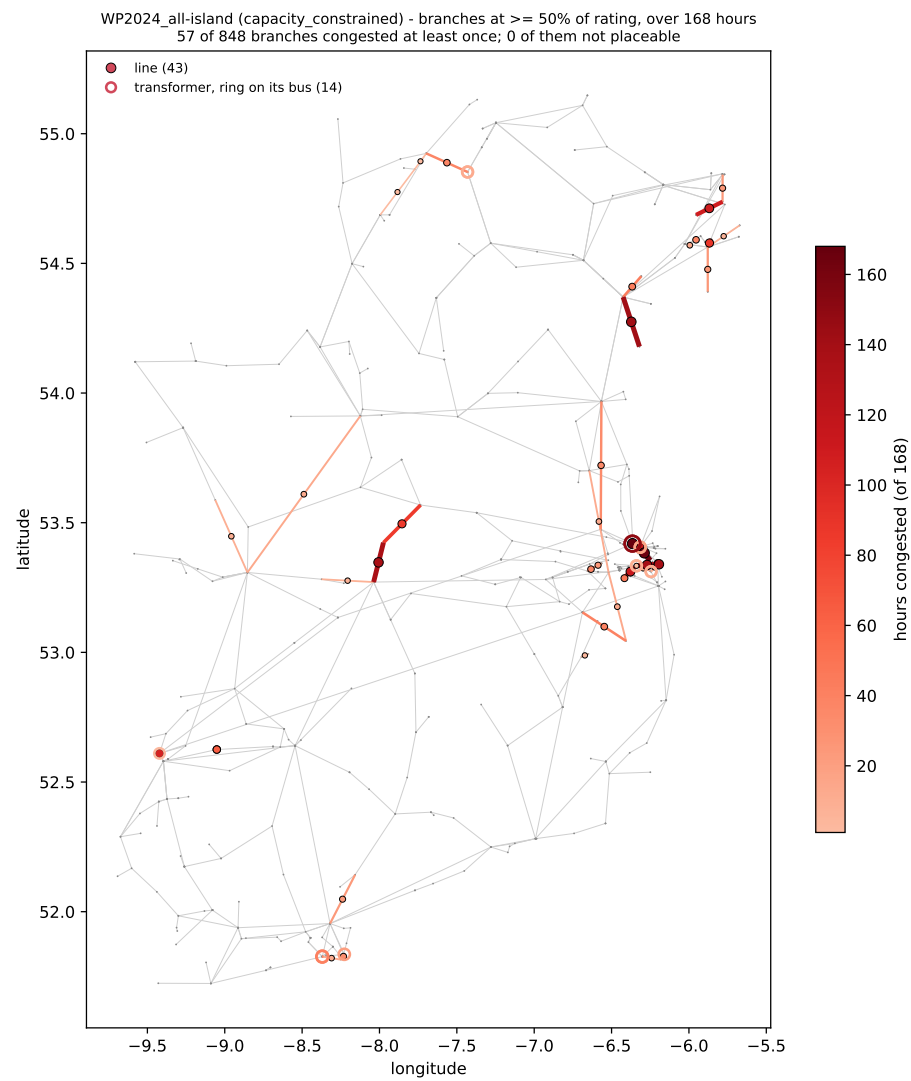
Method

1. Grid as a weighted graph: buses and generators are nodes.
2. Normalised Laplacian $L_{\text{sym}} = I - D^{-1/2} A D^{-1/2}$.
3. Its k smallest eigenvectors as coordinates per node.
4. k -means on those coordinates.

Edge weights

susceptance	$1/x_e$ — static network
headroom	$s_{\text{nom}} - F_e $ — one hour of dispatch

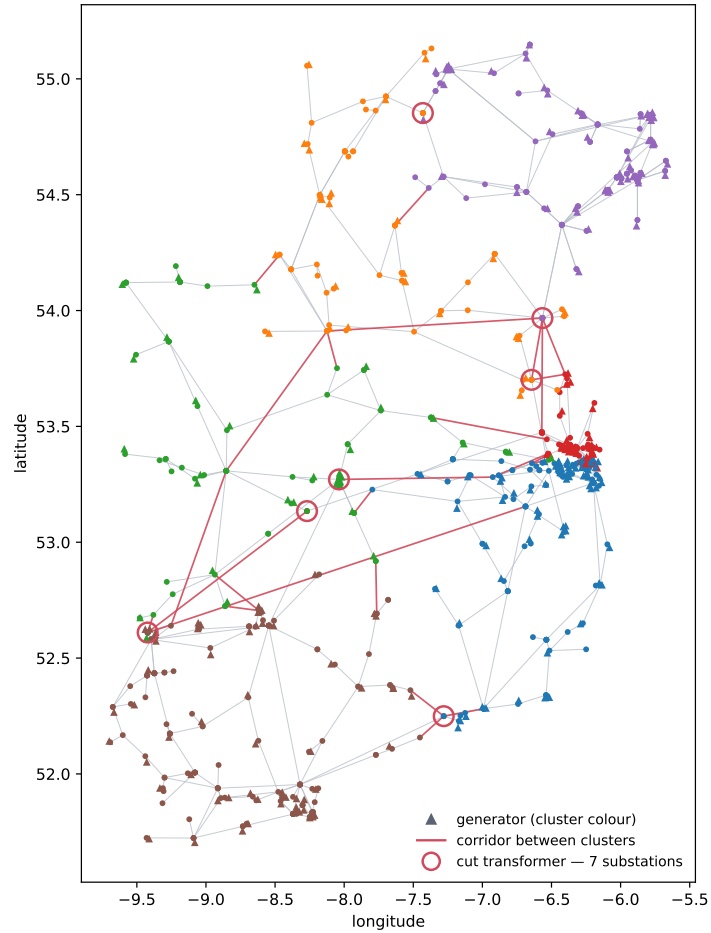
WP2024 — where the grid is loaded



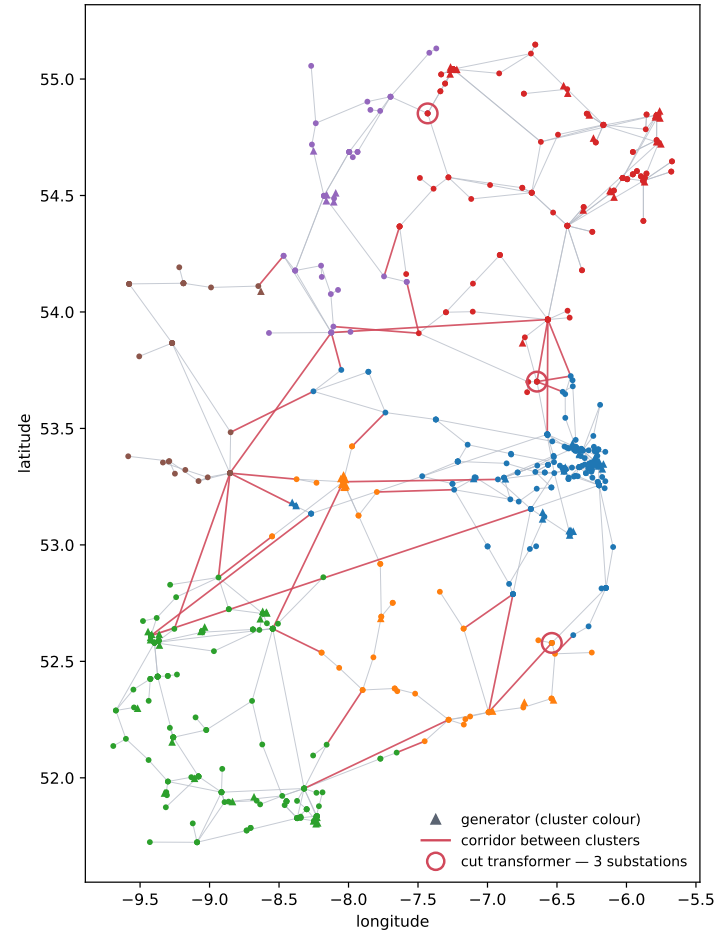
Branches at $\geq 50\%$ of rating over one week.
57 of 848 branches.

WP2024 — clustering, $k = 6$: headroom and susceptance

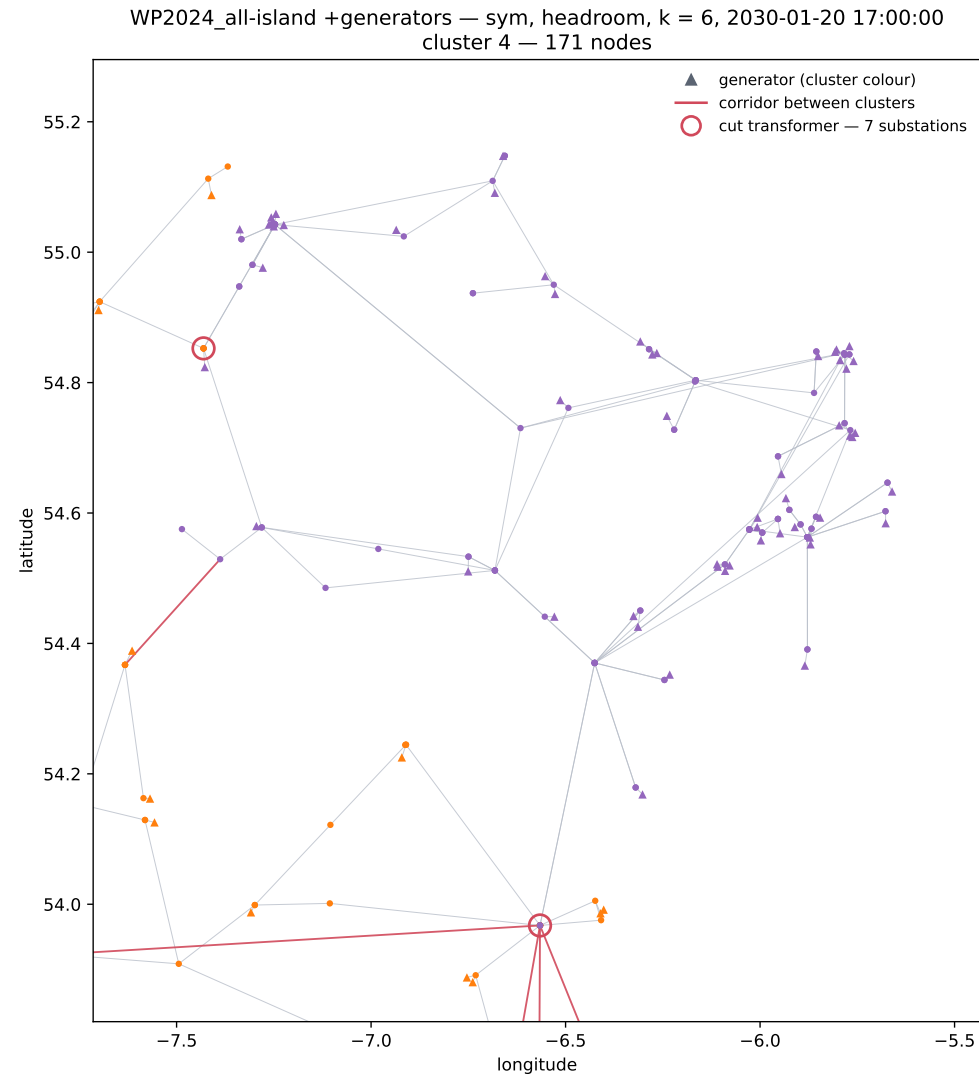
WP2024_all-island +generators — sym, headroom, $k = 6$, 2030-01-20 17:00:00
643/643 buses placed, 40/1097 drawn corridors cut



TYTFS2024_WP2024_V35_transmission +generators — sym, susceptance, $k = 6$
643/643 buses placed, 35/895 drawn corridors cut

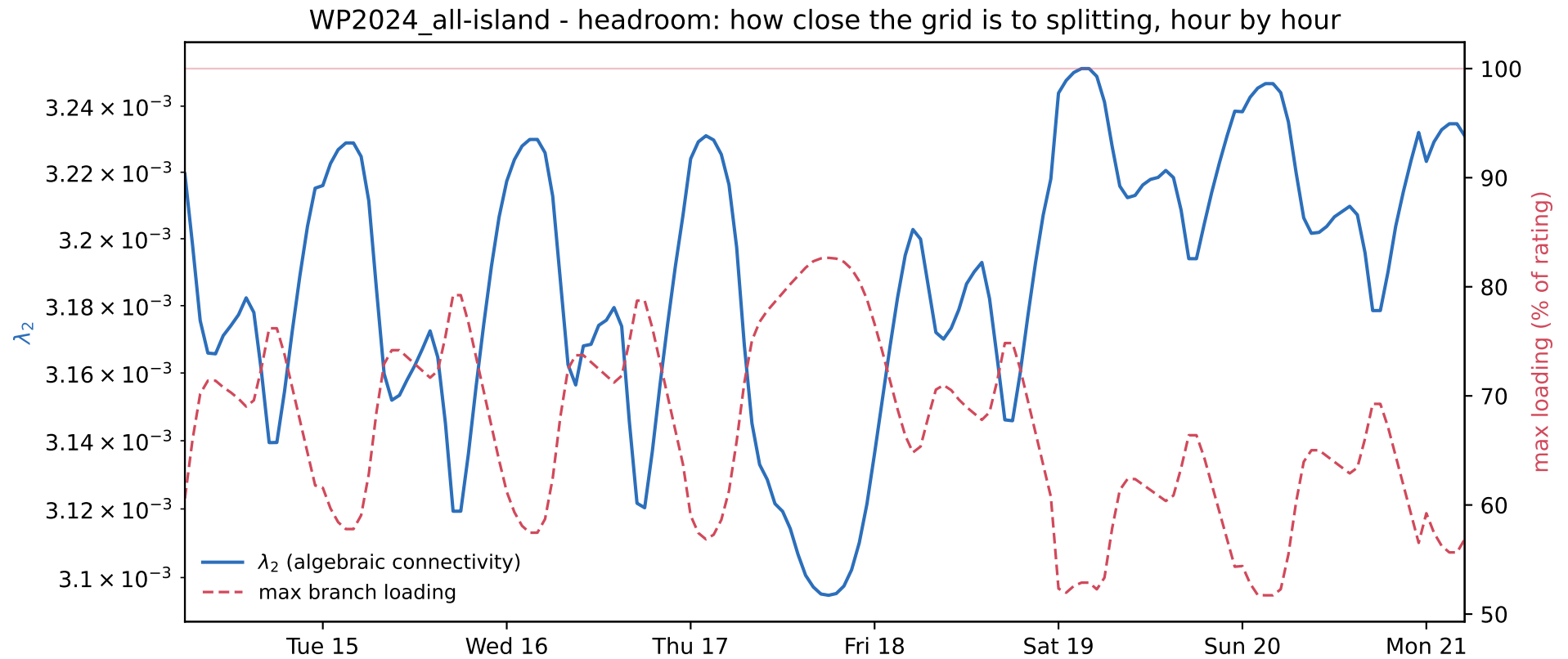


WP2024 — the north-east cluster



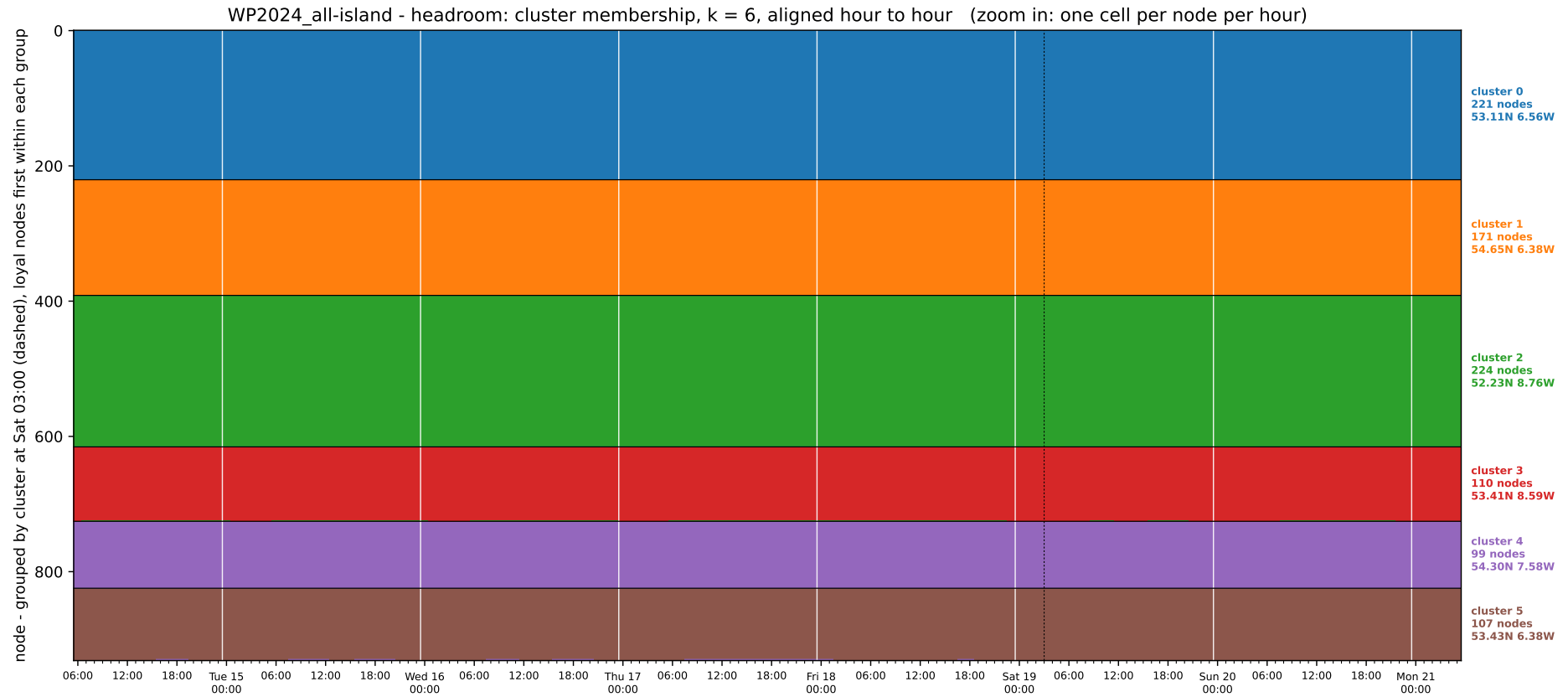
171 nodes, linked to the rest through a few corridors.

WP2024 — the week: how close the grid is to splitting



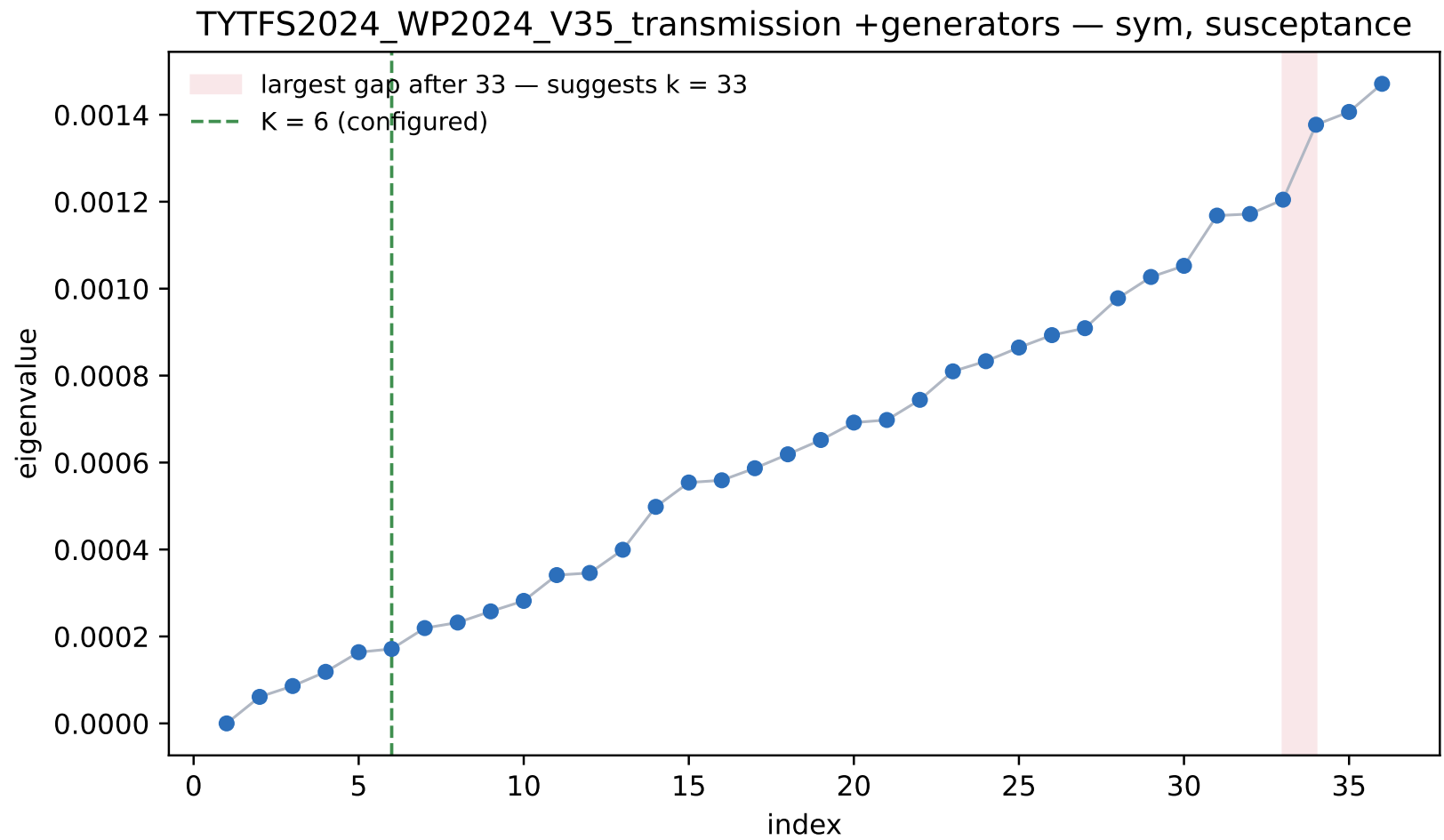
λ_2 dips whenever the maximum branch loading peaks.

WP2024 — the week: does the partition move?



No node changes cluster over the 168 hours.

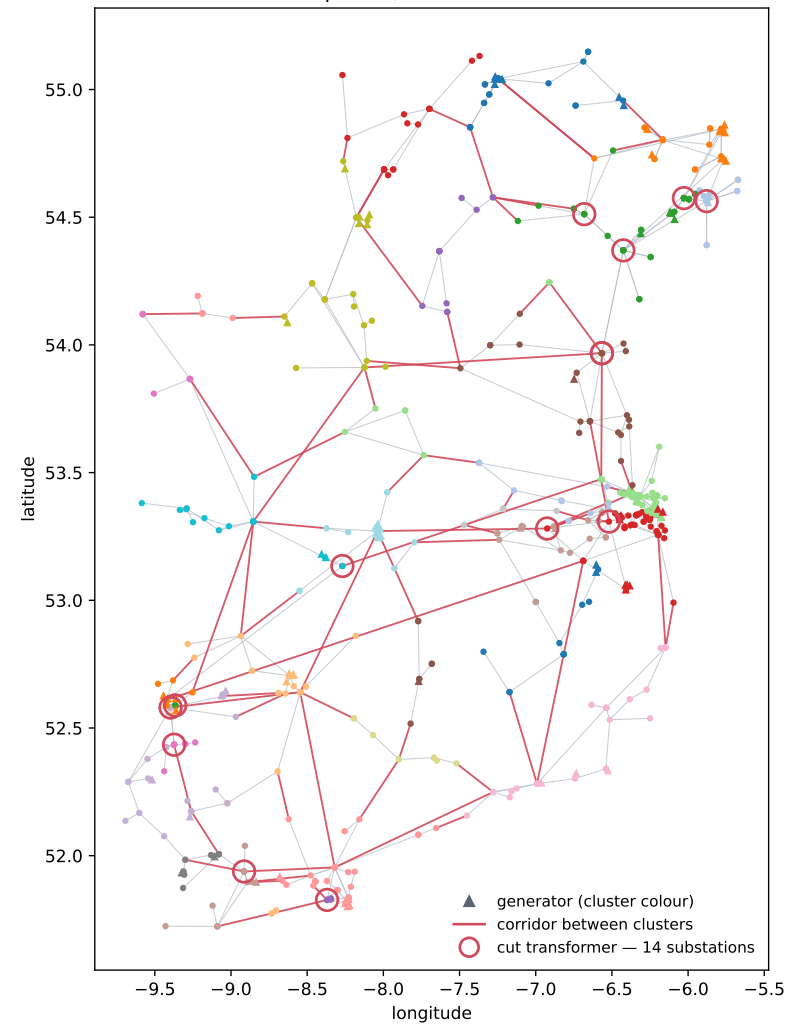
WP2024 — the eigenvalues: how many clusters?



No gap at $k = 6$; the largest gap is after 33.

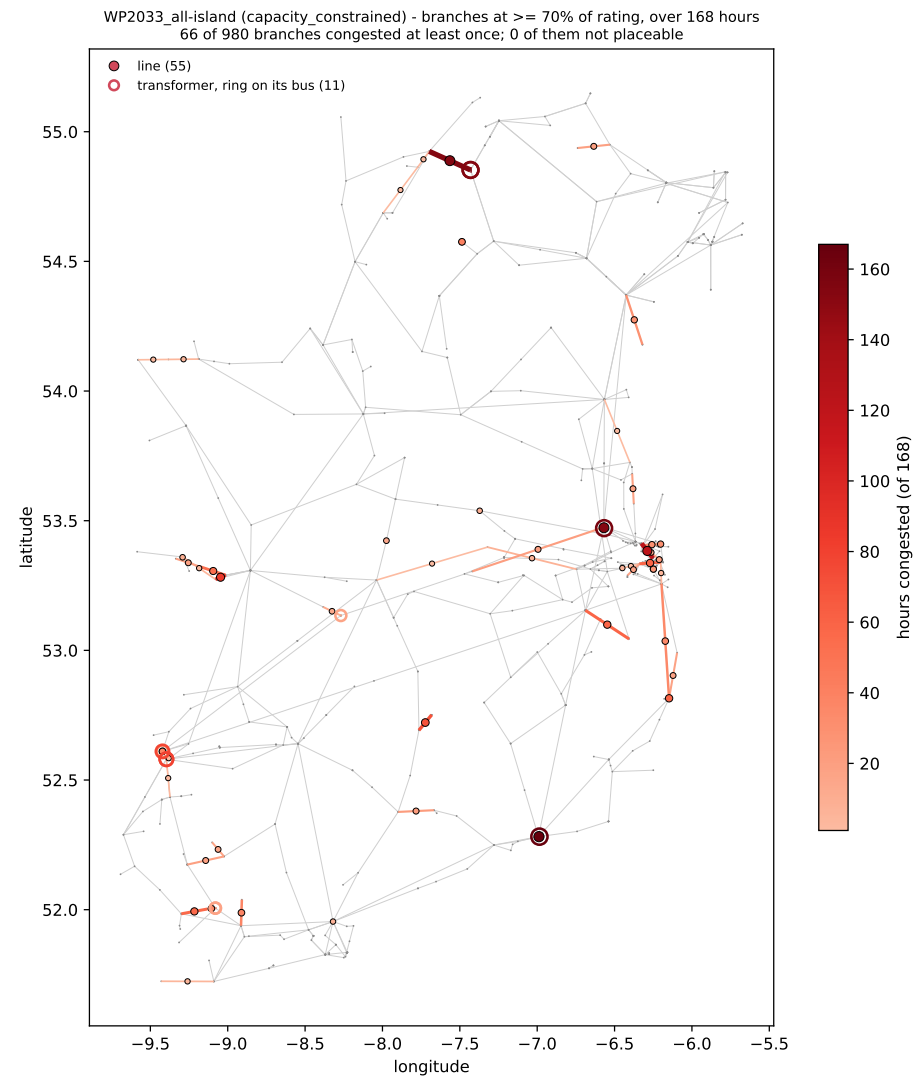
WP2024 — susceptance clustering, $k = 33$

TYTFS2024_WP2024_V35_transmission +generators — sym, susceptance, k = 33
643/643 buses placed, 103/895 drawn corridors cut



k taken from the eigengap.

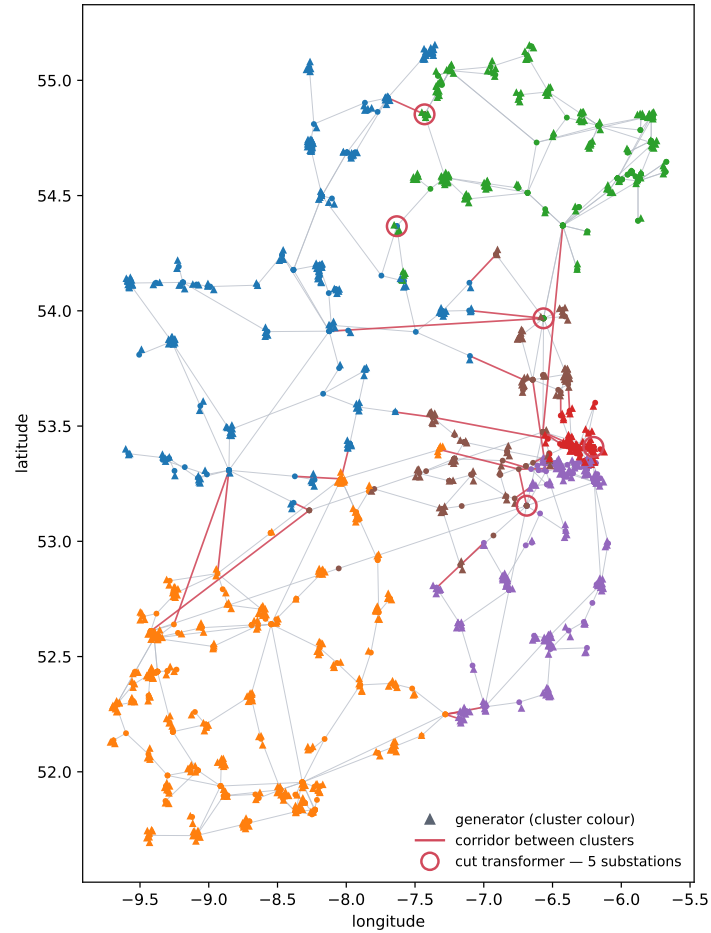
WP2033 — where the grid is loaded



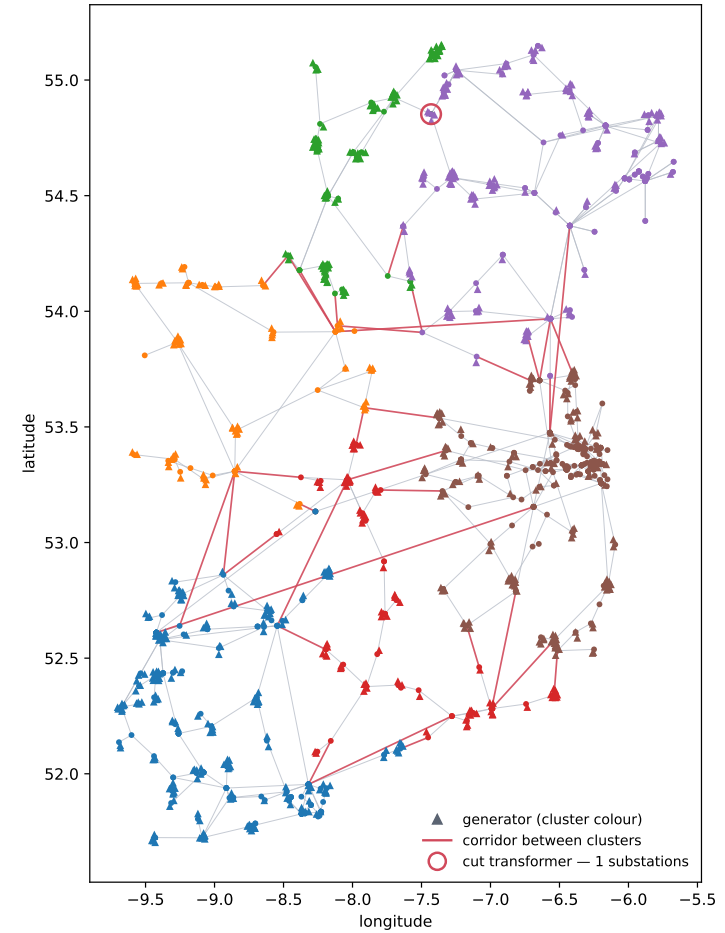
Branches at $\geq 70\%$ of rating over one week.
66 of 980 branches.

WP2033 — clustering, $k = 6$: headroom and susceptance

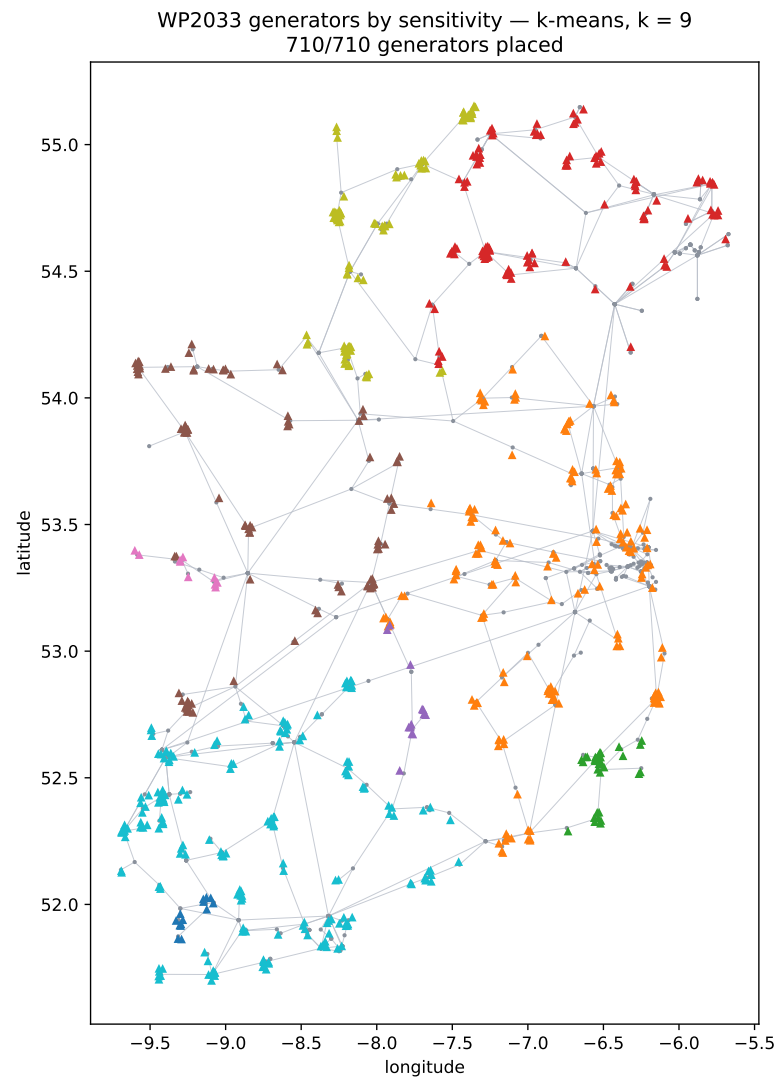
WP2033_all-island +generators — sym, headroom, $k = 6$, 2030-01-20 17:00:00
751/751 buses placed, 42/1858 drawn corridors cut



TYTFS2024_WP2033_V35_transmission +generators — sym, susceptance, $k = 6$
746/751 buses placed, 35/1637 drawn corridors cut



WP2033 — generators grouped by shift factor, $k = 9$



k -means on each generator's shift factors across all 980 branches.

NEXT



Hackathon

Studying the Irish electric grid; improving constraint groups and incentive structures

Team 1

September 11, 2026

Section 0: Introduction

What our goals are

"By 2030, 80% of electricity generation must come from renewables."

— EirGrid Policy Objective

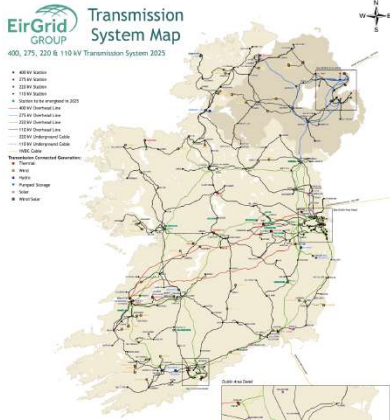
- In 2020, EirGrid met its target of sourcing 40% of electricity generation from renewables; the new objective is a considerably more ambitious 80% by 2030.

- We can represent the grid as a graph with nodes for generators/busses and transmission lines as edges.
- Some of the lines are more overloaded than the others, which causes overheating and cascading blackouts.
- Therefore, generators' outputs are being capped. As controlling each generator is not convenient, they are grouped into **Constarint Groups**, with the same limitations set, also called **Dispatch-Down**

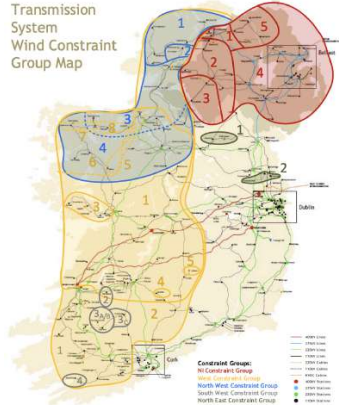
Main Objective

Reduce the renewable energy loss due to dispatch-down.

Constraint Groups



Excerpt from the TPSA 2026 Hackathon
Information Sheet



Excerpt from EirGrid's 2025 Transmission System
Map

- Currently, generators operate under firm and non-firm connection agreements with EirGrid
- Firm connection agreements guarantee generators compensation if dispatched down equal to the profit they would have made otherwise; under non-firm agreements, generators connect faster but accept uncompensated dispatch-down.
- Reducing dispatch-down **decreases** consumer and taxpayer **costs**

Section 1: Mathematical Framework

Metrics and grouping algorithms

Aim and scope

- Identify generators with **similar responses on stressed lines**.
- Use DC optimal power flow to define dispatch and line flows.
- Test intact and every single-line-outage state separately.
- Replace hundreds of individual controls with a handful of **group controls**.
- Study, but do not physically interpret, correlations between stochastic optimal dispatches.

Important

Response groups are not physical or electrical clusters.

Method overview

- ① Assemble the distance matrix
- ② Apply recursive complete-linkage clustering
- ③ Optimize DC network approximation under fixed conditions and conditions
- ④ Test under N-1 and variations
- ⑤ Apply and analyze stochastic behavior on model
- ⑥ Define correlation matrix and spectral cluster into correlation groups

Distance matrix

For each generator i , inject 1 MW at its bus, withdraw from loads proportionally to mean demand, solve

$$\widehat{B} \widehat{\theta}^{(i)} = \widehat{p}^{(i)}.$$

For a branch-state $r = (c, e)$ on a stressed line $e = (a, b)$,

$$A_{ir} = b_e (\theta_a^{(i)} - \theta_b^{(i)}).$$

Screen branch-states with $|f_{tr}^0| > F_r$ in some hour, weight by relative overload:

$$u_r = \sum_t \frac{\max\{0, |f_{tr}^0| - F_r\}}{F_r}, \quad w_r = \frac{u_r}{\sum_{r'} u_{r'}},$$

$$M_{ij} = \sqrt{\sum_{r \in \mathcal{R}} w_r (A_{ir} - A_{jr})^2}.$$

Row A_i : flow-response fingerprint. No congestion $\Rightarrow$ one group.

Complete-linkage clustering

Merge heights $h_1 \leq \dots \leq h_{q-1}$, gaps $d_t = h_{t+1} - h_t$. First largest gap:

$$t^* = \min\{t : d_t = \max_s d_s\}, \quad k = q - t^*.$$

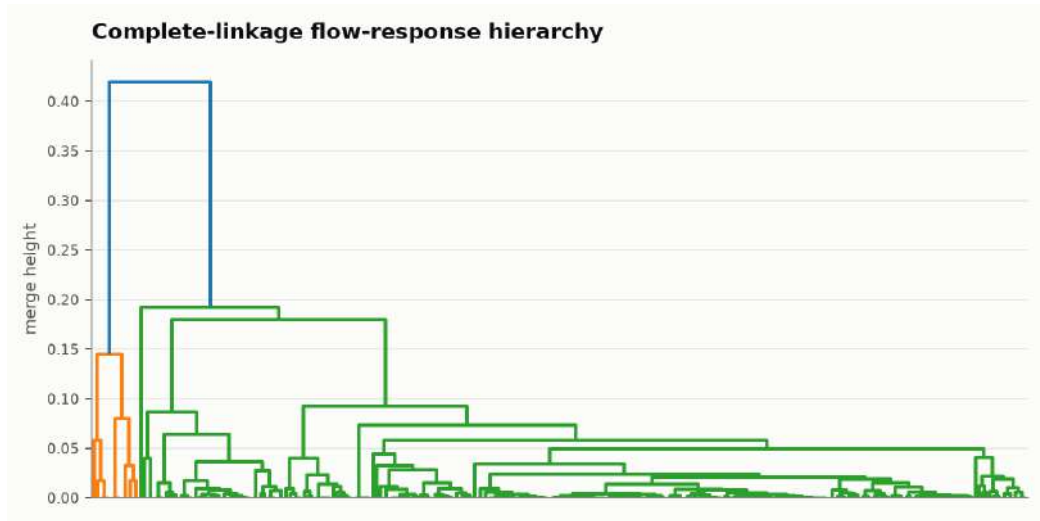
Collapse zero-distance classes $Q_a = \{i, j : M_{ij} = 0\}$. For clusters U, V ,

$$\Delta(U, V) = \max_{i \in U, j \in V} M_{ij}, \quad \Delta(U \cup V, C) = \max\{\Delta(U, C), \Delta(V, C)\}.$$

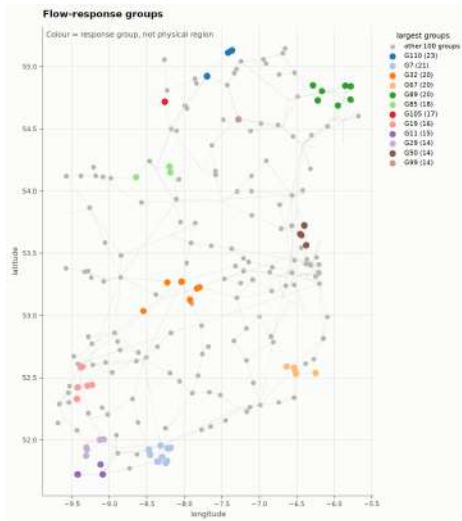
Merge the pair with smallest Δ ; ties by fixed rule.

- Same rule inside each global group $\rightarrow$ local operational groups.
- Optional size cap $S_{\max}$: recurse on oversized groups.

What the hierarchy looks like



Flow-response groups on the grid



DC power-flow formulation

For a fixed network state, generation availability, and demand, let the net injection at bus i be

$$P_i = p_{g,i} m_i - d_i, \quad 0 \leq m_i \leq 1.$$

The DC power-flow equation is

$$B\theta = P$$

where B is the bus susceptance matrix and θ is the vector of bus voltage angles.

For a transmission line connecting buses i and j ,

$$f_{ij} = \frac{\theta_i - \theta_j}{x_{ij}}$$

where x_{ij} is the line reactance.

Collecting all line flows gives

$$F = X\theta$$

Construction of B and X

The bus susceptance matrix B define as follows

$$B_{ij} = -\frac{1}{x_{ij}}, \quad B_{ii} = \sum_{i \neq j} \frac{1}{x_{ij}}.$$

Since B is Laplacian and not invertible, θ is usually calculated with a fixed reference angle:

$$\theta_{ref} = 0$$

To find the flow, the line-flow matrix X is naturally defined as follows

$$X_{km} = \begin{cases} \frac{1}{x_{ij}}, & m = i, \\ -\frac{1}{x_{ij}}, & m = j, \\ 0, & \text{otherwise.} \end{cases}$$

Incorporating the operating groups

Suppose the clustering procedure produces groups

$$\mathcal{G} = \{G_1, \dots, G_K\}.$$

Generators belonging to the same operating group are assigned a common utilisation level:

$$i, j \in G_k \quad \Rightarrow \quad m_i = m_j = m_{G_k}.$$

Introduce the group-membership matrix H , so that

$$m = Hm_G.$$

The nodal power-flow equation consequently becomes

$$B\theta = P_g Hm_G - d.$$

The resulting transmission flows subject to the line-flow constraints are

$$F = X\theta, \quad f_{\min} \leq X\theta \leq f_{\max}$$

Complete optimisation problem for fixed inputs

For fixed P_g , d , B , X , $f_{\min}$ and $f_{\max}$, the optimisation problem is

$$\begin{aligned} & \underset{m_G, \theta}{\text{maximize}} && W^T P_g H m_G \\ & \text{subject to} && B\theta = P_g H m_G - d, \\ & && \mathbf{1}^T P_g H m_G = \mathbf{1}^T d, \\ & && f_{\min} \leq X\theta \leq f_{\max}, \\ & && 0 \leq m_{G_k} \leq 1, \quad k = 1, \dots, K, \\ & && \theta_{\text{ref}} = 0. \end{aligned}$$

The weighting vector is defined by

$$W_i = \begin{cases} 5, & \text{wind or hydro,} \\ 1, & \text{otherwise.} \end{cases}$$

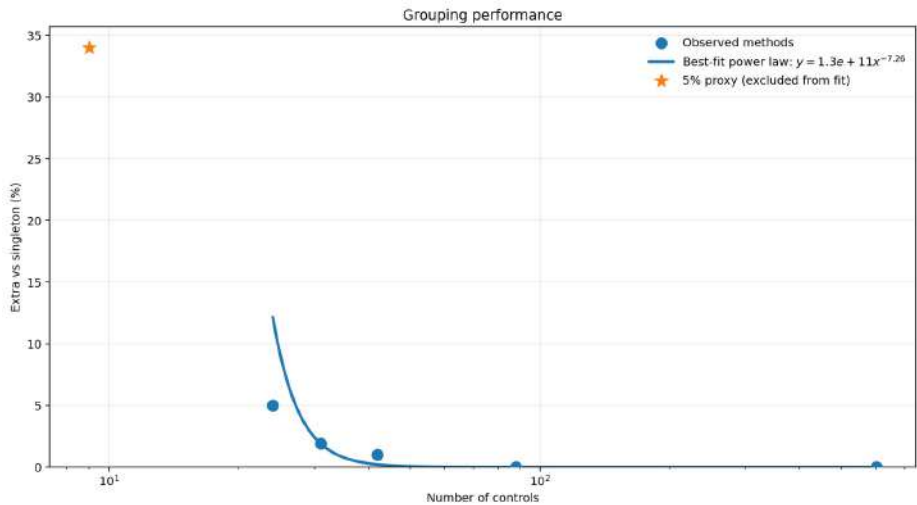
How good is the grouping?

Held-out all-island, intact network, 48 test hours.

Method	Controls	Max size	Group failures	Extra vs singleton
cap 200	24	150	0	4.98%
cap 100	31	64	0	1.91%
cap 50	42	43	0	0.99%
cap 25	88	23	0	0.0005%
5% proxy	9	223	0	34.0%
singleton	603	1	0	0%

Cap 25 matches singleton almost exactly, with $7\times$ fewer controls.

The performance curve



Stochastic generation and demand

Instead of considering only one operating condition, we consider a set of scenarios

$$s = 1, \dots, S.$$

For each scenario,

$$P_g^{(s)} = \text{diag} \left(p_{g,1}^{(s)}, \dots, p_{g,n}^{(s)} \right), \quad d^{(s)} = \begin{bmatrix} d_1^{(s)} & \dots & d_n^{(s)} \end{bmatrix}^T.$$

The uncertain operating state is represented by

$$\xi \sim \mathcal{N}(\mu, \Sigma) \mid \xi \in \Omega,$$

where Σ captures dependence between the stochastic variables.

Physical feasibility is enforced by

$$0 \leq p_{g,i}^{(s)} \leq \bar{p}_{g,i}^{(s)}, \quad d_i^{(s)} \geq 0.$$

Here $\bar{p}_{g,i}^{(s)}$ denotes the available generation in scenario s ; therefore,

Optimisation across scenarios

For each scenario, the same fixed-input optimisation is solved:

$$\left(P_g^{(s)}, d^{(s)}\right) \longrightarrow \text{optimisation} \longrightarrow m^{(s)}$$

where $m^{(s)}$ is the optimal generation utilisation vector.

The network parameters remain fixed:

$$B, \quad X, \quad s_{\text{nom}}.$$

The resulting set of optimal solutions is

$$\mathcal{U} = \left\{ m^{(1)}, m^{(2)}, \dots, m^{(S)} \right\}.$$

The optimisation is solved independently for each scenario, using the corresponding generation availability and demand.

Node response vectors

For each node i , we collect its optimal utilisation across all scenarios:

$$\mathbf{n}_i = \begin{bmatrix} m_i^{(1)} \\ m_i^{(2)} \\ \vdots \\ m_i^{(S)} \end{bmatrix}$$

Thus, $\mathbf{n}_i$ records how the optimal utilisation of node i changes across the scenario set.

The empirical mean and the corresponding variance is

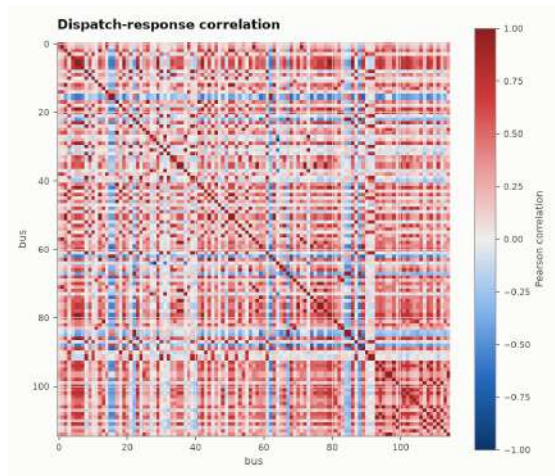
$$\bar{m}_i = \frac{1}{S} \sum_{s=1}^S m_i^{(s)}, \quad \sigma_{m_i}^2 = \frac{1}{S-1} \sum_{s=1}^S \left(m_i^{(s)} - \bar{m}_i \right)^2.$$

The collection $\{\mathbf{n}_1, \mathbf{n}_2, \dots, \mathbf{n}_n\}$ provides a statistical description of the optimal dispatch behaviour generated by the stochastic optimisation.

Correlation structure

For nonzero-variance buses I :

$$R_{ij} = \text{corr}(m_i, m_j) \\ = \frac{\sum_s (m_i^{(s)} - \bar{m}_i)(m_j^{(s)} - \bar{m}_j)}{\sqrt{\sum_s (m_i^{(s)} - \bar{m}_i)^2} \sqrt{\sum_s (m_j^{(s)} - \bar{m}_j)^2}}$$



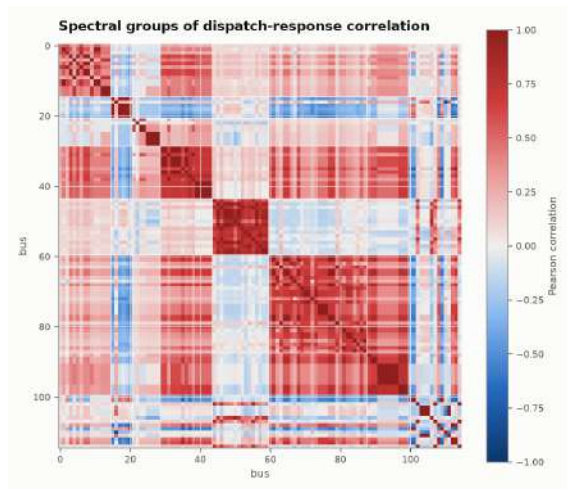
Linear similarity of optimal responses. Not proximity, not causation.

Spectral clustering of correlation groups

Let $D = \text{diag}(\sum_j R_{ij})$ and form

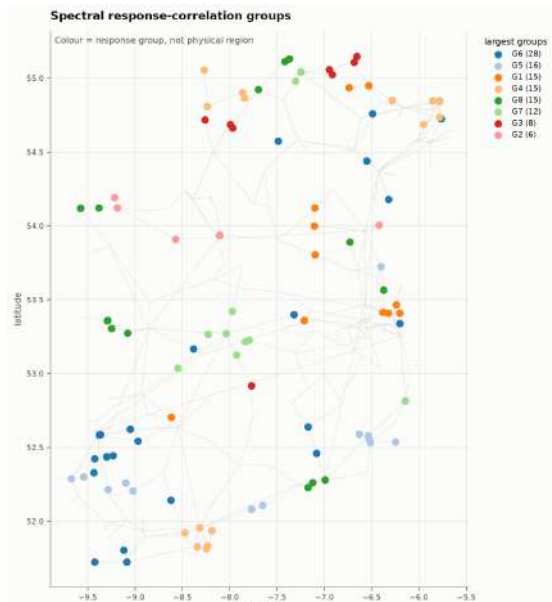
$$L_{\text{corr}} = I - D^{-1/2} R D^{-1/2}.$$

Order eigenvalues as $0 \leq \lambda_1 \leq \dots \leq \lambda_q$;
 k -means on eigenvectors based on eigengaps



Exploratory grouping of optimisation responses
only.

Spectral groups on the grid



Section 2: Policy Side

Strategy of Dispatch-Down prevention

Incentive Structures for generators

3.3 Dispatch Down Strategy

Constraint groups are used to dispatch down a set of generation nodes at once. Is there an optimal strategy to ensure a safe and effective incentive structure is enforced?

Without new incentives, new developers continue clustering in these congested zones—exacerbating grid bottlenecks, forcing deeper dispatch-down cuts on neighboring generators, and driving up system compensation costs.

What is our strategy

- Our approach is to create a policy that distinguishes between preexisting renewable generators and new generators and fines new generators the exact cost of the additional MegaWatt Hours that are dispatched-down due to operation of new farms.
- This fine will be distributed back to the neighboring developers so the cost of the constraint payments and bid prices burdened by taxpayers does not rise due to the new generators.
- It will also create an incentive for prospective developers to develop in new areas with fewer preexisting generators and circuit lines that are not overburdened, thereby not only reducing dispatch-down but also increasing the absolute output of renewable generators.

Incentive Structure Algorithm

Phase 1: Calculating Dispatch-Down

- 1 At the beginning of each fixed time interval, as decided by the Transmission Service Operator (TSO), we run the current dispatch-down model including all the (new and preexisting) generators and calculate how much dispatch-down each preexisting generator must endure (D_i).
- 2 Then, we run the dispatch-down model once more (under identical weather, demand, network conditions, as well as any other constraints currently considered) but removing all the new generators. Calculate the new dispatch-down for each preexisting generator (D_{ic}).
- 3 To find how much additional dispatch-down the new generators must endure due to the addition of the new generators, we calculate

$$\Delta D_i = \max(0, D_i - D_{ic})$$

- 4 The total reduction in power for each generator due to the new nodes is

$$P_i^{\text{rc}} = (P_i)(\Delta D_i)$$

Incentive Structure Algorithm

Phase 2: Calculating the harm caused by the new generators

- 1 Having calculated the time rate of power by which each preexisting generator's output must be reduced (P_i^{reduced}), we can calculate the raw financial damage done as

$$C_i = (\Delta P_i^{rc})(\Delta t)(p)$$

- Δt - time between each execution
- p - price per MegaWatt Hour

- To ensure we do not process fines that are smaller than the cost of administering the fine, we correct C_i to get

$$C_i = \begin{cases} C_i^{\text{prov}}, & \text{if } C_i^{\text{prov}} \geq \varepsilon \\ 0, & \text{if } C_i^{\text{prov}} < \varepsilon \end{cases}$$

- The total cost of dispatch-down over all the preexisting generators is therefore

$$C_{\text{total}} = \sum_i C_i$$

Phase 3: Calculating the fine for each new generator

- ① Using the currently existing process, we hold the values of each new generator's impact (shift factor) on a given congested line, given as $S_{j,\ell}$
- ② By definition of the shift factor, the exact megawatts forced into the congested line ℓ by new generator j is $I_{j,\ell} = \max(0, S_{j,\ell})(A_j)$
 - A_j - Actual Power of the j th new generator (MW)
- ③ The fraction of congestion on line ℓ caused by the new generators specifically caused by generator j is

$$f_{j,\ell} = \frac{I_{j,\ell}}{\sum_j I_{k,\ell}}$$

Incentive Structure Algorithm

Phase 3: Calculating the fine for each new generator

- ④ It immediately follows that the fine on new generator j for causing congestion on line ℓ should be the fraction of the congestion it caused times the cost of dispatch-down on all the preexisting generators, so

$$F_{j,\ell} = f_{j,\ell} C_{total}$$

- ⑤ We propose embedding our algorithm directly into SEMO's weekly ex-post settlement software. Each week, the algorithm calculates the exact incremental dispatch-down caused by the new generator. SEMO's software would automatically deduct this congestion fee from the new farm's weekly energy payout and credit the affected legacy farms. Leveraging SEMO's existing infrastructure eliminates the need for new administrative agencies or outsourcing invoicing; it compensates preexisting generators without passing congestion costs onto taxpayers or consumers.

References to 1st section

-  F. Murtagh and P. Contreras, “Algorithms for hierarchical clustering: an overview,” *WIREs Data Mining and Knowledge Discovery*, vol. 2, no. 1, pp. 86–97, 2012. doi:10.1002/widm.53.
-  B. Stott, J. Jardim, and O. Alsac, “DC Power Flow Revisited,” *IEEE Transactions on Power Systems*, vol. 24, no. 3, pp. 1290–1300, 2009. doi:10.1109/TPWRS.2009.2021235.
-  F. Capitanescu et al., “State-of-the-art, challenges, and future trends in security constrained optimal power flow,” *Electric Power Systems Research*, vol. 81, no. 8, pp. 1731–1741, 2011. doi:10.1016/j.epsr.2011.04.003.
-  K. Pearson, “Note on regression and inheritance in the case of two parents,” *Proceedings of the Royal Society of London*, vol. 58, pp. 240–242, 1895. doi:10.1098/rspl.1895.0041.
-  A. Y. Ng, M. I. Jordan, and Y. Weiss, “On spectral clustering: analysis and an algorithm,” in *Advances in Neural Information Processing Systems 14*, pp. 849–856, 2002.
-  U. von Luxburg, “A tutorial on spectral clustering,” *Statistics and Computing*, vol. 17, pp. 395–416, 2007. doi:10.1007/s11222-007-9033-z.

References to 2nd section

-  The Problem Solving Association CLG, TF Wind, and TPSA of Ireland, *Hackathon_2026_Problem_Sheet-2_2.pdf*, 2026. [Online]. Available: <https://github.com/farrencc/Hackathons>
-  EirGrid, *Ireland's Grid: How the Grid Works*, 2026. [Online]. Available: <https://www.eirgrid.ie/grid/how-grid-works/irelands-grid>
-  EirGrid, *Explore Ireland's Transmission System Map* (Video). YouTube, 2025. [Online]. Available: <https://www.youtube.com/watch?v=P8re8N5bq1U>
-  Single Electricity Market Operator (SEMO), *Markets Overview*, EirGrid and SONI, 2026. [Online]. Available: <https://www.sem-o.com/markets>
-  EirGrid, *Renewable Electricity Support Scheme (RESS)*, 2026. [Online]. Available: <https://www.eirgrid.ie/industry/renewable-electricity-support-scheme-ress>

Questions & Discussion

We welcome your feedback and comments

Thank You

for your attention!

Team 1

On the Electric Grid: Framework & Policy Strategy

Hackathon Policy Document

Jose Cossich, Mykyta Grytensko

September 11, 2026

Contents

1	What problem did we tackle?	2
1.1	Background	2
1.2	Physical incentives to reduce dispatch-down	2
1.3	Economic incentives to reduce dispatch-down	2
2	Constraint Groups	3
3	Incentive Structures	3
3.1	Incentive Structures for Generators	3
3.2	Incentive Structure Algorithm	4

1 What problem did we tackle?

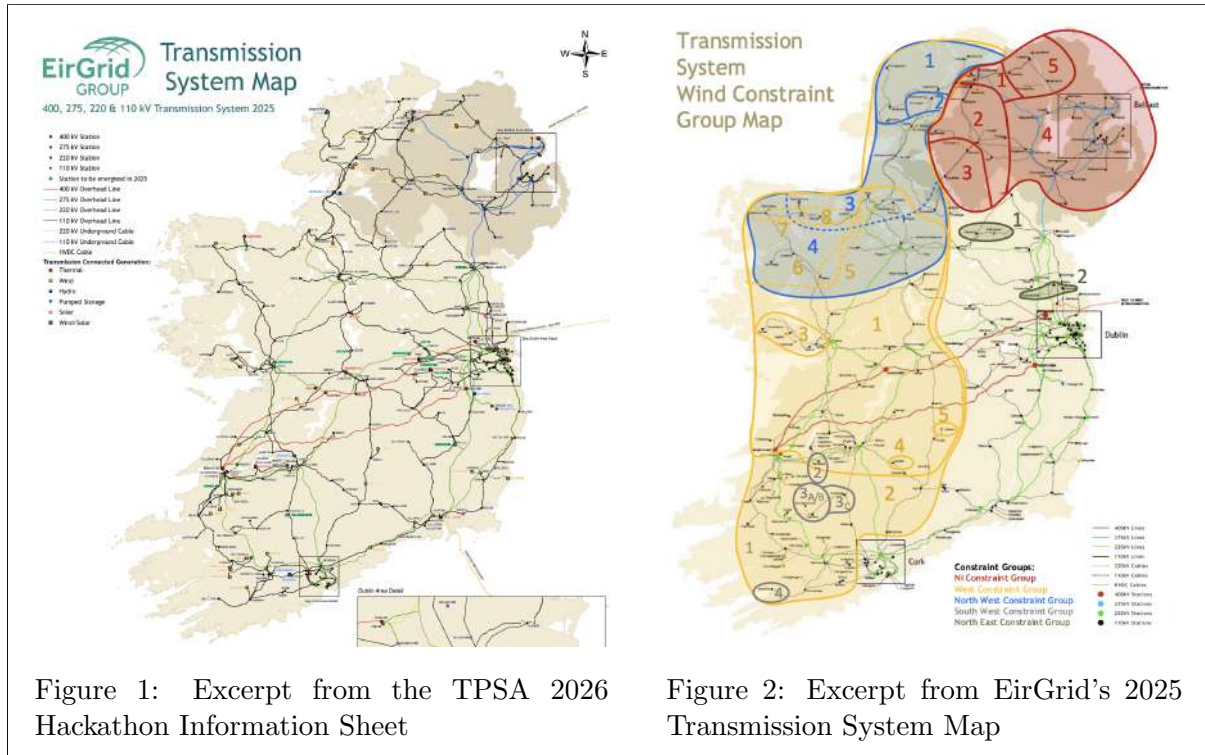
1.1 Background

In 2020, EirGrid met its target of sourcing 40% of electricity generation from renewables; the new objective is a considerably more ambitious 80% by 2030 [1]. The primary barrier is developing the physical and market infrastructure required to integrate substantial overcapacity, thereby ensuring supply security during periods of wind and solar scarcity.

1.2 Physical incentives to reduce dispatch-down

We can represent the electrical grid as a graph with generators as nodes and edges as the circuit system composed of Underground Cables (UGC), Overhead Lines (OHL), and High-Voltage Direct Current (HVDC) Subsea Interconnectors [2]. Some edges are more central than others, that is, more current wants to go through specific circuit lines to supply electricity demand than through others.

This causes complications because each line has a thermal capacity rating that describes how much total current can flow on any given edge at any given time before the lines overheat, sag, and can short-circuit. Circuit lines that exceed their thermal limit whose currents are not reduced in time force their current through adjacent circuits, thereby causing cascading blackouts. This serious risk, particularly given the antiquated state of many lines, makes it especially desirable to discover which nodes have the highest impact on central edges, so as to reduce their generation (called *dispatch-down*) and not overload the circuit lines. The groups of nodes with the highest impact on the central edges are called *constraint groups* and targeting them for dispatch-down means not needing to dispatch-down other generators with less of an impact on congested lines.



1.3 Economic incentives to reduce dispatch-down

Currently, generators operate under firm and non-firm connection agreements with Eirgrid. Firm connection agreements guarantee generators compensation if dispatched down equal to the

profit what they would have made otherwise; under non-firm agreements, generators connect faster but accept uncompensated dispatch-down.

Constraint payments for firm agreements are recovered through the Imperfections Charge levied on electricity suppliers by the Single Electricity Market Operator (SEMO) [4], which is added directly to the cost paid by electricity supply companies before reselling to consumers. Reducing constraint events directly shrinks this cost.

Similarly, though non-firm generators do not receive direct constraint payments when turned off, developers forecast their expected dispatch-down losses before bidding in the Renewable Electricity Support Scheme (RESS) public auctions where they bid for 15-20 year-long contracts with a guaranteed price per MegaWatt Hour. If a developer expects their energy to be dispatched-down and be uncompensated, they submit higher bids; since the RESS is funded by the Public Service Obligation (PSO) levy added to consumer electricity bills, less dispatch-down risk lowers auction bids and PSO levies (taxes).

2 Constraint Groups

3.2 Constraint Group Generation

Build a correlation map Complex systems as they exist on graph networks, as in the continuous limit, often have bizarre or obscure non-local effects. That is to say, high correlations in quantities measured at geographic distances. It is the goal of this problem to investigate and potentially build a map which shows these non-local effects in the grid. Currently one metric EirGrid uses in this domain is the *shift factor*. It is worth questioning if this is the best tool for the job. ¹

Investigate methods of generating groups There are multiple ways that constraint groups can be generated in principle. One of them also uses this shift factor as a metric. One might apply clustering or community detection methods as another means of generation. But these are far from exhaustive. It is important to develop these with the mindset of N , $N - 1$, $N - 1 - 1$ fault tolerance that EirGrid abides by.

3.4 Full Grid Analysis

Extend your analysis of the North-West region to cover the entire grid. Although a more computational intensive problem there are other features to consider as well such as interconnectors.

3 Incentive Structures

3.3 Dispatch Down Strategy

Constraint groups are used to dispatch down a set of generation nodes at once. Is there an optimal strategy to ensure a safe and effective incentive structure is enforced?

Figure 3: Excerpt from the TPSA 2026 Hackathon Information Sheet [1]

3.1 Incentive Structures for Generators

In Ireland, wind generation is geographically concentrated far from major demand centers, overloading local transmission corridors and creating a tragedy of the commons on the power grid. Without new incentives, new developers continue clustering in these congested zones—exacerbating grid bottlenecks, forcing deeper dispatch-down cuts on neighboring generators, and driving up system compensation costs.

Our approach is to create a policy that distinguishes between preexisting renewable generators and new generators and fines new generators the exact cost of the additional MegaWatt Hours that are dispatched-down due to operation of new farms. This fine will be distributed back to the neighboring developers so the cost of the constraint payments and bid prices burdened by taxpayers does not rise due to the new generators. It will also create an incentive for prospective developers to develop in new areas where less preexisting generators are and circuit lines are not overburdened, thereby not only reducing dispatch-down but also increasing the absolute output of renewable generators.

This will be enforced legally when that developers apply for new generators to be built and connected to the grid.

3.2 Incentive Structure Algorithm

We define the following terms to calculate the appropriate fine for each new generator imputed into the grid.

ε	Cost of implementing each fine (€)
D_i	Dispatch-down of the i th preexisting generator in the system with all new nodes (%)
D_{ic}	Dispatch-down of the i th preexisting generator in the counterfactual (%)
P_i	Theoretical Maximum Power (MW) of the i th preexisting generator
Δt	Settlement interval duration (hours)
p	RESS Contract Strike Price (€/MWh)
C_{total}	Total financial compensation owed to legacy farms (€)
$S_{j,\ell}$	Shift factor of the j th new generator on circuit line ℓ (%)
A_j	Actual Power of the j th new generator (MW)
$I_{j,\ell}$	Power added to congested line ℓ by the j th new node (MW)
$f_{j,\ell}$	Fraction of impact on congested line ℓ caused by the j th new node
$F_{j,\ell}$	Fine on the j th new generator for causing congestion on circuit line ℓ (€)

Before running the algorithm, we calculate the administrative cost of fining a developer for each new generator ε .

Phase 1: Calculating Dispatch-Down

1. At the beginning of each fixed time interval, as decided by the Transmission Service Operator (TSO), we run the current dispatch-down model including all the (new and preexisting) generators and calculate how much dispatch-down each preexisting generator must endure (D_i).
2. Then, we run the dispatch-down model once more (under identical weather, demand, network conditions, as well as any other constraints currently considered) but removing all the new generators. Calculate the new dispatch-down for each preexisting generator (D_{ic}).
3. To find how much additional dispatch-down the new generators must endure due to the addition of the new generators, we calculate $\Delta D_i = \max(0, D_i - D_{ic})$
4. The total reduction in power for each generator due to the new nodes is $P_i^{\text{rc}} = (P_i)(\Delta D_i)$

Phase 2: Calculating the harm experienced by each preexisting generator

1. Having calculated the time rate of power by which each preexisting generator's output must be reduced (P_i^{rc}), we can calculate the raw financial damage done to each preexisting

generator by multiplying this by the time between each execution of the algorithm (Δt) as well as the price per MegaWatt Hour p to get a provisional $C_i = (\Delta P_i^{rc})(\Delta t)(p)$

2. To ensure we do not process fines that are smaller than the cost of administering the fine, we correct C_i to get

$$C_i = \begin{cases} C_i^{\text{prov}}, & \text{if } C_i^{\text{prov}} \geq \varepsilon \\ 0, & \text{if } C_i^{\text{prov}} < \varepsilon \end{cases}$$

The total cost of dispatch-down over all the preexisting generators is therefore

$$C_{\text{total}} = \sum_i C_i$$

Phase 3: Calculating the fine for each new generator and administration

1. Using the currently existing process, we hold the values of the each new generator's impact (shift factor) on a given congested line $S_{j,\ell}$
2. By definition of shift factor, the exact megawatts forced into the congested line ℓ by new generator j is $I_{j,\ell} = \max(0, S_{j,\ell})(A_j)$
3. The fraction of congestion on line ℓ caused by the j th new generators is

$$f_{j,\ell} = \frac{I_{j,\ell}}{\sum_j I_{k,\ell}}$$

4. It immediately follows that the fine on new generator j for causing congestion on line ℓ should be the fraction of the congestion it caused times the cost of dispatch-down on all the preexisting generators, so

$$F_{j,\ell} = f_{j,\ell} C_{\text{total}}$$

5. We propose embedding our algorithm directly into SEMO's weekly ex-post settlement software. Each week, the algorithm calculates the exact incremental dispatch-down caused by the new generator. SEMO's software would automatically deduct this congestion fee from the new farm's weekly energy payout and credit the affected legacy farms. Leveraging SEMO's existing infrastructure eliminates the need for new administrative agencies or outsourcing invoicing; it compensates preexisting generators without passing congestion costs onto taxpayers or consumers.

References

- [1] The Problem Solving Association CLG, TF Wind, and TPSA of Ireland, *Hackathon_2026_Problem_Sheet-2_2.pdf*, 2026. [Online]. Available: <https://github.com/farrenc/Hackathons>
- [2] EirGrid, *Ireland's Grid: How the Grid Works*, 2026. [Online]. Available: <https://www.eirgrid.ie/grid/how-grid-works/irelands-grid>
- [3] EirGrid, *Explore Ireland's Transmission System Map* (Video). YouTube, 2025. [Online]. Available: <https://www.youtube.com/watch?v=P8re8N5bqlU>
- [4] Single Electricity Market Operator (SEMO), *Markets Overview*, EirGrid and SONI, 2026. [Online]. Available: <https://www.sem-o.com/markets>
- [5] EirGrid, *Renewable Electricity Support Scheme (RESS)*, 2026. [Online]. Available: <https://www.eirgrid.ie/industry/renewable-electricity-support-scheme-ress>

On the electric grid

Syrus Chan, Michal Mogielnicki

September 11, 2026

Abstract

We investigate operational group clustering, linear programming optimisation, and stochastic analyzation under network constraints to develop an alternative methodology for studying the behaviour and correlations of the electric grid at a higher system level.

Contents

1	Clustering	2
1.1	Distance matrix	2
1.2	Complete-linkage clustering	2
1.3	Global groups and local subgroups	3
2	DC power-flow optimisation	3
2.1	DC power-flow model	3
2.2	Transmission line flows	4
2.3	Global power balance	5
2.4	Generation weighting	5
2.5	Operational-group coupling	5
2.6	Group-membership representation	6
2.7	Optimisation formulation	7
2.8	Interpretation of the optimal response	7
2.9	N–1 security	8
2.10	Summary of the optimisation framework	9
3	Stochastic optimisation and correlation analysis	9
3.1	Stochastic generation and demand	9
3.2	Scenario-based optimisation	9
3.3	Node response vectors	9
3.4	Correlation structure	10
3.5	Spectral clustering of correlation groups	10
	References	11

1 Clustering

1.1 Distance matrix

Let N be the number of generators. For each intact or N-1 network state c from Section 2.9, write the susceptance matrix from Section 2.1 as B and initialize $B = 0$. For every oriented edge $e = (a, b)$ with reactance $x_e > 0$, define

$$b_e = \frac{1}{x_e}$$

and update

$$\begin{aligned} B_{aa} &\leftarrow B_{aa} + b_e, & B_{bb} &\leftarrow B_{bb} + b_e, \\ B_{ab} &\leftarrow B_{ab} - b_e, & B_{ba} &\leftarrow B_{ba} - b_e. \end{aligned}$$

For each connected component, choose one reference bus, set its angle to zero, and delete its row and column, giving $\hat{B}$.

For each generator i , let $p^{(i)}$ inject 1 MW at its bus and withdraw 1 MW across loads in the same component, proportional to mean demand. Delete reference entries and calculate

$$\hat{\theta}^{(i)} = \hat{B}^{-1} \hat{p}^{(i)}.$$

Restore zero at each reference bus.

For every branch-state $r = (c, e)$ with $|f_{tr}^0| > F_r$ in some hour, where $e = (a, b)$, record

$$A_{ir} = b_e (\theta_a^{(i)} - \theta_b^{(i)}).$$

Obtain target flow f_{tr}^0 from the line-unconstrained version of (16) with independent utilisation ($H = I$), and let $F_r > 0$ be the screening rating. Calculate

$$u_r = \sum_t \frac{\max\{0, |f_{tr}^0| - F_r\}}{F_r}, \quad w_r = \frac{u_r}{\sum_{r'} u_{r'}}.$$

If no branch-state exceeds its screening rating, the method returns one group. Otherwise, for every generator pair define

$$M_{ij} = \sqrt{\sum_r w_r (A_{ir} - A_{jr})^2}. \quad (1)$$

1.2 Complete-linkage clustering

Apply complete linkage [1] to M from (1). Let $\mathcal{P}_0 = \{Q_1, \dots, Q_q\}$ be the zero-distance partition, where $i \sim j$ if $M_{ij} = 0$. For distinct clusters $U, V \in \mathcal{P}_t$, define

$$\Delta(U, V) = \max_{i \in U, j \in V} M_{ij}.$$

While $|\mathcal{P}_t| > k$, choose distinct $U_{t+1}, V_{t+1} \in \mathcal{P}_t$ such that

$$\Delta(U_{t+1}, V_{t+1}) = \min_{\substack{U, V \in \mathcal{P}_t \\ U \neq V}} \Delta(U, V),$$

and set

$$\mathcal{P}_{t+1} = (\mathcal{P}_t \setminus \{U_{t+1}, V_{t+1}\}) \cup \{U_{t+1} \cup V_{t+1}\}.$$

For every remaining cluster C ,

$$\Delta(U_{t+1} \cup V_{t+1}, C) = \max\{\Delta(U_{t+1}, C), \Delta(V_{t+1}, C)\}.$$

After $q - k$ merges, $\mathcal{P}_{q-k}$ is the resulting partition into k clusters. Ties in the minimizing pair are resolved by a fixed rule.

1.3 Global groups and local subgroups

For $q \geq 3$, using the merges from Section 1.2, let $h_t = \Delta(U_t, V_t)$ denote the height of merge t , with

$$h_1 \leq \dots \leq h_{q-1}.$$

Define the height gaps

$$d_t = h_{t+1} - h_t, \quad 1 \leq t < q - 1.$$

Choose the first largest gap:

$$t^* = \min \left\{ t : d_t = \max_{1 \leq s < q-1} d_s \right\}, \quad k = q - t^*.$$

Cut after merge t^* . Apply the same rule within each global group to obtain local subgroups. Groups with fewer than three zero-distance classes remain unsplit. The local subgroups are used as operational groups.

With an optional size bound $S_{\max}$, instead start with all generators and recursively apply the same split only to oversized groups. Unsplit oversized groups remain.

2 DC power-flow optimisation

The operational clustering described in Section 1 provides groups of generators with similar electrical effects across the studied branch states. In this section, these operational groups are introduced directly into a constrained DC power-flow optimisation.

The purpose is not to assume that generators within a group are physically identical. Rather, group membership is used as a modelling assumption: generators assigned to the same operational group are required to share a common generation-utilisation factor. This allows the optimisation to study the behaviour of the grid when electrically similar generators are treated as a coordinated operational group.

2.1 DC power-flow model

Let n be the number of buses and let $m_i \in [0, 1]$ denote the generation utilisation factor at bus i . Let the available generation be represented by the diagonal matrix

$$P_g = \text{diag}(p_{g,1}, \dots, p_{g,n}),$$

where $p_{g,i}$ is the available generation at bus i . Let $d \in \mathbb{R}^n$ denote the demand vector.

The net injection at each bus is

$$P = P_g m - d, \tag{2}$$

where

$$m = [m_1 \quad \dots \quad m_n]^T.$$

Under the DC power-flow approximation [2], net power injections and bus phase angles satisfy

$$B\theta = P, \tag{3}$$

where

$$B \in \mathbb{R}^{n \times n}$$

is the network susceptance matrix and

$$\theta \in \mathbb{R}^n$$

is the vector of bus phase angles.

Combining (2) and (3) gives

$$B\theta = P_g m - d. \quad (4)$$

For a transmission line connecting buses i and j , with reactance $x_{ij} > 0$, define

$$b_{ij} = \frac{1}{x_{ij}}.$$

The entries of the susceptance matrix are

$$B_{ij} = \begin{cases} \sum_{k \in \mathcal{N}_i} \frac{1}{x_{ik}}, & i = j, \\ -\frac{1}{x_{ij}}, & i \neq j \text{ and } i, j \text{ are connected,} \\ 0, & \text{otherwise,} \end{cases}$$

where $\mathcal{N}_i$ contains the buses adjacent to bus i .

Consequently,

$$B\mathbf{1} = 0, \quad \mathbf{1}^T B = 0^T.$$

Since the susceptance matrix is singular before fixing a reference angle, one reference bus is selected in each connected component. Its phase angle is fixed to zero.

2.2 Transmission line flows

Let ℓ be the number of oriented transmission lines. Define

$$X \in \mathbb{R}^{\ell \times n}.$$

If row k of X represents a line directed from bus i to bus j , then

$$X_{km} = \begin{cases} \frac{1}{x_{ij}}, & m = i, \\ -\frac{1}{x_{ij}}, & m = j, \\ 0, & \text{otherwise.} \end{cases}$$

The vector of line flows is therefore

$$F = X\theta. \quad (5)$$

Transmission-line flows are restricted by lower and upper bounds:

$$f_{\min} \leq X\theta \leq f_{\max}, \quad (6)$$

where $f_{\min}, f_{\max} \in \mathbb{R}^\ell$ are vectors of flow bounds in MW. The simulation uses $f_{\min} = -\rho s_{\text{nom}}$ and $f_{\max} = \rho s_{\text{nom}}$.

These bounds are chosen operating limits within the DC model. They are therefore modelling parameters rather than a claim about operational limits used by EirGrid.

2.3 Global power balance

Premultiplying (4) by $\mathbf{1}^T$ gives

$$\mathbf{1}^T B \theta = \mathbf{1}^T P_g m - \mathbf{1}^T d.$$

Since

$$\mathbf{1}^T B = 0^T,$$

the total generation must equal the total demand:

$$\boxed{\mathbf{1}^T P_g m = \mathbf{1}^T d.} \quad (7)$$

For a disconnected network state, this balance must hold independently within every connected component. If C denotes a connected component, then

$$\boxed{\mathbf{1}_C^T P_g m = \mathbf{1}_C^T d.} \quad (8)$$

This prevents the optimisation from satisfying the overall balance by transferring power between disconnected components.

2.4 Generation weighting

To give greater preference to renewable generation, a weighting vector $W \in \mathbb{R}^n$ is introduced. The weight at bus i is defined as

$$W_i = \begin{cases} 5, & \text{if bus } i \text{ contains wind or hydro generation,} \\ 1, & \text{otherwise.} \end{cases}$$

The optimisation therefore maximises

$$W^T P_g m. \quad (9)$$

The ratio 5 : 1 is a modelling choice. It is intended to give renewable generation greater priority in the optimisation and should not be interpreted as an EirGrid dispatch rule.

2.5 Operational-group coupling

The clustering procedure described in Section 1 partitions the generators into operational groups

$$\mathcal{G} = \{G_1, G_2, \dots, G_K\}.$$

Identify each group with its generators' buses; co-located generators remain together.

These groups are obtained from the electrical-effect distance defined in Section 1.1. In particular, generators are clustered according to the similarity of their electrical effects across the studied branch states.

The resulting group membership is then introduced explicitly into the optimisation. If two buses i and j belong to the same operational group,

$$i, j \in G_k,$$

their generation-utilisation variables are constrained to be equal:

$$\boxed{m_i = m_j = m_{G_k}.} \quad (10)$$

More generally, every bus belonging to group G_k satisfies

$$m_i = m_{G_k}, \quad i \in G_k, \quad (11)$$

where

$$m_{G_k} \in [0, 1]$$

is the common utilisation factor of group G_k .

This means that the clustering is not used only as a descriptive label. Instead, it becomes an explicit coupling constraint in the optimisation. The optimisation determines one utilisation level for each operational group rather than independently determining the utilisation of every generator.

Importantly, this does not imply that generators within the same group have identical available capacities. If $p_{g,i}$ and $p_{g,j}$ are different, their dispatched generation is still different in magnitude:

$$p_{g,i}m_{G_k} \neq p_{g,j}m_{G_k}$$

in general. What is shared is the utilisation factor.

For example, if

$$G_k = \{i, j, p, q\},$$

then

$$m_i = m_j = m_p = m_q = m_{G_k},$$

while the total generation from this group is

$$\sum_{i \in G_k} p_{g,i}m_i = \left(\sum_{i \in G_k} p_{g,i} \right) m_{G_k}. \quad (12)$$

2.6 Group-membership representation

The group constraints can be represented compactly using a group-membership matrix.

Define

$$H \in \mathbb{R}^{n \times K}$$

by

$$H_{ik} = \begin{cases} 1, & i \in G_k, \\ 0, & \text{otherwise.} \end{cases}$$

Let

$$m_G = [m_{G_1} \quad m_{G_2} \quad \cdots \quad m_{G_K}]^T$$

contain the group-level utilisation variables.

The bus-level utilisation vector is then

$$\boxed{m = Hm_G.} \quad (13)$$

Substituting (13) into the DC power-flow equation gives

$$\boxed{B\theta = P_g H m_G - d.} \quad (14)$$

Similarly, the line flows become

$$F = X\theta, \quad (15)$$

with θ determined by the group-level dispatch through (14).

The total generation of group G_k can be written as

$$P_{G_k} = \sum_{i \in G_k} p_{g,i} m_{G_k} = \left(\sum_{i \in G_k} p_{g,i} \right) m_{G_k}.$$

Thus, the operational clustering reduces the number of independent generation-utilisation decisions from the individual bus level to the group level.

2.7 Optimisation formulation

Let $\mathcal{N}_r$ contain all buses except one reference bus from each connected component.

Using the operational-group mapping

$$m = H m_G,$$

the computational problem is

$$\begin{array}{ll} \max_{m_G, \theta} & W^T P_g H m_G \\ \text{subject to} & (B\theta)_i = p_{g,i}(H m_G)_i - d_i, \quad i \in \mathcal{N}_r, \\ & \mathbf{1}_C^T P_g H m_G = \mathbf{1}_C^T d, \quad \text{for every component } C, \\ & f_{\min} \leq X\theta \leq f_{\max}, \\ & \theta_{r_C} = 0, \quad \text{for every component } C, \\ & 0 \leq m_{G_k} \leq 1, \quad k = 1, \dots, K. \end{array} \quad (16)$$

The objective maximises weighted dispatched generation while the constraints enforce DC nodal balance, component-wise power balance, transmission-line limits, reference angles and feasible group utilisation levels.

The key difference from an uncoupled bus-level dispatch is the condition

$$m_i = m_j \quad \text{whenever} \quad i, j \in G_k.$$

Consequently, the electrical-effect clustering directly influences the feasible set of the optimisation.

2.8 Interpretation of the optimal response

The group constraint also determines how optimal responses are represented under changing operating conditions.

Let

$$s = 1, \dots, S$$

index stochastic operating scenarios. For each scenario s , the optimisation produces a group-level optimal utilisation vector

$$m_G^{(s)} = \begin{bmatrix} m_{G_1}^{(s)} & \dots & m_{G_K}^{(s)} \end{bmatrix}^T.$$

The corresponding bus-level response is

$$m^{(s)} = H m_G^{(s)}. \quad (17)$$

Therefore, if

$$i, j \in G_k,$$

then for every scenario

$$\boxed{m_i^{(s)} = m_j^{(s)} = m_{G_k}^{(s)}}. \quad (18)$$

The response vectors of two members of the same operational group are therefore identical:

$$n_i = \begin{bmatrix} m_i^{(1)} \\ m_i^{(2)} \\ \vdots \\ m_i^{(S)} \end{bmatrix}, \quad n_j = \begin{bmatrix} m_j^{(1)} \\ m_j^{(2)} \\ \vdots \\ m_j^{(S)} \end{bmatrix},$$

and

$$\boxed{n_i = n_j}. \quad (19)$$

Thus, the subsequent stochastic response analysis builds on the operational grouping rather than being independent of it. The first clustering determines which generators are coupled operationally; the scenario optimisation determines how those coupled groups respond to changing grid conditions.

2.9 N-1 security

For transmission-line N-1 analysis [3], let $\mathcal{C}_1$ contain the intact network and every state formed by removing one transmission line.

For each contingency state

$$c \in \mathcal{C}_1,$$

the network matrices are rebuilt using the surviving transmission lines. The corresponding optimisation problem is then solved using the same operational-group structure.

Let E_c denote the set of transmission lines surviving in contingency state c . For each studied hour

$$t \in \mathcal{T},$$

let the optimal phase angles be

$$\theta^{(t,c)}.$$

The network passes this test when every corresponding optimisation is feasible and

$$f_{\min,e} \leq \left(X^{(c)} \theta^{(t,c)} \right)_e \leq f_{\max,e}$$

for every

$$t \in \mathcal{T}, \quad c \in \mathcal{C}_1, \quad e \in E_c.$$

Equivalently, the complete N-1 feasibility condition is

$$\boxed{f_{\min,e} \leq \left(X^{(c)} \theta^{(t,c)} \right)_e \leq f_{\max,e}, \quad \forall t \in \mathcal{T}, c \in \mathcal{C}_1, e \in E_c}. \quad (20)$$

Each contingency state permits independent redispatch of the operational groups. Thus, the utilisation vector

$$m_G^{(t,c)}$$

may differ between contingency states.

This is an independent-redispatch N-1 feasibility analysis. It is not a preventive common-dispatch N-1 formulation, and it does not provide an AC or dynamic security guarantee.

2.10 Summary of the optimisation framework

The complete relationship between the operational clustering and the DC optimisation can be summarised as

$$\boxed{\begin{array}{l} \text{electrical-effect clustering} \longrightarrow \text{operational groups} \longrightarrow m = Hm_G \\ \longrightarrow \text{DC-constrained optimisation} \longrightarrow m_G^{(s)} \end{array}}$$

The clustering therefore determines the structure of the decision variables, while the DC optimisation determines the feasible and generation-preferred response of those groups under the network constraints.

The resulting scenario-dependent group responses provide the basis for the stochastic response analysis in Section 3.

3 Stochastic optimisation and correlation analysis

The deterministic formulation (16) assumes that available generation P_g and demand d are known. Replace them with stochastic scenarios.

3.1 Stochastic generation and demand

Let $s = 1, \dots, S$ index operating scenarios, with

$$P_g^{(s)} = \text{diag} \left(p_{g,1}^{(s)}, \dots, p_{g,n}^{(s)} \right), \quad d^{(s)} = \begin{bmatrix} d_1^{(s)} & \dots & d_n^{(s)} \end{bmatrix}^T.$$

Let

$$\xi = (p_{g,1}, \dots, p_{g,n}, d_1, \dots, d_n)^T \sim \mathcal{N}(\mu, \Sigma) \mid \xi \in \Omega,$$

where Σ contains cross-bus dependence and $\Omega = \{0 \leq p_{g,i} \leq \bar{p}_{g,i}, d_i \geq 0\}$ enforces physical bounds.

3.2 Scenario-based optimisation

For each scenario, substitute $P_g^{(s)}$ and $d^{(s)}$ into (16); let $m^{(s)}$ be a selected optimal utilisation vector. Network parameters B , X , and line ratings remain fixed. Infeasible scenarios must be reported rather than assigned a response. If an optimum is non-unique, correlations may depend on the stated tie-breaking rule. The feasible solutions form the indexed sample

$$\mathcal{U} = \left(m^{(1)}, \dots, m^{(S)} \right).$$

3.3 Node response vectors

For bus i , define

$$\mathbf{n}_i = \begin{bmatrix} m_i^{(1)} \\ m_i^{(2)} \\ \vdots \\ m_i^{(S)} \end{bmatrix} \in \mathbb{R}^S.$$

For $S \geq 2$, its empirical mean and variance are

$$\bar{m}_i = \frac{1}{S} \sum_{s=1}^S m_i^{(s)}, \quad \sigma_{m_i}^2 = \frac{1}{S-1} \sum_{s=1}^S (m_i^{(s)} - \bar{m}_i)^2.$$

These formulas assume all S scenarios are feasible; otherwise use and report the feasible subset consistently.

3.4 Correlation structure

Let $I = \{i : \sigma_{m_i}^2 > 0\}$ and $q = |I|$. For $i, j \in I$, define the Pearson response correlation [4]

$$R_{ij} = \frac{\sum_{s=1}^S (m_i^{(s)} - \bar{m}_i)(m_j^{(s)} - \bar{m}_j)}{\sqrt{\sum_{s=1}^S (m_i^{(s)} - \bar{m}_i)^2} \sqrt{\sum_{s=1}^S (m_j^{(s)} - \bar{m}_j)^2}}.$$

Then $R \in \mathbb{R}^{q \times q}$. It measures linear similarity of optimal responses, not physical connectivity or causation. Within-group correlations are one whenever response variance is positive, by construction.

3.5 Spectral clustering of correlation groups

Assuming $q \geq 2$ and positive degrees, use normalised spectral clustering [5, 6]. Convert correlation to affinity

$$S_{ij} = \begin{cases} \frac{1 + R_{ij}}{2}, & i \neq j, \\ 0, & i = j. \end{cases}$$

Let $D = \text{diag}(\sum_j S_{ij})$ and form

$$L_{\text{corr}} = I - D^{-1/2} S D^{-1/2}.$$

Order its eigenvalues as

$$0 \leq \lambda_1 \leq \dots \leq \lambda_q$$

Fix $k_{\max} < q$ before analysis and choose the first largest admissible eigengap:

$$k^* = \min \left\{ k : \lambda_{k+1} - \lambda_k = \max_{1 \leq r \leq k_{\max}} (\lambda_{r+1} - \lambda_r) \right\}.$$

Let $v_1, \dots, v_{k^*}$ be orthonormal eigenvectors for the smallest eigenvalues and set

$$Y = [v_1 \ \dots \ v_{k^*}], \quad z_i = \frac{Y_{i,:}}{\|Y_{i,:}\|_2}.$$

Apply k^* -means to the rows z_i . The result groups similar optimisation responses only; it is not a partition by physical proximity, electrical connectivity, or causation. The eigengap choice is a heuristic.

References

- [1] F. Murtagh and P. Contreras, “Algorithms for hierarchical clustering: an overview,” *WIREs Data Mining and Knowledge Discovery*, vol. 2, no. 1, pp. 86–97, 2012. doi:10.1002/widm.53.
- [2] B. Stott, J. Jardim, and O. Alsac, “DC Power Flow Revisited,” *IEEE Transactions on Power Systems*, vol. 24, no. 3, pp. 1290–1300, 2009. doi:10.1109/TPWRS.2009.2021235.
- [3] F. Capitanescu et al., “State-of-the-art, challenges, and future trends in security constrained optimal power flow,” *Electric Power Systems Research*, vol. 81, no. 8, pp. 1731–1741, 2011. doi:10.1016/j.epsr.2011.04.003.
- [4] K. Pearson, “Note on regression and inheritance in the case of two parents,” *Proceedings of the Royal Society of London*, vol. 58, pp. 240–242, 1895. doi:10.1098/rspl.1895.0041.
- [5] A. Y. Ng, M. I. Jordan, and Y. Weiss, “On spectral clustering: analysis and an algorithm,” in *Advances in Neural Information Processing Systems 14*, pp. 849–856, 2002.
- [6] U. von Luxburg, “A tutorial on spectral clustering,” *Statistics and Computing*, vol. 17, pp. 395–416, 2007. doi:10.1007/s11222-007-9033-z.